

16- and 18-carbon acyl chains, a primary role in the formation of long acyl chains for membrane synthesis is questionable.

Specific roles for elongation in peroxisomes have not been defined but this organelle may produce the very long chain saturated and polyenoic fatty acids of 24–36 carbons. Import of very long chain fatty acids into peroxisomes is defective in patients with the inherited disease, X-linked adrenoleukodystrophy, and this results in increased cellular and serum levels of very long chain fatty acids. The gene whose mutation leads to the degenerative disease apparently codes for a transporter required for importing very long chain fatty acids into peroxisomes for subsequent activation by a peroxisomal very long chain fatty acyl-CoA synthase. The fate of long chain acyl-CoAs formed in peroxisomes is not yet understood.

Studies using the yeast *Saccharomyces cerevisiae* resulted in the isolation of three genes required for elongation of fatty acyl-CoAs — *ELO1*, *ELO2*, and *ELO3* [7]. *ELO1* encoded protein prefers to elongate 14-carbon fatty acyl-CoAs, the *ELO2* encoded protein elongates 20- and 22-carbon chains, and the *ELO3* gene product elongates a broad range of fatty acyl CoA-chains to 26-carbon products. Inactivation of the *ELO2* or *ELO3* gene results in defective sphingolipid synthesis. In both yeast and mammals, sphingolipids store fatty acids greater than 24 carbons indicating that *ELO2* and *ELO3* elongation enzymes are coupled directly to synthesis of sphingolipids. Inactivation of *ELO2* or *ELO3* genes gives yeast cells the ability to survive in the absence of v-SNARES required for targeting of membrane vesicles from the Golgi to the plasma membrane. This suggests a role for long chain fatty acids or sphingolipids, mediated by *ELO2* and *ELO3* encoded proteins, in regulated fusion of biological membranes.

3. Formation of monounsaturated fatty acids by oxidative desaturation

The spectrum of fatty acyl chains needed to meet requirements of lipid storage, membrane synthesis and maintenance, and lipid regulation of cellular processes cannot be provided by diet or de novo fatty acid synthesis and chain elongation alone. Unsaturated fatty acids also must be synthesized in cells, supplemented by essential fatty acids in the diet.

3.1. Nomenclature to describe double bonds

Before discussing desaturation enzymes, abbreviations to describe the number and position of double bonds in acyl chains (Table 1) will be outlined using linoleic acid as an example.

- (1) To indicate that linoleic acid is an 18-carbon fatty acid with two double bonds, the shorthand 18:2 is used. The number before the colon denotes the number of carbon atoms and the number following refers to the number of double bonds.
- (2) To assign the position of an individual double bond or specificity of an enzyme inserting it, the delta (Δ) nomenclature is used. This describes a bond position relative to the carboxyl (number one) carbon of the acyl chain. For linoleic acid, the double bonds are in the $\Delta 9$ and $\Delta 12$ positions, between carbons 9–10 and

12–13, and are introduced into the 18-carbon chain by $\Delta 9$ and $\Delta 12$ desaturase enzymes.

- (3) To designate an individual fatty acid within a 'family' of structurally related acids, the *n*- nomenclature is used. Here, the position of the first double bond from the methyl end is described. Thus, 18:2 *n*-6 indicates that the double bond closest to the methyl end is 6 carbons from the methyl group and in the $\Delta 12$ position. This convention is particularly useful in designating groups of fatty acids derived from the same parent compound as metabolic reactions do not occur on the methyl side of an existing double bond. Among other conventions, the $\omega 6$ designation is still widely used to describe the position of a double bond from the methyl end (ω -carbon). It is similar to the *n*- nomenclature. Linoleic acid is an $\omega 6$ fatty acid.
- (4) To indicate the geometric configuration of a double bond, the designation is preceded by a *c*- for *cis* or *t*- for *trans*. Thus, *c,c*-18:2 *n*-6 distinguishes linoleic acid from a *trans*-containing isomer, such as a conjugated linoleic acid (CLA; e.g., 9*c*,11*t*-18:2). Generally, double bond configuration is *cis*.

3.2. Characteristics of monounsaturated fatty acid-forming desaturation enzymes

Monounsaturated fatty acids are formed in mammalian systems by direct oxidative desaturation (removal of two hydrogens resulting in the introduction of a double bond) of a preformed long chain saturated fatty acid. The oxygenase type of enzyme is associated with the ER of liver, mammary gland, brain, testes and adipose tissue. The $\Delta 9$ desaturase is the predominant, if not exclusive, desaturation enzyme for saturated acids in these tissues and is rate-limiting in the formation of 18:1 *n*-9. $\Delta 9$ desaturase acts on fatty acyl-CoA. For most tissues, 14- to 18-carbon saturated fatty acyl chains are good substrates, with stearoyl-CoA being most active. Reduced pyridine nucleotide is required and generally NADH is more active than NADPH. The $\Delta 9$ desaturase has an absolute requirement for molecular oxygen which acts as an electron acceptor for two pairs of hydrogens — one from NADH and the other from the fatty acyl-CoA.

The $\Delta 9$ desaturation system consists of three major protein components: (1) NADH-cytochrome *b5* reductase, (2) cytochrome *b5*, and (3) a desaturase domain that is cyanide-sensitive and rate-limiting in double bond formation (Fig. 3). In some systems, such as those in insects, cytochrome *P*-450 may substitute for cytochrome *b5*. Integral association of $\Delta 9$ desaturase with the ER has retarded characterization of the complex. Loss of activity during solubilization was largely overcome through use of controlled ratios of detergents to protein and combinations of mild extraction solvents. The purified desaturase component has one atom of non-heme iron per molecule as the prosthetic group and 62% non-polar amino acid residues. Thus, the desaturase is largely within the microsomal membrane, with the active centre exposed to the cytosol. Interaction with specific reagents suggests that arginyl residues play a role at the binding site for the negatively charged CoA moiety of the substrate and tyrosine residues are involved in chelation of the iron prosthetic group. Isolation, characterization and site-directed mutagenesis of the cDNA coding for mammalian $\Delta 9$ desaturase has confirmed the essential catalytic role of histidine residues.

NADH-cytochrome *b5* reductase and cytochrome *b5* are more readily solubilized

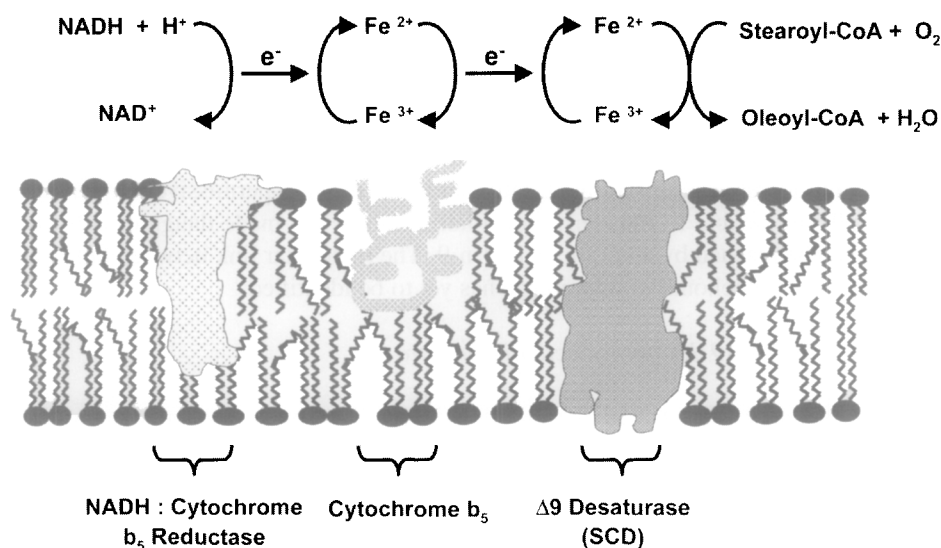


Fig. 3. Representation of the $\Delta 9$ (stearoyl-CoA) desaturase (SCD) complex, including electron transport proteins.

than the desaturase. Cytochrome b_5 has a hydrophilic region of 85 residues (including the NH₂-terminal) and the protein terminates in a hydrophobic COOH-terminal tail of 40 amino acids that attaches the protein to the membrane. It is uncertain whether one or two cytochrome b_5 molecules per complex are required to transfer two electrons from NADH to oxygen.

Stereochemical studies have shown that the *D*-hydrogens at positions 9 and 10 are removed concertedly to give a *cis* double bond without involving an oxygen-containing intermediate. Attempts to demonstrate a hydroxyacyl intermediate have been negative and hydroxyacyl-CoAs are not readily desaturated. In insect systems oxygen free radical is involved.

$\Delta 9$ desaturase is translated on soluble cytoplasmic polysomes with post-translational binding of iron and insertion into the ER. Plasmid expression vectors have been used to synthesize desaturase peptides that can be reconstituted with cytochrome b_5 and cytochrome b_5 reductase to give $\Delta 9$ desaturase activity. Dietary induction of $\Delta 9$ desaturase facilitated isolation of poly(A⁺) RNA and preparation of cDNA for stearoyl-CoA desaturase (SCD). mRNA increased 40- to 60-fold following refeeding of fasted animals and during induced differentiation of preadipocytes. Two genes (*SCD-1* and *SCD-2*) encoding isoforms of $\Delta 9$ desaturase with 87% homology were identified. Whereas liver exclusively expresses SCD-1, brain, spleen, heart and lymphocytes express only SCD-2; both SCD-1 and SCD-2 are expressed in adipose tissue, lung and kidney.

A structural gene for $\Delta 9$ desaturase from yeast (*OLE1*) has been isolated and characterized [16]. The *OLE1* gene encodes the only fatty acid desaturase in yeast and *OLE1* protein is found in the ER. The predicted protein amino acid sequence has a

high degree of similarity to mammalian SCD. Indeed, expression of rat SCD in yeast can substitute for yeast *OLE1*. If the *OLE1* gene is genetically inactivated, yeast are no longer viable, although life can be rescued in medium containing 14:1, 16:1, 18:1, or 18:2 fatty acids. A temperature-sensitive allele of the *OLE1* gene results in intact protein at 25°C but inactive protein at 37°C; at the latter temperature, there is an increase in saturated, and decrease in unsaturated, fatty acids. At the non-permissive temperature, dividing cells are unable to transfer mitochondria into newly made daughter cells, but the defect can be rescued by addition of 18:1 n-9. The molecular link between fatty acid desaturation and mitochondrial inheritance has yet to be identified.

3.3. *Modification of $\Delta 9$ desaturase activities in vitro*

Cyanide completely inhibits $\Delta 9$ desaturation in rat liver by acting on the terminal desaturase component (cyanide-sensitive factor), apparently related to accessibility to the non-heme iron. Some $\Delta 9$ desaturases (e.g., in insects and yeast) are not inhibited by cyanide.

Cyclopropenoid fatty acids found in sterula and cotton seeds are potent inhibitors of $\Delta 9$ desaturase. Sterculoyl- and malvaloyl-CoA (18- and 16-carbon derivatives with cyclopropene rings in the $\Delta 9$ position) specifically inhibit $\Delta 9$ desaturation. Hens fed meal containing cyclopropene fatty acids have decreased 18:1/18:0 ratios in their egg yolks. Cyclopropene acids have been used in vitro to differentially alter $\Delta 9$ and $\Delta 6$ desaturase activities, distinguishing relative contributions of these two enzymes in perinatal brain development.

3.4. *Age-related, dietary and hormonal regulation of $\Delta 9$ desaturase*

Extreme response to dietary alterations is a remarkable feature of $\Delta 9$ desaturase. Levels of *SCD-1* and *SCD-2* encoded $\Delta 9$ desaturase are highly regulated at the levels of transcription and of transcript and protein stability [6,8,9,15]. When rats are not fed for 12–72 h, liver $\Delta 9$ desaturase (*SCD-1*) activity declines to less than 5% of control values (Fig. 4). After refeeding, $\Delta 9$ desaturase activity increases dramatically to more than 2-fold above normal. The restoration has been termed ‘super-induction’ as levels of enzyme activity can be more than 100-fold above the fasted state, particularly when the rats are refed a fat-free diet enriched in carbohydrate or protein. Protein synthesis inhibitors and immunological techniques have been used to show that synthesis of the desaturase component is altered quantitatively. Liver *SCD-1* mRNA is dramatically altered by dietary changes; for example, a nearly 100-fold increase occurs in rat pups nursed by mothers on an essential fatty acid deficient diet compared to controls, and a 45-fold increase occurs in liver upon refeeding fasted mice with a fat-free, high-carbohydrate diet. Responses of liver enzyme to dietary intake probably explain the ‘circadian changes’ in $\Delta 9$ desaturase, where liver activities can fluctuate 4-fold over a 24-h period; highest activity (around midnight) corresponds to maximal food intake in the nocturnal rat.

In contrast to the liver enzyme, brain $\Delta 9$ desaturase (*SCD-2*) is little altered by dietary restrictions. This ensures continuing activity during crucial stages of brain

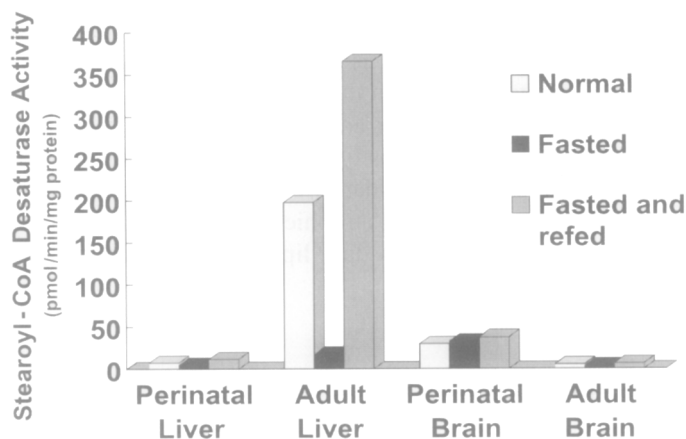


Fig. 4. Effects of fasting for 48 h and subsequent refeeding with a normal diet for 24 h on in vitro $\Delta 9$ desaturase activities of brain and liver from 10-day-old and adult rats. Adapted from Cook and Spence [29].

development. Brain $\Delta 9$ desaturase activity is greatest during the perinatal and suckling period in rats and is generally higher than in liver. However, when rats are weaned, brain $\Delta 9$ desaturase activity slowly declines. In contrast to the extreme changes in liver SCD-1 noted above, brain SCD-2 mRNA increases only 2-fold in neonates that are suckling mothers on an essential fatty acid deficient diet.

Transcription of *SCD-1* and *SCD-2* is regulated through specific elements in the promoter of each gene [8]. Although there are differences in the promoter region of *SCD-1* and *SCD-2* that lead to tissue-specific expression, shared regions appear to be responsible for decreased transcription in response to cholesterol (or conversely, increased transcription in response to sterol depletion). The regulation of SCD-1 and SCD-2 in response to cholesterol is through the binding of the transcription factor SREBP-1a and nuclear factor-Y (that heterodimerize) to three conserved regions within the *SCD-1* and *SCD-2* promoters. Addition of PUFA, particularly 18:2 n-6 or 20:4 n-6 [17], also decreases the level of SCD-1 and SCD-2 mRNA; whether this is due to decreased transcription or increased mRNA degradation, or both, is still unclear. SCD proteins also turn over rapidly in cells, possibly to ensure that transcriptional responses quickly alter SCD protein levels. The N-terminal 33-amino acids of SCD are required for turnover. The protease system responsible for degradation of SCD protein is not known.

Hormonal regulation of $\Delta 9$ desaturase is complex and not understood fully. Rats with genetic diabetes, or made diabetic by destruction of pancreatic β -cells, have depressed $\Delta 9$ desaturase activity in liver, mammary gland and adipose tissue. Insulin restores activity in vivo but is without effect in an in vitro assay. Insulin appears essential for basal transcription of the *SCD-1* gene and markedly induces transcription through a process requiring synthesis of a protein regulator. Significant changes in cytochrome *b5* and the reductase are not elicited by insulin. Other hormones and effectors, such as glucagon and cyclic AMP, do not alter $\Delta 9$ desaturase activity, whereas epinephrine and thyroxine do.

Studies of SCD mutations in mice support an integration of monoene formation with neutral lipid and lipoprotein metabolism as well as unpredicted effects on the skin and eyes [17–19]. Mice containing a natural mutation that inactivates SCD-1 display a dramatic decrease in synthesis of liver triacylglycerols and cholesterol esters, leading to lower levels of these lipids in very low and low density lipoproteins. Specific genetic inactivation of the mouse SCD-1 gene produces similar effects on triacylglycerols. In addition, these mice develop skin abnormalities and atrophied sebaceous glands (that secrete oil into the skin) and meibomian glands (that produce lipid secretions for the eye), resulting in an inability of these mice to blink.

4. Formation of polyunsaturated fatty acids

4.1. Characteristics in animal systems

All eukaryotic organisms contain polyenoic fatty acyl chains in their membrane lipids, and most mammalian tissues modify acyl chain composition by introducing more than one double bond.

- (1) The first double bond introduced into a saturated acyl chain is generally in the $\Delta 9$ position so that substrates for further desaturation contain either a $\Delta 9$ double bond or one derived from the $\Delta 9$ position by chain elongation. Relatively large amounts of 16:1 n-7 and 18:1 n-7 in neonatal rat brain are exceptions but the qualitative significance during brain development is unclear.
- (2) Like $\Delta 9$ desaturation that inserts the first double bond, further desaturation is an oxidative process requiring molecular oxygen, reduced pyridine nucleotide and electron transfer involving a cytochrome and related reductase.
- (3) Animal systems cannot introduce double bonds beyond the $\Delta 9$ position. Thus, second and subsequent double bonds are always inserted between an existing bond and the carboxyl end of the acyl chain, never on the methyl side (Fig. 5). Plants, on the other hand, introduce second and third double bonds between the existing double bond and the terminal methyl group (Chapter 4). Diatoms, *Euglena*, insects, snails and slugs can desaturate on either side of an existing bond; at least 15 insect species form 18:2 n-6 by de novo synthesis and some produce 20:4 n-6. Consequently, double bonds are found at the $\Delta 9$, $\Delta 6$, $\Delta 5$ and $\Delta 4$ positions as a result of desaturation in animals, at the $\Delta 9$, $\Delta 12$ and $\Delta 15$ positions in plants, and at the $\Delta 5$, $\Delta 6$, $\Delta 9$, $\Delta 12$ and $\Delta 15$ positions in insects and other invertebrates. Well-established evidence confirms distinct $\Delta 9$, $\Delta 6$ and $\Delta 5$ desaturases in many animal tissues. In contrast, the longstanding assumption that a $\Delta 4$ desaturase exists in the classical pathways of PUFA metabolism (Fig. 6) has not been confirmed by direct enzyme characterization. The abundance of long chain polyenoic acids containing $\Delta 4$ double bonds in tissues such as brain, retina and testes, and their formation in vivo and in vitro from more saturated precursors, indicate that mammalian tissues can produce PUFA with a $\Delta 4$ double bond. A new dimension to the classical pathway, erroneously thought to include direct $\Delta 4$ desaturation, is now recognized. In rat liver, fish and human monocytes (and probably other

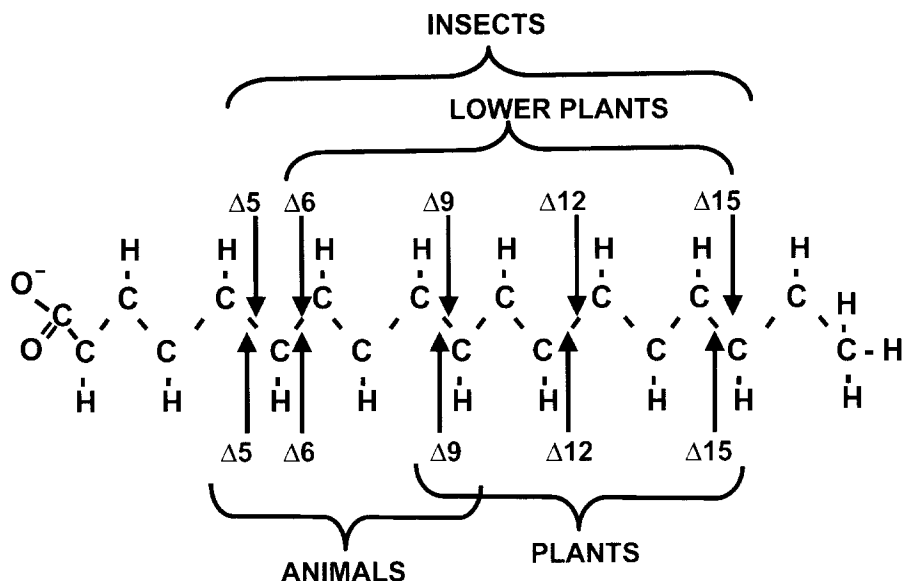


Fig. 5. Positions of fatty acyl chain desaturation by enzymes of animals, plants, insects and lower plants.

human tissues), formation of 22:6 n-3 from 22:5 n-3 or of 22:5 n-6 from 22:4 n-6 involves direct chain elongation of these respective precursors and then $\Delta 6$ desaturation followed by a 2-carbon chain shortening, the latter by β -oxidation in peroxisomes. This alternative to direct $\Delta 4$ desaturation has become known as the Sprecher pathway [20]. Comparative studies with tissue explants, primary cultures and neoplastic cells suggest that introduction of the $\Delta 4$ bond into 22:6 n-3 is characteristic of differentiated tissue and may be restricted in undifferentiated cells.

- (4) In most organisms, and certainly in higher animals, methylene interruption between double bonds is usually maintained with conjugated double bonds being rare.
- (5) All bonds introduced by oxidative desaturation in animals are in the *cis* geometric configuration.

4.2. Essential fatty acids — a contribution of plant systems

Requirements for PUFA cannot be met by *de novo* metabolic processes in mammalian tissues. Animals are absolutely dependent on plants (or insects) for providing double bonds in the $\Delta 12$ and $\Delta 15$ positions of the two major precursors of the n-6 and n-3 fatty acids, linoleic and linolenic acids. In animal tissues these acyl chains are converted to fatty acids containing 3–6 double bonds. Even very long chain PUFA of 28–36 carbons apparently do not have more than 6 double bonds.

4.3. Families of fatty acids and their metabolism

Relationships among fatty acids in the metabolic pathways can be evaluated by considering groups or families of fatty acids based on the parent unsaturated acid in the

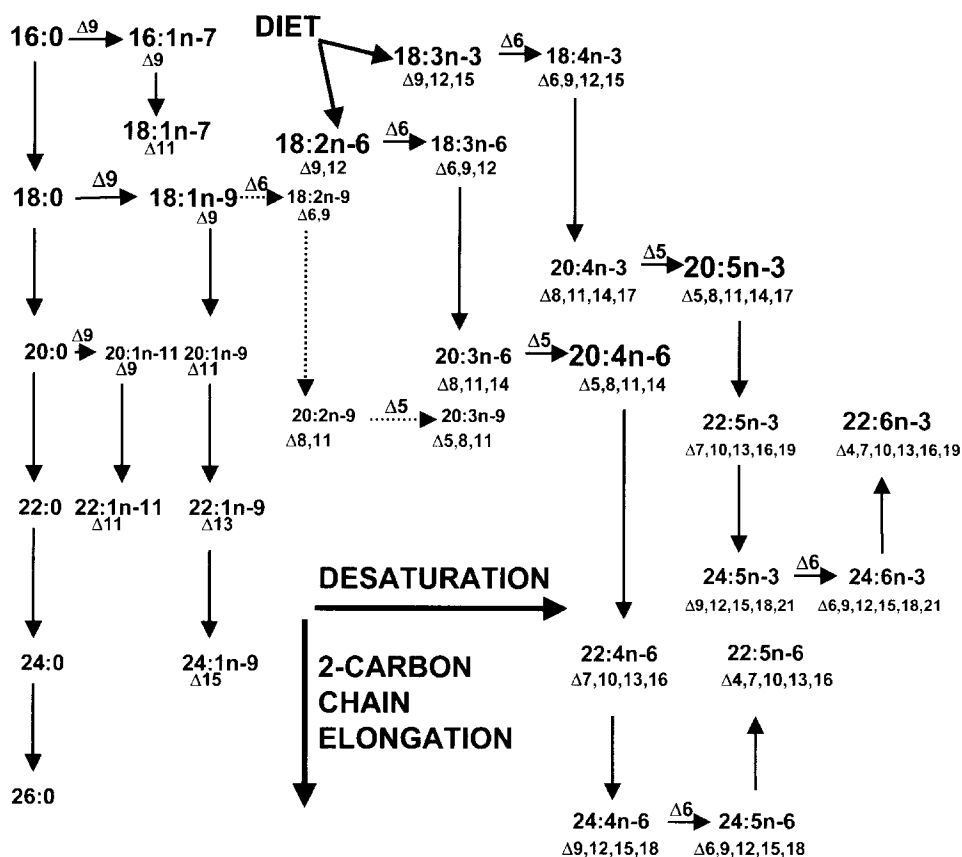


Fig. 6. Major pathways of fatty acid desaturation and chain elongation in animal tissues. Note the alternating sequence of desaturation in the horizontal direction and chain elongation in the vertical direction in the formation of polyunsaturated fatty acids from dietary essential fatty acids.

sequence. The predominant fatty acid families (Fig. 6) are:

- n-6 acids derived from 18:2 n-6
- n-3 acids derived from 18:3 n-3
- n-9 acids derived from 18:1 n-9, and
- n-7 acids derived from 16:1 n-7.

4.3.1. The n-6 family

Arachidonate, 20:4 n-6, is an abundant polyenoic acyl chain found in most animal tissues. 20:4 n-6 can be derived from 18:2 n-6 by the alternating sequence of Δ⁶ desaturation, chain elongation of the 18:3 n-6 intermediate, and Δ⁵ desaturation of 20:3 n-6 (Fig. 6). Δ⁶ desaturation is considered rate-limiting in most situations. 20:4 n-6 is a component of phospholipids contributing to the structural integrity of membranes and is the primary precursor of important biological signalling molecules (e.g., prostaglandins and leukotrienes) with a variety of biological activities (Chapter 13). Frequently, 20:4

n-6 is referred to as an essential fatty acid as there is an absolute requirement for it; however, 18:2 n-6 can be converted to adequate 20:4 n-6 by most mammalian cell types. Neutrophils are a major exception, requiring 20:4 (-6) for leukotriene production but being unable to synthesize it from 18:2 n-6.

In liver and most other tissues of animals in a normal, balanced state, only 18:2 n-6 and 20:4 n-6 accumulate in relatively large quantities; much lower levels of the intermediates, 18:3 n-6 and 20:3 n-6, are detected. Such observations support a rate-limiting role for $\Delta 6$ desaturase in the sequence. When $\Delta 6$ desaturase activity is limited, this can be by-passed by providing 18:3 n-6 from enriched sources (such as evening primrose oil) for subsequent elongation and $\Delta 5$ desaturation to 20:4 n-6.

The genes and cDNAs coding for mammalian $\Delta 6$ and $\Delta 5$ desaturase enzymes have been isolated and characterized [4,5]. In humans, both desaturases are on chromosome 11. Mouse and human $\Delta 6$ desaturases are comprised of 444 amino acids with 87% homology between species. A surprising feature in analysis of the amino acid sequences is that both $\Delta 6$ and $\Delta 5$ desaturases have a cytochrome *b5* domain at the N-terminus of the polypeptide chain with a C-terminal catalytic domain very similar to that found in $\Delta 9$ desaturase, including the positioning of specific catalytic histidine residues. Yeast $\Delta 9$ desaturase also has an N-terminus cytochrome *b5* domain; genetic inactivation of the yeast cytochrome *b5* gene still allows for functional $\Delta 9$ desaturation, while deletion of the *OLE1* encoded $\Delta 9$ desaturase cytochrome *b5* domain results in inactive $\Delta 9$ desaturase. Thus, cellular cytochrome *b5* can not substitute for the covalently attached N-terminal cytochrome *b5* domain. By analogy, the N-terminal cytochrome *b5* domain also is likely functional and essential for mammalian $\Delta 6$ and $\Delta 5$ desaturases.

mRNA transcript levels for $\Delta 6$ and $\Delta 5$ desaturases, determined for various mammalian tissues, correlate reasonably with detectable enzyme activities. Highest levels of transcript and activity for both $\Delta 6$ and $\Delta 5$ desaturases are found in liver, followed by lower amounts in heart, brain, placenta and lung. Liver $\Delta 6$ and $\Delta 5$ desaturase mRNA levels and enzyme activities are affected by fasting and refeeding with the level of PUFA in the diet being highly influential; high PUFA diets result in decreased liver $\Delta 6$ and $\Delta 5$ desaturase mRNA and enzyme activity levels and diets lacking PUFA result in high $\Delta 6$ and $\Delta 5$ desaturase mRNA and enzyme activities. Enhancement of desaturase activity observed upon refeeding after fasting can be suppressed by glucagon or dibutyryl cAMP for $\Delta 6$ desaturase activity but not for $\Delta 5$ desaturase. In rat liver, $\Delta 6$ desaturase activity is low during fetal and neonatal development, increases dramatically around weaning and remains high throughout adulthood. Brain $\Delta 6$ desaturase activity is relatively high in the fetus and neonates and markedly declines by weaning, remaining low throughout adulthood. $\Delta 5$ desaturase follows similar trends. Liver microsomes from human neonates have $\Delta 6$ and $\Delta 5$ desaturases at lower levels than for adult humans or rodents.

Relationships between PUFA metabolism and insulin are complex: PUFA formation is impaired in diabetes, insulin increases the activities of $\Delta 6$ and $\Delta 5$ desaturases, essential fatty acid deficiency prevents induced diabetes and supplementation with fish-oil (containing n-3 fatty acids) worsens hyperglycemia in diabetic patients. Epinephrine suppresses $\Delta 6$ and $\Delta 5$ desaturases through β -receptors. Triiodothyronine inhibits both $\Delta 6$ and $\Delta 5$ desaturases. Generally, $\Delta 6$ and $\Delta 5$ desaturases respond similarly to

glucocorticoids, other steroids and adrenocorticotrophic hormone. Potential roles of protein kinases as mediators of hormone action in control of PUFA biosynthesis remain to be defined. Vitamin E supplementation increases brain $\Delta 6$ desaturase and vitamin B6 deficiency markedly decreases liver $\Delta 6$ desaturation. Some cultured cell lines have lost $\Delta 6$ desaturase but retain $\Delta 5$ desaturase activity. The seed extract sesamin inhibits $\Delta 5$, but not $\Delta 6$, desaturase and the hypocholesterolemic drug simvastatin specifically increases $\Delta 5$ desaturation. These differences in activity are likely mediated through alterations in mRNA levels, although whether this is due to changes in the level of transcription or mRNA stability has yet to be determined.

In some tissues, 22:4 n-6 and 22:5 n-6 are quantitatively significant. Although numerous earlier reports concluded that there is $\Delta 4$ desaturation, the Sprecher pathway appears to be responsible primarily for production of 22:5 n-6 [20]. Exogenous 22:4 n-6 also is a substrate for 'retroconversion'. This process of partial degradation involves loss of either a 2-carbon fragment or a double bond and 2 or 4 carbons by β -oxidation. In general, retroconversion utilizes fatty acids of 20 carbons or greater. Since only double bonds in the $\Delta 4$ position are lost, this process could provide fatty acids with the first double bond in the $\Delta 5$ position. The quantitative significance of this process is not established. Deficiency of retroconversion of 22:6 n-3 to 20:5 n-3 in fibroblasts from a Zellweger patient with defective assembly of peroxisomes supports a role for peroxisomal β -oxidation (Chapter 5) in retroconversion.

A physical relationship of desaturases with chain elongation enzymes in ER membranes remains to be demonstrated. Indirect evidence supports a sequence of reactions (including esterification of products into phospholipids) that proceeds in a concerted manner without release of free fatty acyl intermediates [21].

4.3.2. *The n-3 family*

The most abundant n-3 acyl chains are 20:5 n-3 and 22:6 n-3. These fatty acids are esterified to phospholipids in cerebral cortex, retina, testes, muscle and liver. In retinal rod outer segments, the fatty acids in phospholipids are 40–60% 22:6 n-3. Many animal tissues convert dietary 18:3 n-3 to 20:5 n-3 and 22:6 n-3 by desaturation and elongation combined with peroxisomal chain shortening (Fig. 6). Whether the $\Delta 6$ desaturase in the Sprecher pathway converting 22:5 n-3 to 22:6 n-3 (or 22:4 n-6 to 22:5 n-6) is the same as the $\Delta 6$ desaturase acting directly on 18:3 n-3 (or 18:2 n-6) is unresolved with evidence for and against a common enzyme. Alternative sequences for initial steps of 18:3 n-3 (and 18:2 n-6) metabolism involving chain elongation followed by sequential $\Delta 5$ and $\Delta 8$ desaturation have been detected but their significance is unclear.

Potential benefits of increased consumption of fish and fish oil products abundant in 20:5 n-3 and 22:6 n-3 have been investigated extensively in animals and humans [22]. Enzymes in phytoplankton consumed by fish, or in fish themselves, produce these fatty acids from 18:3 n-3. Consumption of n-3 fatty acids leads to altered acyl chain composition of phospholipids in plasma, platelets, neutrophils and red cells of mammals, with further enrichment of n-3 fatty acids in peripheral body tissues taking longer. Generally, increases in n-3 content are at the expense of n-6 acids. Some populations, such as Greenland Inuit, routinely consume high levels of n-3 fatty acids and have a lower incidence of ischemic heart disease and longer bleeding times

related to a reduction in platelet aggregation. Clinical trials support positive effects of n-3 fatty acids in decreasing platelet aggregation, lowering blood pressure, reducing circulating triacylglycerols and producing modest beneficial changes in cholesterol and lipoproteins.

Some studies indicate that n-3 PUFA reduce severity of arthritis or impaired kidney functions involving abnormal immune and inflammatory responses. Other areas of potential n-3 fatty acid involvement include enhanced insulin sensitivity in the diabetic state, countering n-6 fatty acids in chemically induced, transplanted or metastatic tumors, and altered visual acuity and response to learning tests. Each fatty acid family has a role in overall nutritional balance believed largely to be manifested at the level of eicosanoid production (Chapter 13). The high concentration of n-3 fatty acids in some body tissues with specialized functions (e.g., brain and retina) and tenacious retention of n-3 acids during dietary deprivation suggest important structural and physical roles.

Suggestions that 0.2–0.3% of energy as n-3 fatty acids is adequate for adults and 0.5% during pregnancy, lactation and infancy must be matched with the need for n-3 to n-6 ratios of 1:4 to 1:8 to promote normal growth and development. Reported deficiencies in humans on long-term intravenous or gastric tube feeding have been corrected by supplementation with n-3 fatty acids. Interest in potential positive benefits of concentrated fish oils in the diet should be balanced by attention to potential negative or toxicological side effects.

Competition among fatty acids of the n-3 and n-6 families for desaturation and chain elongation enzymes is extensively documented from a variety of in vivo and in vitro experiments. 18:3 n-3 is a better substrate for $\Delta 6$ desaturase than is 18:2 n-6; accordingly, abundance of 18:3 n-3 can effectively decrease formation of 20:4 n-6. In humans, conversion of 18:3 n-3 to 20:5 n-3 and 22:6 n-3 is much greater than conversion of 18:2 n-6 to 20:4 n-6.

4.3.3. *The n-9 family*

The prominent acyl chain in this family is 18:1 n-9. Generally, competition from 18:2 n-6 and 18:3 n-3 for $\Delta 6$ desaturase prevents formation and accumulation of more unsaturated n-9 acids. However, in animals on a diet deficient in essential fatty acids, competition by 18:2 n-6 is removed and 18:1 n-9 is utilized as a substrate for the rate-limiting $\Delta 6$ desaturase (Fig. 6). Further chain elongation of 18:2 n-9 to 20:2 n-9 and $\Delta 5$ desaturation results in accumulation of 20:3 n-9. While 20:3 n-9 may partially substitute for some physical functions within membranes, it is not a precursor of prostaglandins and cannot alleviate the signs of essential fatty acid deficiency.

Since deficiency of essential fatty acids markedly reduces 20:4 n-6 while increasing 20:3 n-9, a ratio of triene to tetraene (20:3 n-9/20:4 n-6) of less than 0.2 in tissues and serum usually indicates essential fatty acid deficiency. However, use of this ratio has limitations as inhibition of $\Delta 6$ desaturase will reduce formation of both 20:3 n-9 and 20:4 n-6 and result in a deficiency state without altering the ratio. Total amount of n-6 acids may be a better reflection of essential fatty acid deficiency.

4.3.4. The n-7 family

The primary n-7 acid in membranes and circulating lipids is 16:1 n-7. As most analyses do not distinguish specific 18:1 isomers, the contribution of 18:1 n-7 to the 18:1 fraction is seldom appreciated even though in developing brain 18:1 n-7 comprises 25% of total 18-carbon monoene. Potentially, 20:4 n-7, with only a single carbon shift of the double bonds compared to 20:4 n-6, could be formed from 16:1 n-7; however, high levels of PUFA derived from 16:1 n-7 are not detected even on a fat-free diet, although increased levels of 16:1 n-7 frequently accompany deficiency of essential fatty acids.

5. Unsaturated fatty acids with *trans* double bonds

5.1. General properties

Trans-unsaturated fatty acids, the geometric isomers of naturally occurring *cis*-acids, are not produced by mammalian enzymes but are formed by microorganisms in the gastrointestinal tract of ruminant mammals and chemically during commercial partial hydrogenation of fats and oils [23,24]. *Trans*-acids have been inaccurately labelled as unnatural, foreign or non-physiological. From diets containing beef fat, milk fat, margarines and partially hydrogenated vegetable oils, *trans*-acids are ingested, incorporated and modified in animal tissues.

Early studies suggested selective exclusion of *trans*-acids from metabolic processes and incorporation into membrane lipids, particularly in the central nervous system. Collectively, further studies have indicated little selectivity for absorption, esterification or β -oxidation compared to *cis*-isomers. Discrimination against specific positional isomers of *trans*-acids (e.g., $\Delta 13$ *trans*) may occur. In general, *trans*-monoenoic acids are recognized as a distinct class with properties intermediate between saturated and *cis*-monounsaturated acids, particularly in specificity for esterification to phospholipids. Accumulation of *trans*-acids in tissues generally is proportional to dietary levels. Lack of preferential accumulation over the long term suggests that *cis* and *trans* isomers turn over similarly. Nonetheless, potential negative influences of *trans*-unsaturated fatty acids in biological systems continue to be controversial.

Specific interactions of *trans*-acids with desaturation and chain elongation enzymes of animal tissues have been reported. Positional isomers of *t*-18:1 (except for the $\Delta 9$ isomer) are substrates for $\Delta 9$ desaturase of rat liver microsomes, resulting in *cis,trans*-dienoic isomers that, in some cases, are desaturated again by $\Delta 6$ desaturase to unusual polyunsaturated structures.

5.2. *Trans*-polyenoic fatty acids

Isomers of *trans,trans*-dienoic fatty acids, including *t,t*-18:2 n-6, are substrates for $\Delta 6$ desaturation in liver and brain, albeit at a lower rate than for *c,c*-18:2 n-6. Dienoic *trans*-isomers of 18:2 clearly lack properties of essential fatty acids and interfere with normal conversion of 18:2 n-6 to 20:4 n-6, primarily at the $\Delta 6$ desaturase. Diets containing *trans*-acids intensify signs of mild essential fatty acid deficiency; conversion

of available 18:2 n-6 to 20:4 n-6 is reduced as is 18:1 n-9 to 20:3 n-9. Although complex interactions are possible, *trans*-dienes are minor components (usually less than 1%) of hydrogenated vegetable oils used in human food products.

Many investigators conclude that *trans*-acids are undesirable dietary components. The actual influence of increased *trans* fatty acid intake during early development, on genetic disorders or on cardiovascular disease remains controversial and unresolved. It is reasonable to attempt to reduce their human consumption but continued investigations of *trans*-acids as a distinct and significant class of fatty acids are necessary.

5.3. Fatty acids with conjugated double bonds

While research interests in *trans* unsaturated fatty acids have oscillated somewhat over the last few decades, much recent attention has been directed to a specific class of dietary conjugated fatty acids, some with *trans* bonds, known as conjugated linoleic acid isomers (CLA; e.g., 9c,11t; 9t,11t; 10t,12c; 10t,12t; 10c,12c isoforms — compared to 9c,12c-linoleic acid) [24,25]. Intriguing, but poorly understood, health benefits have been implicated for CLAs including suppression of cancer cell growth, weight gain, diabetes and atherosclerosis. CLA isoforms can be metabolized to isoforms of 20:4 n-6 and some interfere with $\Delta 9$ desaturation or 20:4 n-6 and prostaglandin production. However, information on their relative metabolism, interference with normal *cis* isomers such as 18:1 n-9 and 18:2 n-6 or alterations by deficiency and disease states is limited.

6. Abnormal patterns of distribution and metabolism of long chain saturated and unsaturated fatty acids

Despite the diversity of enzymes involved in unsaturated fatty acyl chain formation, documented clinical defects specific to unsaturated acyl chain metabolism are few. Most alterations of acyl chain patterns reflect dietary deficiency states or impaired activities but not absence of specific fatty acid metabolizing enzymes. This indicates that desaturation and elongation enzymes are essential in supporting life-sustaining cellular processes. However, defects or deficiencies resulting in abnormal patterns of unsaturated fatty acid distribution have been documented.

6.1. Deficiency of essential fatty acids and related nutrients

In experimental animals, severe effects are observed in the absence of dietary essential fatty acids. These include a dramatic decrease in weight, dermatosis and increased permeability to water, enlarged kidneys and reduced adrenal and thyroid glands, cholesterol accumulation, impaired reproduction, and ultimate death. Substantial changes in fatty acid composition of tissue and circulating lipids occur. Brain is exceptionally resistant to loss of essential fatty acids, but modification of acyl patterns can be achieved if a deficient diet is started at an early age and continued long enough. The four n-6 acids in the sequence from 18:2 n-6 to 20:4 n-6 individually have similar potency in reversing

effects of deficiency, whereas the capacity of 18:3 n-3 is much lower. Collective assessment of nutritional studies indicates that 2–4% 18:2 n-6 and 0.2–0.5% 18:3 n-3 are adequate, but such requirements depend on the demands of specific tissues and the stage of growth, development and metabolism.

Inadequate supply of essential fatty acids resulting in deficiency signs described for rats is rare in humans. Normal diets contain enough 18:2 n-6 and 18:3 n-3, or their metabolic products, to meet tissue demands. Adipose stores provide a protective buffer against temporary limited intake. However, severe deficiency states have been observed in humans (especially premature infants with restricted adipose stores) on prolonged intravenous feeding or artificial milk formulations without adequate lipid supplements. Marked alterations of serum fatty acid patterns characterized by depletion of n-6 acids and a major increase in the 20:3 n-9 to 20:4 n-6 ratio are accompanied by severe skin rash, loss of hair and irritability. These signs are reversed rapidly by supplementation with lipid emulsions containing 18:2 n-6.

Functions for 18:2 n-6, in addition to a role as precursor of 20:4 n-6, seem likely. Some fatty acids that cannot serve as prostaglandin precursors prevent signs of essential fatty acid deficiency. Cats apparently require both 18:2 n-6 and 20:4 n-6 in their diets but may have relatively low, rather than absence of, $\Delta 6$ desaturase activity.

Gross signs of zinc deficiency are similar to those observed in essential fatty acid deficiency. Possible relationships between zinc, PUFA and eicosanoids have been proposed, but direct connections at the metabolic level have not been shown.

In several human diseases, abnormal patterns of PUFA, attributable to insufficient dietary 18:2 n-6 or to abnormal metabolism, have been described [26]. Some, including acrodermatitis enteropathica, cystic fibrosis, Crohn disease, peripheral neuropathy, and congenital liver disease have diminished capabilities for desaturation or chain elongation of PUFA or conversion to eicosanoids. Alcoholism, cirrhosis, Reye's syndrome, and chronic malnutrition are accompanied by significantly abnormal patterns of essential fatty acids in serum phospholipids. As high intake of n-6 may be proinflammatory, a countering interaction of n-6 PUFA could protect in situations such as rheumatoid arthritis, autoimmune disease (e.g., AIDS) or malignant tumor progression. Balance between PUFA families may be particularly important during stress of recovery from surgery or burn injury.

6.2. Relationships to plasma cholesterol

Considerable evidence supports a correlation between high intake of dietary saturated fats, relative to PUFA, and the occurrence of atherosclerosis and coronary disease. Risk of coronary disease is proportional to serum cholesterol levels and total serum cholesterol can be decreased following dietary intake of lipids enriched in PUFA [22]. Factors such as platelet aggregation, blood pressure and vascular obstruction may be influenced through some of the oxygenated derivatives of PUFA.

7. Regulation through sensors and receptors

7.1. Membrane sensing factors and response elements

Cells closely regulate membrane content of unsaturated fatty acids, although precisely how the level is sensed and a response is initiated is not understood. Genetic studies in yeast resulted in discovery of an unsaturated fatty acids sensing system. A transcription factor encoded by the yeast *STP23* gene is required for efficient transcription of the $\Delta 9$ desaturase gene, *OLE1*. Addition of exogenous unsaturated fatty acids results in decreased *OLE1* transcription and synthesis of unsaturated fatty acids. The regulatory *SPT23* protein contains a C-terminal, membrane-spanning domain that anchors the protein to the ER. Upon addition of exogenous unsaturated (but not saturated) fatty acids, *SPT23* protein is ubiquitinated, resulting in targeted cleavage of the C-terminus. The N-terminal *SPT23* transcription factor moves to the nucleus and shuts off transcription of the *OLE1* gene [27,28]. Thus, this ER-bound protein is a precursor of a nuclear transcription factor and is similar to SREBP that is derived from an ER precursor protein (Chapter 15).

Based on similarities of the yeast system to mammalian SREBP processing in response to altered sterol levels (Chapter 15), a system for sensing of unsaturated fatty acids levels in mammalian cells also may exist. Both SCD1 and SCD2 are regulated by sterols through SREBP. In transgenic mice, truncated forms of SREBP give a 3- to 4-fold increase in both SCD1 and 2. Both SREBP and PUFA regulation of SCDs through promoter reporter regions are dependent on sterol- and adjacent-response elements.

7.2. Peroxisomal proliferator-activated receptors (PPARs)

Fatty acids and their oxygenated derivatives are among the known activators of members of the PPAR family of nuclear receptors [14]. PPAR γ is a transcription factor and critical modulator of fat cell differentiation and function, thus providing a direct link between fatty acid concentrations and regulation of gene transcription in adipocytes. Drugs that act as agonists or antagonists of PPAR γ can be used to alter lipid metabolism, insulin resistance and diabetes and to influence related diseases such as obesity, atherosclerosis and hypertension. Although paradoxes remain in the complex interactions mediated by PPARs, studies with gene knockouts or specifically designed modulator molecules are shedding new light on PPARs in relation to uptake, storage and oxidation of fatty acids.

8. Future directions

Progress in understanding desaturation and chain elongation of fatty acyl chains and the influence of fatty acids on metabolic functions has been exciting over the last half century. At the same time, expansion of knowledge of mechanisms, regulation and functions of these processes, particularly the extent to which information applies to humans, is needed. Appropriate levels and balance of n-6 and n-3 fatty acids and the modulating role of n-3 acids, especially relating to fetal and neonatal development, to

stresses such as surgery, accidental injury or incompetent immunity, and to inherited or acquired diseases require continuing study. Surprisingly, potential deleterious effects of *trans* unsaturated fatty acids still has not been resolved fully. Are monenoic fatty acids a logical substitution if other dietary fatty acids are reduced? In establishing nutritional standards for classes of fatty acids, both individual and species variability and the complex interplay of a variety of physiological functions must be considered.

Application of cloning, molecular probes and genetic technologies to desaturation and chain elongation enzymes has substantially progressed recently but the full power of molecular probes and techniques is yet to be realized in this realm. Specific molecular probes for $\Delta 6$ desaturase will be beneficial for determining whether the enzyme(s) acting on 18:2 n-6, 18:3 n-3, 24:4 n-6 and 24:5 n-3 are identical. Modulation of $\Delta 5$ desaturase relative to preceding desaturation and elongation steps could be enlightening. Studies based on mutant cell lines or transgenic and gene disrupted mice, deficient in or overexpressing one or more components of the desaturation-chain elongation sequences, should provide exciting insights into regulation. To what extent are the components coordinately regulated and is the sequence of activities physically associated to provide channelled conversion to the major end products? Is the location of desaturation and elongation confined to the ER or might there be significant activities within peroxisomes or nuclear and plasma membranes?

Intracellular signalling roles for PUFA, in addition to being precursors of eicosanoids, are recognized but require more definition particularly in relation to PPAR (or other orphan receptors) classes and functions. How do PUFA functions relate to diseases and the type and amount of dietary fatty acids we ingest? Details of alternatives to classical pathways of n-3 and n-6 fatty acid metabolism must be pursued.

Abbreviations

ER	endoplasmic reticulum
PPAR	peroxisomal proliferator activated receptor
PUFA	polyunsaturated fatty acid
SCD	stearoyl-CoA desaturase
SREBP	sterol response element binding protein

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Phospholipid biosynthesis in eukaryotes

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1. Introduction

The objective of this chapter is to provide an overview of eukaryotic phospholipid biosynthesis at an advanced level. Phospholipids make up the essential milieu of cellular membranes and act as a barrier for entry of compounds into cells. Phospholipids also function as precursors of second messengers such as diacylglycerol (DG) and inositol-1,4,5-P₃ which is covered in Chapter 12. A third, and usually overlooked function of phospholipids, is storage of energy in the form of fatty acyl components. This function is probably quantitatively important only under extreme conditions such as starvation.

2. Phosphatidic acid biosynthesis and conversion to diacylglycerol

Phosphatidic acid (PA) is an intermediate that occurs at a branch point in glycerolipid biosynthesis as shown in Fig. 1. Significant developments in elucidation of the biosynthetic pathway occurred in the 1950s when Kornberg and Pricer demonstrated that fatty acids are activated to acyl-CoA prior to reaction with glycerol-3-P. Subsequent studies from the laboratories of Kennedy, Shapiro, Hübscher and others delineated the biosynthetic pathway for PA.

An important step in PA biosynthesis is the activation of fatty acids by acyl-CoA synthetases to yield acyl-CoA (Fig. 2). Five different forms of rat acyl-CoA synthetase have been cloned, each encoded by a separate gene (R.A. Coleman, 2001). Different forms of the enzyme have been found on endoplasmic reticulum (ER), mitochondria and mitochondrial associated membrane (MAM), a subfraction of the ER (J.E. Vance, 1990). Hence, these synthetases may function differently in providing substrate for phospholipid and triacylglycerol (TG) biosynthesis.

2.1. Glycerol-3-P acyltransferase

This enzyme catalyzes the first committed reaction in the biosynthesis of PA. The relative importance of this acyltransferase in regulation of phospholipid biosynthesis has not been clearly established. In mammals, two glycerol-3-P acyltransferases have been identified, one associated with mitochondria and the other on the ER (Fig. 2). The ER acyltransferase is inhibited by *N*-ethylmaleimide whereas the mitochondrial

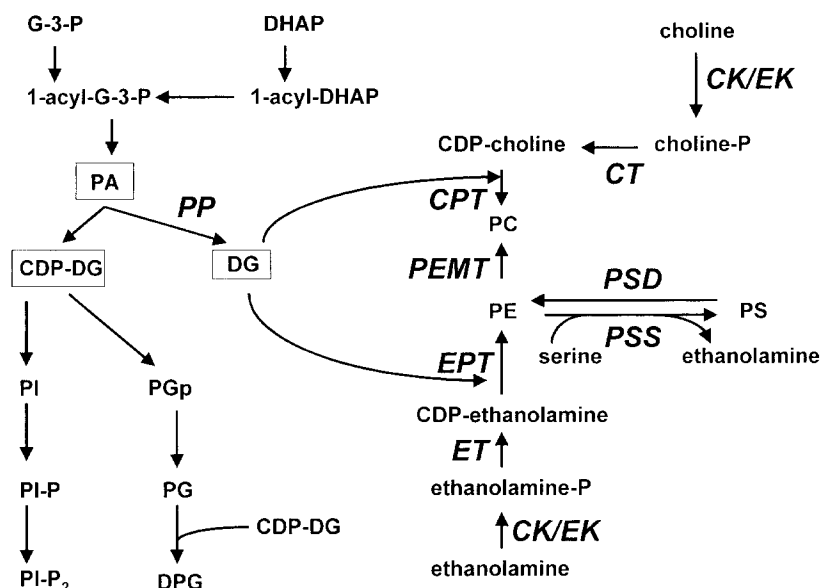


Fig. 1. Phospholipid biosynthetic pathways in animal cells. The abbreviations are: DHAP, dihydroxyacetone phosphate; G-3-P, glycerol-3-phosphate; PA, phosphatidic acid; DG, diacylglycerol; CDP-DG, cytidine diphosphodiacylglycerol; PI, phosphatidylinositol; PG, phosphatidylglycerol; PGp, phosphatidylglycerol phosphate; DPG, diphosphatidylglycerol; PP, phosphatidic acid phosphatase; PE, phosphatidylethanolamine; PC, phosphatidylcholine; PEMT, phosphatidylethanolamine *N*-methyltransferase; CT, CTP:phosphocholine cytidyltransferase; PS, phosphatidylserine; CK/EK, choline/ethanolamine kinase; CPT, CDP-choline:1,2-diacylglycerol cholinephosphotransferase; EPT, CDP-ethanolamine:1,2-diacylglycerol ethanolaminephosphotransferase; ET, CTP:phosphoethanolamine cytidyltransferase; PSD, phosphatidylserine decarboxylase; PSS, phosphatidylserine synthase.

enzyme is not. The mitochondrial acyltransferase prefers palmitoyl-CoA as an acyl donor compared to oleoyl-CoA, whereas the ER enzyme does not show a preference for saturated versus unsaturated acyl-CoAs. For this and other reasons the mitochondrial enzyme is thought to be primarily responsible for the abundance of saturated fatty acids in the SN-1 position of glycerophospholipids (D. Halder, 1994). Mitochondrial glycerol-3-P acyltransferase has been localized to the outer mitochondrial membrane with the active site facing the cytosol (R.A. Coleman, 2001). Mitochondrial glycerol-3-P acyltransferase has been purified and the cDNA cloned.

Transcription of the mitochondrial acyltransferase gene is decreased by starvation and glucagon, and increased by a high carbohydrate diet [1]. These responses make physiological sense since animals do not need to make triacylglycerols and to a lesser extent phospholipids when energy supply is limited. The microsomal activity is not significantly altered by these treatments. The 5' flanking region of the murine gene for the mitochondrial acyltransferase was linked to a luciferase reporter plasmid and expressed in 3T3-L1 pre-adipocytes. Deletion analysis on the promoter indicated that sequences between -86 and -55 bp were important for the expression of luciferase activity. Subsequent studies identified -78 to -55 bp as sites that bind sterol response

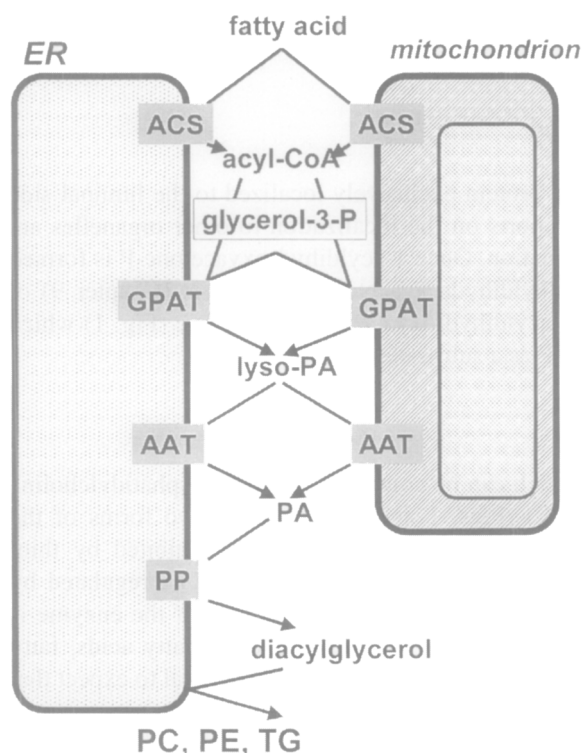


Fig. 2. Biosynthesis of phosphatidic acid (PA) can occur on both the endoplasmic reticulum and the outer membrane of mitochondria. The abbreviations are: ACS, acyl-CoA synthetase; GPAT, glycerol-3-P acyltransferase; AAT, 1-acylglycerol-3-P acyltransferase; PP, phosphatidic acid phosphatase; PC, phosphatidylcholine; PE, phosphatidylethanolamine; TG, triacylglycerol.

element binding protein (SREBP) and NF-Y transcription factors (P.A. Edwards, 1997). As indicated in Chapters 6, 7 and 15, SREBP is a key regulator of cholesterol and fatty acid synthesis, and fatty acid desaturation. Thus, it makes physiological sense that SREBP would also enhance glycerolipid, particularly TG, biosynthesis by increasing the expression of glycerol-3-P acyltransferase. Disruption of the mitochondrial glycerol-3-P acyltransferase gene in mice caused a decrease in the TG and cholesterol content in liver and striking changes in glycerolipid fatty acid composition (R.A. Coleman, 2002).

2.2. 1-Acylglycerol-3-P acyltransferase

Much less is known about the second step in the PA biosynthetic pathway (Fig. 2). The activity of this acyltransferase is much lower in mitochondria than in ER. It is presumed that much of the lyso-PA formed in mitochondria is transferred to ER for the second acylation. In vitro studies indicate that a carrier protein is not required (A.K. Hajra, 1992). The esterification at position 2 is specific for unsaturated fatty acids. However, the types of fatty acyl-CoAs available also influence the acyl-CoA selected for transfer

to lyso-PA. Two human isoforms of 1-acylglycerol-3-P acyltransferase have been cloned and expressed [2].

2.3. Dihydroxyacetone-P acyltransferase

This enzyme is an integral membrane protein exclusively localized to the luminal side of peroxisomes (A. Poulos, 1993). Reports on the localization to other organelles are likely a result of peroxisomal contamination. Once 1-acyldihydroxyacetone-P is formed it can be used as a substrate for 1-alkyldihydroxyacetone-P synthesis (Chapter 9) or reduced to lyso-PA by a peroxisomal acyldihydroxyacetone-P reductase (Fig. 1) which also utilizes 1-alkyldihydroxyacetone-P as a substrate.

2.4. Phosphatidic acid phosphatase

This enzyme hydrolyses PA to DG which can be converted to TG, phosphatidylcholine (PC) or phosphatidylethanolamine (PE) (Figs. 1 and 2). There are two forms of the phosphatase. The cytosolic ER form is dependent on Mg^{2+} and inhibited by thiol reagents such as *N*-ethylmaleimide. The activity of this enzyme can be regulated by reversible translocation between cytosol and ER. The cytosolic form of the enzyme is inactive and is translocated to the ER membrane in the presence of fatty acids, fatty acyl-CoAs and PA. Since the substrate, PA, is found on the ER it is logical to expect that the ER is where the enzyme functions in the cell.

The second PA phosphatase is neither inhibited by *N*-ethylmaleimide nor stimulated by Mg^{2+} . The cDNA for this phosphatase was cloned and expressed [3]. It appears to be a glycosylated protein on the plasma membrane with 6 putative transmembrane regions. This lipid phosphatase also hydrolyzes ceramide-1-phosphate, sphingosine-1-phosphate as well as lyso-PA. The active site is predicted to face outside the cell and has activity on lyso-PA but low activity on PA (H. Kanoh, 2000). Thus, the plasma membrane phosphatase probably has no role in intracellular phospholipid biosynthesis. Consistent with a signaling role, the plasma membrane phosphatase shares 34% sequence identity with Wunen protein (also has phosphatase activity) that functions in germ cell migration in *Drosophila* embryos. Furthermore, both of these proteins are very similar to Dri42, a protein that is upregulated in expression during epithelial differentiation in rat intestinal mucosa.

Yeast also has two PA phosphatases, a 104-kDa, microsomal form and a 45-kDa, mitochondrial form [4]. Addition of inositol induces the expression of the 45-kDa enzyme but not the 104-kDa enzyme (see Section 10 for more on regulation of phospholipid biosynthesis in yeast).

A novel DG pyrophosphate phosphatase has also been identified in yeast [4]. The enzyme has PA phosphatase activity but prefers DG pyrophosphate as substrate. This phosphatase has broad specificity but only DG pyrophosphate and PA have been shown to be substrates in vivo. A yeast mutant defective in the pyrophosphatase gene was viable and accumulated DG pyrophosphate (G.M. Carman, 1998). The enzyme expression is induced by inositol and zinc deprivation (G.M. Carman, 2001). The enzyme is similar to the mammalian plasma membrane PA phosphatase which also hydrolyzes DG

pyrophosphate. The function of DG pyrophosphate is unknown. Since it occurs at a very low level in yeast (0.18 mol% of total phospholipids), DG pyrophosphatase may have a signaling function [4].

3. Phosphatidylcholine biosynthesis

3.1. Historical background

PC was first described by Gobley in 1847 as a component of egg yolk and named 'lecithin' after the Greek equivalent for egg yolk (*lekithos*). In the 1860s Diakonow and Strecker demonstrated that lecithin contained two fatty acids linked to glycerol and that choline was attached to the third hydroxyl by a phosphodiester linkage. The first significant advance in understanding PC biosynthesis occurred in 1932 with the discovery by Charles Best that animals had a dietary requirement for choline. In the 1950s the CDP-choline pathway for PC biosynthesis (Fig. 3) was described by Eugene Kennedy and coworkers. A key observation was that CTP, rather than ATP, was the activating nucleotide for PC biosynthesis [5]. CTP is required not only for PC biosynthesis but also for the de novo synthesis of all phospholipids (prokaryotic and

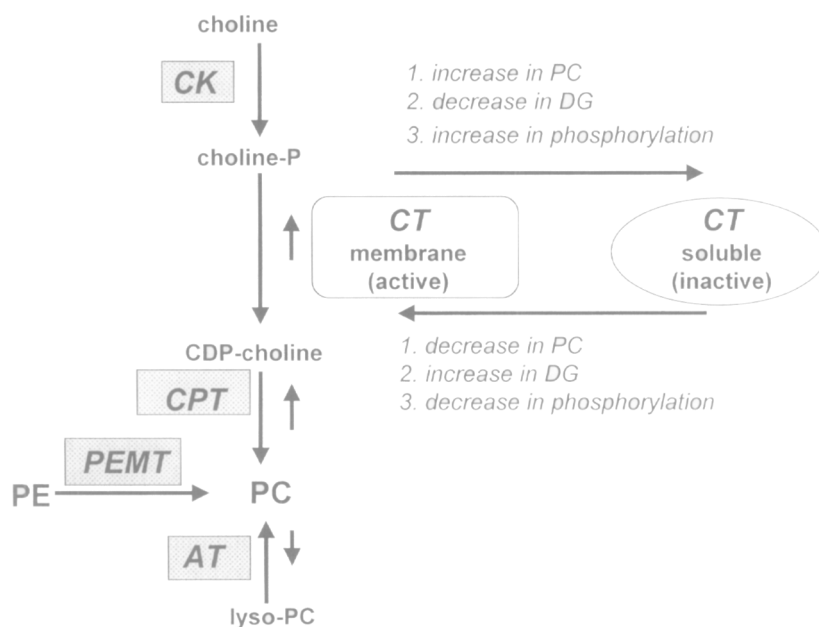


Fig. 3. Regulation of PC biosynthesis via the CDP-choline pathway by modulation of the binding of CTP: phosphocholine cytidyltransferase (CT) to membranes. Three different modes of regulation of CT activity are indicated. The abbreviations are: CK, choline kinase; CPT, CDP-choline:1,2-diacylglycerol cholinephosphotransferase; PEMT, phosphatidylethanolamine N-methyltransferase; AT, lyso-PC acyltransferase; PC, phosphatidylcholine; PE, phosphatidylethanolamine; DG, diacylglycerol.

eukaryotic, excluding PA which can be considered to be an intermediate in glycerolipid biosynthesis).

An alternative pathway for PC biosynthesis, of quantitative significance only in liver, is the conversion of PE to PC via PE methylation (Fig. 3). The first observation of this pathway was in 1941 when Stetten fed [^{15}N]ethanolamine to rats and isolated [^{15}N]choline. Two decades later Bremer and Greenberg detected a microsomal enzyme that converted PE to PC via transfer of methyl groups from *S*-adenosylmethionine.

3.2. Choline transport and oxidation

Choline is not made *de novo* in animal cells except by methylation of PE to PC and subsequent hydrolysis of the choline moiety. Therefore, choline must be imported from extracellular sources. There are two distinct transport mechanisms for choline [6]; a high affinity (K_m or $K_t < 5 \mu\text{M}$), Na-dependent transporter and a lower affinity ($K_t > 30 \mu\text{M}$), Na-independent transporter. Several cDNAs encoding proteins that show high affinity transport of choline have been reported. A human cDNA is predicted to have 13 transmembrane spanning domains (R.D. Blakely, 2000).

Once choline is inside the cell, its normal fate is rapid phosphorylation by choline kinase (Fig. 3). In neurons choline is also converted to the neurotransmitter, acetylcholine. Choline is also oxidized to betaine [$^-\text{OOC}-\text{CH}_2-\text{N}^+(\text{CH}_3)_3$] in the liver and kidney. In liver betaine is an important donor of methyl groups for methionine biosynthesis and the one carbon pool. Betaine is produced in mitochondria into which choline is transported by a specific transporter on the inner membrane. Next, choline is oxidized to betaine aldehyde by choline dehydrogenase on the inner leaflet of the inner mitochondrial membrane. The conversion to betaine is catalyzed by betaine-aldehyde dehydrogenase located in the mitochondrial matrix.

Betaine can be transported into kidney medulla by a betaine transporter. In renal medulla, eubacteria, halotolerant plants, marine invertebrates and cartilaginous fish, betaine accumulates as an osmolyte (a small organic solute that accumulates in response to hypertonicity without adverse affect to the cell or organism) (J.S. Handler, 1992). Hypertonicity of the renal medulla is important for the kidney's ability to concentrate urine.

3.3. Choline kinase

The enzyme was first demonstrated in yeast extracts by J. Wittenberg and A. Kornberg (more famous for his contributions to DNA replication) in 1953. The enzyme was purified by K. Ishidate (1984) from rat kidney and shown also to phosphorylate ethanolamine [6]. This kinase is now referred to as choline/ethanolamine kinase β . The cDNA for a rat liver choline/ethanolamine kinase encoded an enzyme that is now referred to as choline/ethanolamine kinase $\alpha 1$. Northern analyses indicate that the mRNA for choline/ethanolamine kinase $\alpha 1$ is most abundant in testis. Choline/ethanolamine kinase $\alpha 2$ appears to be a splice variant of choline/ethanolamine kinase $\alpha 1$. The choline/ethanolamine kinase α and β genes have been characterized. The length of the gene was 40 kb for the choline/ethanolamine kinase α gene whereas the β gene was only 3.5 kb in length (K. Ishidate, 2000).

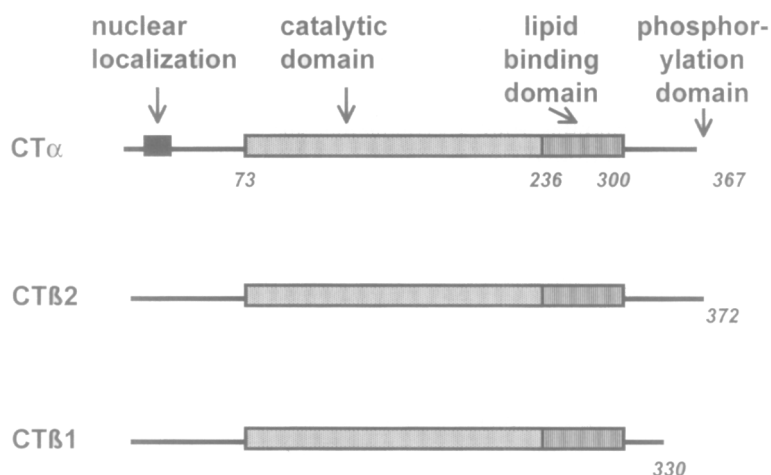


Fig. 4. Domain structures of CTP:phosphocholine cytidylyltransferase (CT) α , β 1 and β 2. CT α contains a nuclear localization signal, a N-terminal catalytic domain, an amphipathic helical (lipid binding) domain and a C-terminal phosphorylation domain. The CT β forms lack the nuclear localization signal but contain catalytic and amphipathic helical domains. CT β 1 is missing the phosphorylation domain whereas CT β 2 has a phosphorylation domain that is different from that of CT α .

Choline is not only required in the diet of animals but also in the medium of animal cells in culture (H. Eagle, 1955). Choline is essential because of the cell's requirement for PC to grow and divide.

There is evidence that the activity of choline kinase might be regulatory for cell division in some cases [6].

3.4. CTP:phosphocholine cytidylyltransferase

This enzyme activity was first described by Kennedy and Weiss in 1955 [5]. Over three decades later CT was finally purified to homogeneity (P.A. Weinhold, 1987). The CT gene was cloned from *S. cerevisiae* (S. Yamashita, 1987) by complementation of a yeast mutant defective in CT activity. The cDNA of rat liver CT was subsequently cloned (R.B. Cornell, 1990). CT is a homodimer in soluble extracts of rat liver and is also found on membranes. In most cells CT is thought to exist in an inactive reservoir in its soluble form and to be active when associated with membranes (Fig. 3).

Two genes encode different forms of CT, α and β . The CT α gene spans approximately 26 kb. Exon 1 is untranslated, exon 2 encodes the translation start site and a nuclear localization signal, exons 4–7 encode the catalytic domain, exon 8 codes for the alpha helical membrane binding domain and exon 9 encodes a C-terminal phosphorylation domain (I. Tabas, 1997) (Fig. 4).

The CT β gene is located on the X chromosome and encodes two isoforms, CT β 1 and CT β 2 (Fig. 4), presumably derived by mRNA splicing. Both isoforms differ from CT α at the amino terminus, lack the nuclear localization signal and are found in the cytoplasm of animal cells [7]. The primary sequences of CT β 1 and CT β 2 are identical except at

the carboxyl terminal. CT β 1 lacks most of the phosphorylation domain that is present in CT β 2. There are significant differences between the sequences of the phosphorylation domains of CT α and CT β .

CT has classically been considered to be a cytoplasmic enzyme since its activity is found in the cytosol and on microsomal membranes in cellular homogenates. However, Kent and coworkers demonstrated that CT was found in the nuclear matrix and associated with the nuclear membrane [8]. The role of the nuclear localization signal in CT α was explored by mutagenesis. Mutated CT α was expressed in a CHO mutant (MT-58) that was temperature-sensitive for CT activity (C. Raetz, 1980). In MT-58 cells, CT activity is present at low levels and the cells grow at 33°C. At the restrictive temperature of 40°C, there was no CT activity and the cells died via apoptosis (F. Tercé, 1996). Expression of CT α in which residues 8–28 (the nuclear localization signal) was deleted resulted in expression of CT largely, but not exclusively, in the cytoplasm [8]. These cells were able to survive at the restrictive temperature. Since some CT α was expressed in the nucleus, the experiment does not yet prove that cells can grow and divide when CT α is present only in the cytoplasm. There is intriguing evidence that CT α migrates into the cytoplasm during the G1 phase of the cell cycle, a time when PC biosynthesis is activated (R.B. Cornell, 1999). Thus, the role the nuclear localization signal of CT plays in cellular PC biosynthesis remains an intriguing question.

The lipid binding domain and the phosphorylated domains are involved in the regulation of CT activity. These domains of CT α have been deleted by either proteolysis with chymotrypsin or by construction of CT α truncation mutants [9]. CT α cDNAs that were truncated in the region of residue 314 (Fig. 4) lacked the phosphorylation segment, and CT truncated at residues 236, 231 or 228 lacked both the phosphorylation and lipid binding domains. When the lipid binding and phosphorylation domains were deleted, CT was a soluble, active enzyme that did not bind to membranes. Thus, the lipid binding domain is regulatory for the binding to membranes and the activation of CT. The binding of phospholipids to CT appears to activate the enzyme by decreasing the apparent K_m value for CTP (S.L. Pelech, 1982; S. Jackowski, 1995).

CT activity is modulated by phosphorylation. Experiments with CT truncation mutants have demonstrated that the phosphorylation domain is not required for lipid binding or CT activity. In vitro, CT is phosphorylated by casein kinase II, cdc2 kinase, cAMP kinase, protein kinase C and glycogen synthase kinase-3 but not by MAP kinase. However, the stoichiometry of phosphorylation is less than 0.2 mol P/mol CT with any of the kinases and in vitro phosphorylation does not affect enzyme activity. Exactly what role phosphorylation of CT plays in a physiologically relevant system remains to be demonstrated (see Section 4.4).

3.5. CDP-choline : 1,2-diacylglycerol cholinephosphotransferase

This enzyme was also discovered by Kennedy and coworkers [5] and is considered to be located on the ER but is also found on the Golgi, MAM and nuclear membranes [10]. Even though the enzyme has been known for more than four decades and despite intense efforts in many laboratories, the cholinephosphotransferase has never been purified. The difficulty is that the enzyme is an intrinsic membrane-bound protein that

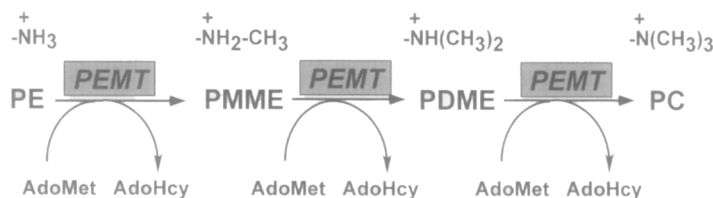


Fig. 5. Reactions catalyzed by phosphatidylethanolamine *N*-methyltransferase (PEMT). AdoMet, *S*-adenosylmethionine; AdoHcy, *S*-adenosylhomocysteine; PMME, phosphatidylmonomethylethanolamine; PDME, phosphatidyltrimethylethanolamine; PC, phosphatidylcholine.

requires detergents for solubilization. Moreover, the detergents complicate purification procedures commonly used such as gel filtration because the protein binds to micelles that are hard to separate on the basis of molecular size. The purification of membrane-bound enzymes has been described as 'masochistic enzymology' (D.E. Vance, 1990).

Yeast genetics and molecular biology have, however, allowed for the cholinephosphotransferase to be cloned. Two genes, *CPT1* and *EPT1*, each account for 50% of the cholinephosphotransferase activity in yeast extracts [10]. By the use of null mutations in these two genes, it has been established that *CPT1* is responsible for 95% of the PC made and *EPT1* gene product accounts for 5%. The *EPT1* gene product utilizes both CDP-choline and CDP-ethanolamine whereas *CPT1* catalyzes only reactions with CDP-choline.

More recently a human choline/ethanolaminephosphotransferase cDNA (hCEPT1) was cloned and expressed (C.R. McMaster, 1999). The open reading frame predicts a protein with 7 membrane-spanning domains. Subsequently, the same lab cloned a human cDNA that encoded for a CDP-choline-specific enzyme (hCPT1) with 60% sequence identity to hCEPT1. hCEPT1 mRNA was detected in all tissues tested whereas the expression of hCPT1 was highest in heart, testis, intestine and colon.

Cholinephosphotransferase acts at a branch point in the metabolism of DG that can also be converted to PE, TG or PA (Fig. 1). Most studies indicate that there is an excess of cholinephosphotransferase in cells, hence, the amount of active enzyme does not limit PC biosynthesis. However, it is clear that the *in vivo* activity of cholinephosphotransferase is regulated by substrate supply. The supply of CDP-choline is regulated by the activity of CT (Section 3.4). The supply of DG in liver seems to be controlled by the supply of fatty acids. Excess DG not utilized for PC or PE biosynthesis is stored in liver as TG.

3.6. Phosphatidylethanolamine *N*-methyltransferase

All nucleated cells contain PC and the CDP-choline pathway. Thus, it was not obvious why the pathway for PE methylation (Fig. 5) survived during evolution. Nor was it obvious why PE methyltransferase (PEMT) activity is mostly found in liver whereas 2% or less of the hepatic PEMT activity is found in other tissues of the body.

PEMT was purified from rat liver microsomes although it is an intrinsic membrane protein (N.R. Ridgway, 1987). Sequence of the amino terminal enabled the cloning of

the cDNA for PEMT (Z. Cui, 1993). Preparation of an antibody to the deduced sequence of the carboxyl terminal peptide permitted subcellular localization of the enzyme. The major activity for PEMT is found on the ER but the antibody only recognized a protein that was exclusively localized to MAM (J.E. Vance, 1990). This isoenzyme of PEMT is referred to as PEMT2 and the activity on the ER is called PEMT1. Both PEMTs catalyze all three transmethylation reactions that convert PE to PC (Fig. 5).

A mouse was generated in which the *Pemt* gene was disrupted and there was no PEMT activity [11]. The *Pemt*^{-/-} mice lived and bred normally and there was a 50% increase in CT activity in their livers. Since the mice retained the CDP-choline pathway, the lack of an obvious phenotype was not surprising. However, when the mice were fed a choline-deficient diet for 3 days, which attenuates PC synthesis via the CDP-choline pathway, the *Pemt*^{-/-} mice exhibited severe liver failure [12]. *Pemt*^{+/+} mice fed a choline-deficient diet were normal with no obvious liver pathology. Thus, it seems that the PEMT pathway has survived in evolution to provide PC at times when the CDP-choline pathway is less active such as might occur during starvation. Moreover, pregnant rats and suckling mothers can also have choline reserves depleted (S.H. Zeisel, 2000), hence, the PEMT pathway might provide an evolutionary advantage in this respect. The structurally related compound, dimethylethanolamine [$\text{HOCH}_2\text{-CH}_2\text{-N}^+(\text{CH}_3)_2$] would not substitute for choline in the *Pemt*^{-/-} mice even though it was converted to phosphatidyl dimethylethanolamine (K.A. Waite, 2002). Thus, it seems that the third methyl group on the phospholipid has a critical function in mice.

Further studies on *Pemt*^{-/-} mice showed that male, but not female, *Pemt*^{-/-} mice fed a high fat/high cholesterol diet have a defect in secretion of very low density lipoproteins that contain apolipoprotein B100 (A. Noga, 2002) (Chapter 19). The mechanism(s) for this sexual dimorphism is not clear.

The human gene encoding PEMT has been cloned and characterized. Whereas only one mRNA transcript has been identified in mice, human liver has three separate mRNAs that differ only at the 5' end, in a non-coding region of the transcript (D.J. Shields, 2001). Thus, the three transcripts encode the same protein. The function of separate PEMT mRNAs is going to be difficult to study in humans.

Yeast also has both the PE methylation pathway and the CDP-choline pathway. In yeast two enzymes are used for the conversion of PE to PC [13]. The methylation of PE to phosphatidylmonomethylethanolamine is catalyzed by the *PEM1/CHO2* gene product whereas the subsequent two methylations are catalyzed by the *PEM2/OPI3* gene product. Deletion of both *PEM1* and *PEM2* genes is lethal unless the yeast is supplied with choline. Yeast normally grows in the absence of choline and depend on the PEMT pathway. Thus, the CDP-choline pathway and the PE methylation pathway can compensate for each other in yeast.

Bacteria generally do not contain PC but *Rhodobacter sphaeroides* make PC by methylation of PE. Interestingly, this enzyme is soluble and has virtually no homology to PEMT or the yeast enzymes (V. Arondel, 1993). Also in one bacterium, a novel choline-dependent pathway was recently discovered in *Sinorhizobium meliloti* in which choline reacts with CDP-DC to form PC (O. Geiger, 2000).

4. Regulation of phosphatidylcholine biosynthesis

4.1. The rate-limiting reaction

The CT reaction usually limits the rate of PC biosynthesis. The first evidence in favor of this conclusion was measurement of pool sizes of the aqueous precursors (in rat liver, choline = 0.23 mM, phosphocholine = 1.3 mM, CDP-choline = 0.03 mM). These values assume that 1 g wet tissue is 1 ml and there is no compartmentation of the pools. The second assumption may not be valid as there is evidence for compartmentation of PC precursors (M. Spence, 1989). Nevertheless, the relative amounts of these compounds might be correct in the biosynthetic compartment(s). The concentration of phosphocholine is 40-fold higher than CDP-choline which is consistent with a 'bottleneck' in the pathway at the reaction catalyzed by CT.

Pulse-chase experiments demonstrate this bottleneck more vividly. After a 0.5 h pulse of hepatocytes with [*methyl*-³H]choline, more than 95% of the radioactivity in the precursors of PC was in phosphocholine, the remainder in choline and CDP-choline. When the cells were chased with unlabeled choline in the medium, labeled phosphocholine was quantitatively converted to PC (Fig. 6). The radioactivity in CDP-choline remained low during the chase and CDP-choline was rapidly converted to PC. There was minimal radioactivity in choline which suggests that choline is immediately phosphorylated after it enters the cell.

One additional point should be made. If a cell or tissue is in a steady state, pool sizes and reaction rates are not changing. Thus, although the rate of PC synthesis is determined by the CT reaction, the rates of the choline kinase and cholinephosphotransferase reactions will be the same as that catalyzed by CT. Otherwise, changes in the pool sizes of precursors would occur. For example, if the choline kinase reaction were faster than

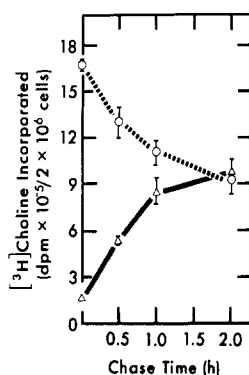


Fig. 6. Incorporation of [³H-*methyl*]choline into phosphocholine and PC as a function of time. Hepatocytes from rat liver were incubated with labeled choline for 30 min. Subsequently, the cells were washed and incubated (chased) for various times with unlabeled choline. The disappearance of radioactivity from phosphocholine (dashed line) and its appearance in PC (solid line) are shown. Adapted from fig. 1 of Pelech et al. (1983), J. Biol. Chem. 258, 6783, with permission.

the CT reaction, there would be an increase in the amount of phosphocholine. Thus, CT sets the pace, but the other reactions proceed at the same rate.

4.2. *The translocation hypothesis*

CT is recovered from cells and tissues in both cytosol and microsomal fractions. However, in the early 1980s evidence from several laboratories suggested a close correlation between CT activity on the microsomal membranes and the rate of PC biosynthesis. The hypothesis was that the active form of the enzyme was on cellular membranes and CT in the cytosol acted as a reservoir (Fig. 3). In agreement with this proposal, cytosolic fractions contain essentially no phospholipid and CT requires phospholipids for activity. Thus, cells have a facile mechanism for altering the rate of PC biosynthesis by a reversible translocation of CT between a soluble, inactive reservoir and cellular membranes. This mechanism for activation applies to CT α and both CT β 1 and CT β 2. This hypothesis remains basically valid except much of the 'cytosolic' CT may originate from the nucleus and activated CT may be associated with the nuclear membrane as well as the endoplasmic reticulum.

Binding of CT to membranes begins by electrostatic adsorption followed by hydrophobic interactions that involve intercalation of the protein into the nonpolar core of the membrane [9] (Fig. 7). When insertion of CT into the membrane lipids is blocked by using viscous gel phase lipids, CT binds electrostatically to the membrane but is not activated. Four properties of membranes promote CT insertion [9]: (1) interfacial packing defects as might occur when lipids with small head groups such as DG are in the membrane; (2) low lateral surface pressure (loose packing) as observed in highly curved compared to planar bilayers; (3) acyl chain disorder that can be caused by oxidation of the fatty acyl chains; (4) curvature strain that would occur when membranes are enriched in hexagonal phase-preferring lipids such as PE and DG. Synthesis of PC would reverse these properties of membranes and form a more stable bilayer.

4.3. *Regulation of phosphatidylcholine biosynthesis by lipids*

As indicated in Fig. 3, the association of CT with membranes and CT activation can be modulated by lipids. Both feed-forward and feed-back mechanisms for regulation of CT activity have been identified. DG may alter the rate of PC biosynthesis both as a substrate and as a modulator of CT binding to membranes. In vitro an increase in the content of DG in membranes enhanced the binding of CT.

Feedback regulation of CT and PC biosynthesis by PC has also been described (H. Jamil, 1990). Regulation of a metabolic pathway by product inhibition is commonly observed. In hepatocytes derived from choline-deficient rats, the rate of PC biosynthesis was inhibited by approximately 70% compared to choline-supplemented rats, the amount of PC declined and there was a corresponding increased binding of CT to membranes. CT appeared to sense a need for increased PC biosynthesis and was poised on the membrane prepared for catalysis. However, in choline-deficient cells there is less substrate, phosphocholine, so increased PC biosynthesis could not occur. When choline-deficient hepatocytes were supplied with choline, there was a positive

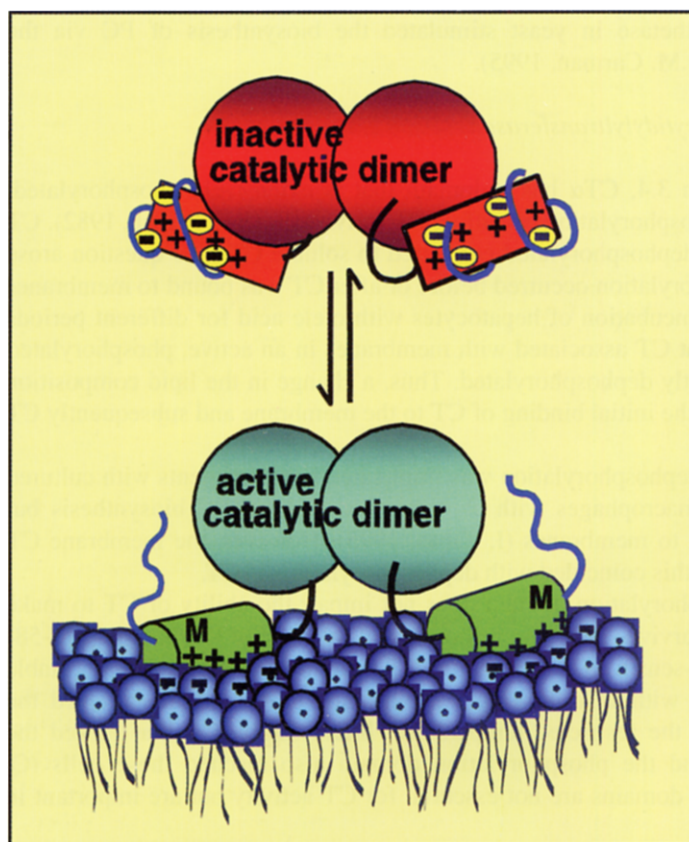


Fig. 7. Translocation of CTP:phosphocholine cytidyltransferase (CT) from an inactive soluble form to a membrane-associated activated form. The reversible interaction of CT with membranes involves the amphipathic helical region lying on the surface of the membrane with the hydrophilic side interacting with the negatively charged lipid head-groups and the hydrophobic side intercalating into the membrane core. From Cornell and Northwood [9] with permission.

correlation between the increased level of PC and the release of soluble CT. Similar correlations were observed when the level of PC was increased, either by providing methionine for enhanced conversion of PE to PC, or by providing lyso-PC which is imported into hepatocytes and acylated to PC.

An elegant feedback regulation of CT has been shown in the yeast *Saccharomyces cerevisiae* (V.A. Bankaitis, 1995). SEC14p is a phospholipid transfer protein that when assayed in vitro prefers phosphatidylinositol (PI) and PC and is an essential gene product (Chapter 17). SEC14p inhibited the CDP-choline pathway when PC was bound to SEC14p. In contrast, when PI was bound to SEC14p, there was minimal inhibition of CT. Thus, in yeast under conditions where PC is abundant, there is a feedback inhibition of CT and the CDP-choline pathway.

CTP, has also been implicated as regulatory in animal systems and yeast. Over-

expression of CTP synthetase in yeast stimulated the biosynthesis of PC via the CDP-choline pathway (G.M. Carman, 1995).

4.4. Phosphorylation of cytidyltransferase

As mentioned in Section 3.4, CT α has a domain that is extensively phosphorylated. Moreover, the state of phosphorylation can affect CT activity [8] (S.L. Pelech, 1982). CT bound to membranes is dephosphorylated compared to soluble CT. The question arose whether or not dephosphorylation occurred before or after CT was bound to membranes (M. Houweling, 1994). Incubation of hepatocytes with oleic acid for different periods of time demonstrated that CT associated with membranes in an active, phosphorylated form and was subsequently dephosphorylated. Thus, a change in the lipid composition of membranes mediated the initial binding of CT to the membrane and subsequently CT was dephosphorylated.

Activation of CT by dephosphorylation was implicated in experiments with cultured macrophages. Loading macrophages with cholesterol stimulated PC biosynthesis but did not alter CT binding to membranes (I. Tabas, 1995). However, the membrane CT increased in activity and this coincided with dephosphorylation of CT.

Deletion of the phosphorylation domain does not impair the ability of CT to make enough PC for cells to survive. This was demonstrated in a line of CHO cells (MT-58) that have a temperature-sensitive defect in the CT reaction (C. Kent, 1995). Stable transfection of the cells with CT α lacking the phosphorylation domain allowed the MT-58 cells to grow at the restricted temperature. Interestingly, CT that lacked the lipid binding domain and the phosphorylation domain also rescued these cells (C. Kent, 1999). Thus, these domains are not essential for CT activity but are important in regulating CT activity.

4.5. Transcriptional and post-transcriptional regulation of CT α

Most studies on CT activity and PC biosynthesis have not indicated regulation at the level of gene expression. The ability of a cell to activate the soluble form of CT would normally satisfy the cell's requirement for PC. Nevertheless, some control over the expression of the genes encoding CT must occur. The proximal promoter of the CT α gene has numerous potential regulatory elements (I. Tabas, 1997). Subsequent studies showed that Sp1, the first mammalian transcription factor purified and cloned (R. Tjian, 1986, 1987), had an important role in regulating the expression of the CT α gene [14]. The related nuclear factor, Sp3, could also activate CT α transcription (M. Bakovic, 2000). DNase protection assays indicated several elements in the proximal promoter bound unidentified nuclear factors. The yeast one hybrid system was utilized to clone the cDNA for one of these factors and transcription enhancer factor 4 (TEF4) was identified as a regulator of CT α transcription (H. Sugimoto, 2001). These initial studies were done in experiments involving transfections with various cDNA constructs. The first experiments to search for transcriptional regulation in a cell physiology-relevant system were on CT α expression in a murine fibroblast cell line as a function of the cell cycle (L. Golfman, 2001). During the G0 to G1 phase of the cell cycle there is an

increase in PC biosynthesis (S. Jackowski, 1996; L. Golfman, 2001) but there was no enhanced transcription of the CT α gene. Instead, increased transcription occurred during the S phase of the cell cycle, possibly to increase the amount of CT α in preparation for mitosis. Obviously, there is much to be done to elucidate the factors and DNA elements involved in transcriptional regulation of the CT α gene. There are no reports on transcriptional regulation of the CT β gene.

SREBPs play a critical role in regulating the expression of genes involved in fatty acid (Chapters 6 and 7) and cholesterol (Chapter 15) metabolism. Thus, several labs explored whether or not SREBPs might alter the expression of the CT α gene. Interestingly, one report indicated there was no direct modulation of CT α transcription (N. Ridgway, 2000) whereas research from another lab implicated sterols and SREBPs in the regulation of CT α transcription (P.A. Edwards, 2001). Further work is required on the relationship between SREBPs and PC biosynthesis.

The level of CT α mRNA can also be regulated by alterations in mRNA stability. When a macrophage cell line was depleted of colony stimulating factor and then repleted there was a 4-fold induction of mRNA for CT α (S. Jackowski, 1991). The stability of CT α mRNA increased after the addition of colony stimulating factor. An increase in CT α mRNA in fetal lung type II cells has also been ascribed to enhanced mRNA stability (M. Post, 1996).

Finally, enhanced turnover of CT α via the ubiquitin–proteosome pathway appears to be the mechanism by which tumor necrosis factor decreases the level of CT α in alveolar type II cells (R. Mallampalli, 2000).

4.6. Transgenic and gene-disrupted murine models of CT α

To determine if enhanced PC biosynthesis would protect macrophages from excess cholesterol-induced toxicity, genetically modified mice have been generated. A truncated version of CT α lacking the phosphorylation domain was expressed specifically in macrophages of mice under control of the scavenger receptor (Chapter 21) promoter (I. Tabas, 1999). These cells were protected from cholesterol-induced toxicity. In another approach, CT α expression was eliminated in macrophages using the Cre-lox method for selective disruption of a gene in specific cells (I. Tabas, 2000). The lack of the CT α gene and hence decreased PC biosynthesis caused enhanced sensitivity to cholesterol loading. In the absence of cholesterol loading, the macrophages without CT α appeared normal, possibly due to increased expression of CT β 2.

5. Phosphatidylethanolamine biosynthesis

5.1. Historical background and biosynthetic pathways

PE was first alluded to in a book published by Thudichum in 1884. He described 'kephalin' as a nitrogen- and phosphorus-containing lipid that was different from lecithin. In 1913, Renall and Baumann independently isolated ethanolamine from

kephalin. In 1930, Rudy and Page isolated the first pure preparation of PE. The structure of PE was established in 1952 by Baer and colleagues.

The biosynthesis of PE in eukaryotes can occur via four pathways (Fig. 8). The route via CDP-ethanolamine constitutes *de novo* synthesis of PE. The other pathways arise as a result of the modification of a pre-existing phospholipid. The CDP-ethanolamine pathway was first described by Kennedy and Weiss in 1956. The decarboxylation of phosphatidylserine (PS) to yield PE (Fig. 8) was shown in 1960 to occur in animal cells. PS decarboxylation is the only route for PE biosynthesis in *E. coli* (Chapter 3). The PE generated by this pathway can react with serine to generate PS and ethanolamine (Fig. 8). This appears to be one mechanism by which ethanolamine is made in cells. The other involves degradation of sphingosine (Chapter 14). The ethanolamine generated by either pathway can be utilized for PE biosynthesis via the CDP-ethanolamine pathway. No one has ever been able to show the decarboxylation of serine to ethanolamine in animal cells. Such a reaction was shown to occur in a plant, *Arabidopsis thaliana* (A.D. Hanson, 2001). PE can also be formed by reacylation of lyso-PE or reaction of ethanolamine with PS (Fig. 8).

5.2. Enzymes of the CDP-ethanolamine pathway

As mentioned in Section 3.3, the phosphorylation of ethanolamine in liver can be catalyzed by choline/ethanolamine kinase (Figs. 1 and 8). The cDNA encoding an ethanolamine kinase was cloned from *Drosophila* (P. Pavlidis, 1994). These scientists did not plan on cloning this cDNA since their approach was to determine the gene responsible for the easily shocked (*eas*) phenotype in this insect. These mutant flies display transient paralysis following a brief mechanical shock. In the *eas* mutant, a 2 base pair deletion caused formation of a premature stop codon in the ethanolamine kinase gene. Analysis of the phospholipids showed a decrease in PE from 59% of the total phospholipid in wild type to 56% in *eas*. How this minor change mediates the paralysis is not known. The difference may reflect a major change in PE content in a particular tissue or subcellular membrane. More recently, the gene that encodes a yeast ethanolamine kinase (G.M. Carman, 1999) and a human cDNA for ethanolamine-specific kinase (S. Jackowski, 2001) were cloned and expressed.

The second step in the CDP-ethanolamine pathway is catalyzed by CTP:phosphoethanolamine cytidyltransferase [15]. The enzyme is distinct from CT and is not activated by lipids. Although the phosphoethanolamine cytidyltransferase is recovered in cytosol from cell extracts, much of the enzyme has been localized to rough ER of rat liver by immunoelectronmicroscopy. Unlike CT α , there is no report of the phosphoethanolamine cytidyltransferase in the nucleus.

CDP-ethanolamine:1,2-diacylglycerol ethanolaminephosphotransferase is an integral membrane protein found on the ER, Golgi and MAM. The enzyme shows a distinct preference for DG species that contain 1-palmitoyl-2-docosaheptaenoyl (22:6) fatty acids. In hepatocytes in culture, nearly 50% of PE made via the ethanolaminephosphotransferase reaction is this species. The purpose of this extraordinary selectivity is unknown. The bovine hepatic enzyme was purified and exhibited both ethanolamine- and choline-phosphotransferase activity (L. Binaglia, 1999).

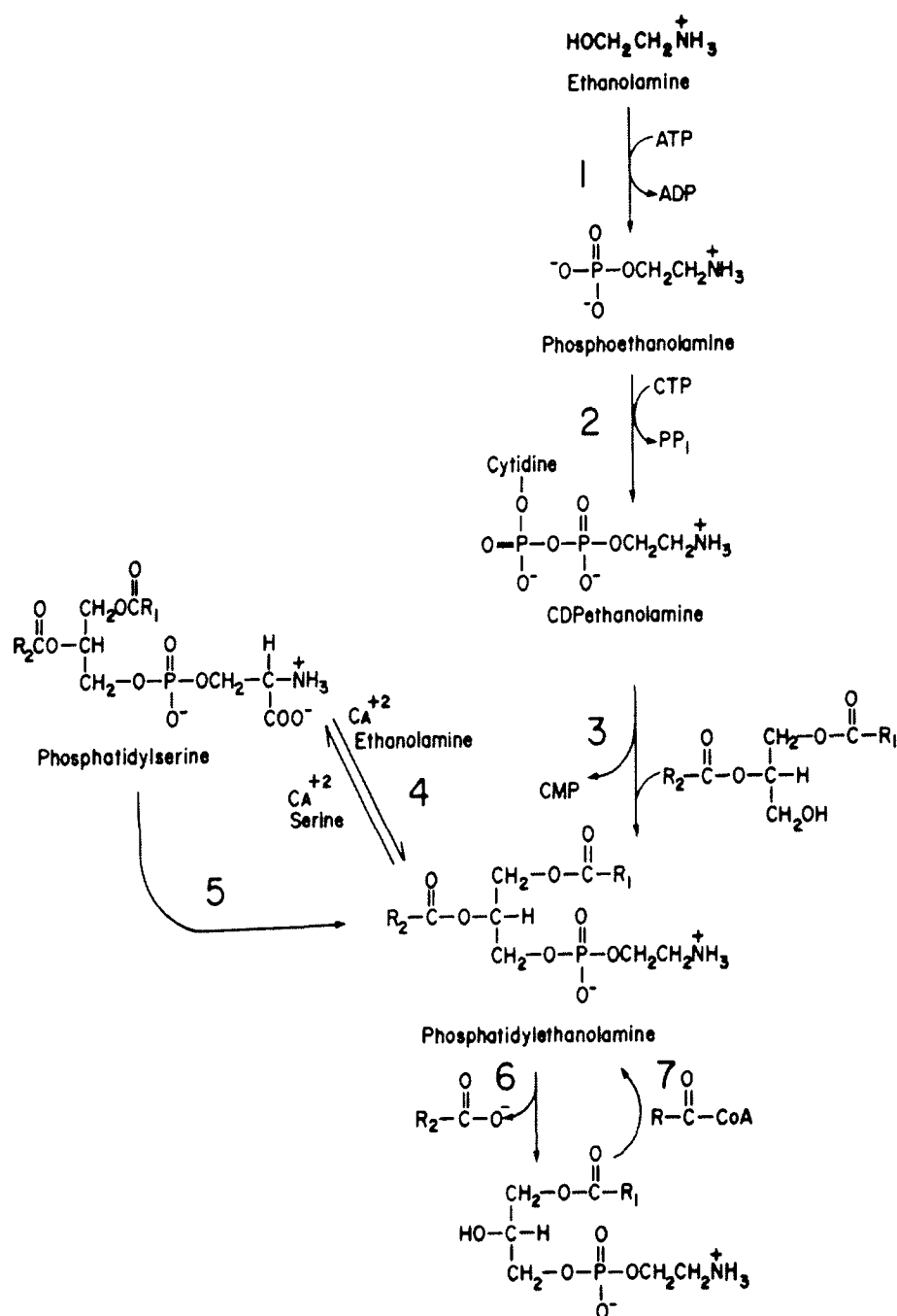


Fig. 8. Pathways for the biosynthesis of PE and PS. The numbers indicate the enzymes involved. 1, ethanolamine(choline) kinase; 2, CTP:phosphoethanolamine cytidyltransferase; 3, CDP-ethanolamine:1,2-diacylglycerol ethanolaminephosphotransferase; 4, PS synthase; 5, PS decarboxylase; 6, phospholipase A2; 7, acyl-CoA: lyso-PE acyltransferase.

A yeast gene (*EPT1*) that encodes an ethanolaminephosphotransferase and a human cDNA that encodes a choline/ethanolaminephosphotransferase have been cloned as discussed in Section 3.5.

5.3. Regulation of the CDP-ethanolamine pathway

Unlike CT, there is minimal literature on the mechanisms that control the activity of phosphoethanolamine cytidylyltransferase. Åkesson and Sundler in the 1970s found that phosphoethanolamine cytidylyltransferase was rate-limiting for PE biosynthesis. However, the supply of DG as a substrate can also limit the rate of PE biosynthesis (L.B.M. Tijburg, 1989). Thus, both the supply of CDP-ethanolamine from the cytidylyltransferase reaction and the supply of DG can regulate PE biosynthesis. Two studies have implicated channeling of intermediates in the biosynthesis of PE in mammalian cells [15].

5.4. Phosphatidylserine decarboxylase

PS decarboxylase is found in both prokaryotes (Chapter 3) and the mitochondria of eukaryotes. The enzyme activity was first described by Kanfer and Kennedy in 1964. The enzyme has not been purified from a eukaryotic source but the gene has been cloned and expressed from CHO cells (M. Nishijima, 1991) and yeast [16]. The yeast gene (*PSD1*) encodes a protein that is localized to mitochondria. However, when *PSD1* was disrupted in yeast, 5% PS decarboxylase activity remained and the yeast continued to grow. Subsequently, a second gene, *PSD2*, was isolated. When both *PSD1* and *PSD2* were disrupted, the yeast became ethanolamine auxotrophs. The PSD2 protein has been localized to the vacuolar and Golgi compartments. The function of *PSD2* is not known other than it can supply enough PS decarboxylase to allow growth of yeast in the absence of *PSD1*. The rate of PS decarboxylation is determined by the rate of PS transport into mitochondria (Chapter 17).

6. Phosphatidylserine biosynthesis

6.1. Historical developments and biosynthesis

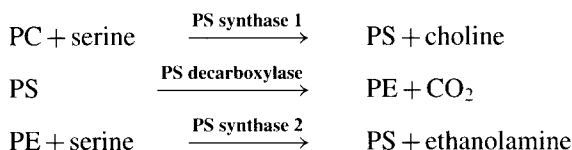
PS accounts for 5–15% of the phospholipids in eukaryotic cells. The lower concentration of PS compared to PC and PE is probably the reason PS was not discovered as a separate component of ‘kephalin’ (originally identified to be only PE in 1930) until 1941 by Folch. The correct structure was proposed by Folch in 1948 and confirmed by chemical synthesis in 1955 by Baer and Maurukas. PS is a required cofactor for protein kinase C and is required for initiation of the blood clotting cascade. In the plasma membrane of cells PS is normally located on the inner monolayer. During apoptosis, exposure of PS on the cell surface (outer monolayer) leads to recognition and removal of these cells by macrophages (V. Fadok, 1992).

PS is made in prokaryotes (Chapter 3), in some plants and yeast (S. Yamashita, 1997) via the CDP-diacylglycerol pathway. This route does not exist in animals. Instead, PS is

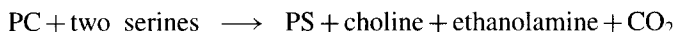
made by a base-exchange reaction catalyzed by PS synthase first described by Hübscher in 1959 (reaction 4 in Fig. 8) in which the head group of a PC or PE is exchanged for serine.

6.2. Chinese hamster ovary cell mutants and regulation

CHO mutants were generated that were auxotrophic for PS and demonstrated that these cells have two PS synthases [17]. PS synthase 1 utilizes PC and serine as substrates whereas PS synthase 2 utilizes only serine and PE. The two PS synthases, when coupled with PS decarboxylase, yield PS at the expense of PC and generate both choline and ethanolamine which could be recycled into the biosynthesis of PC and PE. As a result, PS and PE can both be generated without a decline in the amount of PC.



The sum of the reactions is:



A CHO mutant defective in PS synthase 1 was used to clone by complementation the cDNA for this enzyme [17]. The deduced amino acid sequence for murine PS synthase 1 was >90% identical to the CHO enzyme (S. Stone, 1998). The cDNA for PS synthase 2 from CHO cells was cloned and shown to be 32% identical in amino acid sequence to PS synthase 1 [17]. Immunoblot analysis indicated that both of the murine PS synthases are mainly localized to MAM (S. Stone, 2000). The source of the substantial PS synthase activity in the rough and smooth ER remains unknown, possibly a third PS synthase activity. The mRNAs encoding PS synthases 1 and 2 were found in all murine tissues examined but PS synthase 2 was enriched in testis and kidney (J.E. Vance, 2001).

Our understanding of regulation of PS biosynthesis is in its infancy. Addition of exogenous PS to the medium of CHO cells feedback inhibited the biosynthesis of PS [17]. CHO mutants in which Arg-95 of PS synthase 1, or Arg-97 in PS synthase 2, were replaced by lysine were no longer sensitive to inhibition by PS (M. Nishijima, 1998, 1999). There also appears to be 'cross-talk' between PE biosynthesis via the CDP-ethanolamine pathway and PS synthase 1/PS decarboxylation pathway since over-expression of PS synthase 1 increased production of PE from decarboxylation of PS and decreased PE biosynthesis via the CDP-ethanolamine pathway (S. Stone, 1999). Interestingly, over-expression of PS synthase 2 did not alter the activity of the CDP-ethanolamine pathway.

The murine gene for PS synthase 1 has been cloned and characterized (J.E. Vance, 2001). This is an important step toward the generation of mice with a disrupted gene for PS synthase 1. The gene for PS synthase 2 has been disrupted and the mice are viable (S. Young, 2002).

7. Inositol phospholipids

7.1. Historical developments

A major fate of PA is conversion to DG that is metabolized to PC, PE and TG (Fig. 1). Alternatively, PA can react with CTP to form CDP-DG that is utilized for the biosynthesis of the inositol phospholipids, phosphatidylglycerol (PG) and diphosphatidylglycerol (DPG) (Fig. 1).

Inositol is a cyclohexane derivative in which all 6 carbons are substituted with hydroxyl groups. The most common isoform is *myo*-inositol but other less abundant inositols with different structures also occur. The first report of an inositol-containing lipid was in 1930 from *Mycobacteria* [18] which is ironic since inositol lipids are rarely found in bacteria. Brain is the richest source of these lipids, as first discovered by Folch and Wooley in 1942. In 1949, Folch described PI phosphate (PI-P) which was later found to include PI and PI bisphosphate (PI-P₂). The chemical structures of PI, PI-P and PI-P₂ were determined by Ballou and coworkers between 1959 and 1961. PI (1.7 μ mol/g liver) constitutes around 10% of the phospholipids in a cell or tissue. PI-P and PI-P₂ are present at much lower concentrations (1–3% of PI). Agranoff et al. published the first experiments in 1958 on the incorporation of [³H]inositol into PI. Subsequently, Paulus and Kennedy showed that CTP was the preferred nucleotide donor.

7.2. CDP-diacylglycerol synthase

Regulation of the conversion of PA to CDP-DG is not well understood. The enzyme, CDP-DG synthase, is largely microsomal but is also found in the mitochondrial inner membrane.

A cDNA encoding CDP-DG synthase 1 was cloned from *Drosophila* (C.S. Zuker, 1995). This isoform is specifically located in photoreceptor cells of *Drosophila*. Mutations in this isoform lead to a defect in PI-P₂ biosynthesis. As a result mutant photoreceptor cells show severe defects in their phospholipase C-mediated signal transduction that can be rescued by re-introduction of the CDP-DG synthase cDNA.

CDNAs encoding human and murine CDP-DG synthases 1 and 2 were more recently cloned (S. Jackowski, 1997; B. Franco, 1999). CDP-DG synthase 2 is expressed during embryogenesis in the central nervous system whereas CDP-DG synthase 1 had a high level of expression in adult retina.

Curiously, in *Saccharomyces cerevisiae*, CDP-DG synthase activity is found in microsomes and the mitochondrial inner membrane even though only one gene encodes this activity [19]. Since only a single mRNA species was found, there may not be alternative splicing of the yeast gene. The yeast CDP-DG synthase gene is essential for cell viability as well as germination of spores.

7.3. Phosphatidylinositol synthase

Three potential sources for cellular inositol are: diet, de novo biosynthesis and recycling of inositol. Biosynthesis of inositol from glucose occurs in the brain and testes, and

other tissues to a lesser extent. The rate-limiting step appears to be the synthesis of inositol-3-phosphate from glucose-6-phosphate [20]. Inositol-3-phosphate is hydrolyzed to inositol by a phosphatase.

PI synthase was purified from human placenta [21]. When the cDNAs encoding either CDP-DG synthase 1 or phosphatidylinositol synthase, or both, were over-expressed in COS 7 cells, there was no change in the rate of PI biosynthesis indicating that the level of these enzymes was not limiting for PI biosynthesis (S. Jackowski, 1997).

Disruption of the PI synthase gene in yeast is lethal indicating that PI is essential [22]. Further information on the inositol phospholipids and their functions is covered in Chapter 12.

8. Polyglycerophospholipids

8.1. Historical developments and biosynthetic pathways

Diphosphatidylglycerol (DPG), commonly known as cardiolipin, was discovered in 1942 in beef heart by Pangborn. The correct structure (Fig. 9) was proposed in 1956–1957 and confirmed by chemical synthesis in 1965–1966 by de Haas and van Deenen. Phosphatidylglycerol (PG) was first isolated in 1958 from algae by Benson and Mauro. The structure was confirmed by Haverkate and van Deenen in 1964–1965. The third lipid in this class, bis(monoacylglycerol)phosphate was recovered from pig lung by Body and Gray in 1967. The stereochemistry differs from PG and DPG since bis(monoacylglycerol)phosphate contains *sn*-(monoacyl)glycerol-1-phospho-*sn*-1'-(monoacyl)-glycerol rather than a *sn*-glycerol-3-phospho linkage.

These three lipids (Fig. 9) are widely distributed in animals, plants, and microorganisms. In animals, DPG is found in highest concentration in cardiac muscle (9–15% of phospholipid), hence the name cardiolipin, and is exclusively found in the mitochondria. PG is generally present at a concentration of less than 1% of total cellular phospholipids, except in lung, where it comprises 2–5% of the phospholipid. In pulmonary surfactant and alveolar type II cells, PG is 7–11% of the total lipid phosphorous. Bis(monoacylglycerol)phosphate comprises less than 1% of total phospholipids in animal tissues, except in alveolar (lung) macrophages where it is 14–18% of total phospholipid.

The biosynthesis of PG was elucidated by Kennedy and coworkers in 1963 (Fig. 1). For DPG biosynthesis PA is transferred from CDP-DG to PG to yield DPG. DPG synthesis in *E. coli* differs and involves the condensation of two molecules of PG (Chapter 3).

Understanding the biosynthesis of bis(monoacylglycerol)phosphate has been a particular challenge because the carbon linked to the phosphate residue is the *sn*-1 rather than *sn*-3 configuration. The likely biosynthetic pathway is depicted in Fig. 10 [23].

An intermediate in the biosynthesis of bis(monoacylglycerol)phosphate is 1-acyl-lyso-PG (Fig. 10), also known as lysobis-PA. Recent studies have shown that the inner membranes of late endosomes are enriched in lysobis-PA and that these membranes play an important role in the sorting of insulin growth factor receptor 2 and the mannose-

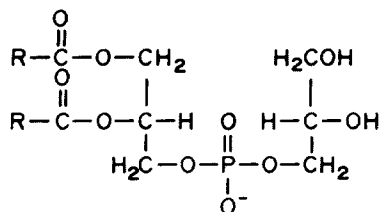
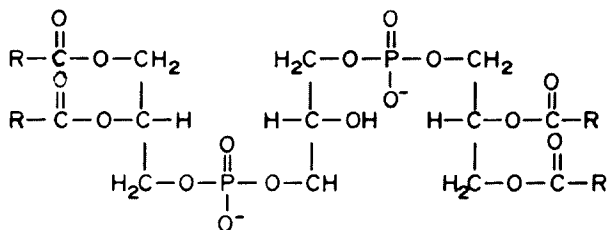
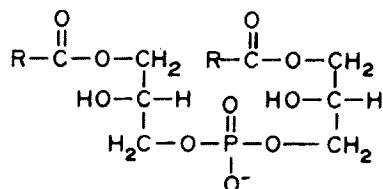
**Phosphatidylglycerol****Diphosphatidylglycerol****Bis(monoacylglycero)phosphate**

Fig. 9. Structures of polyglycerophospholipids.

6-phosphate receptor [24]. Moreover, lysobis-PA cross-reacts with antibodies produced in patients with antiphospholipid syndrome. Possibly, some of the pathological defects in this disease could arise from disruption of endosomal traffic. Moreover, the defect in cholesterol trafficking in Niemann-Pick C disease (Chapter 17) may also involve lysobis-PA (J. Gruenberg, 1999).

8.2. Enzymes and subcellular location

PG can be made in mitochondria and microsomes from various animal cells and, except for lung, appears to be primarily converted to DPG. DPG is biosynthesized exclusively on the matrix side of the mitochondrial inner membrane and is found only in this organelle. DPG synthase requires Co^{2+} for activity (K.Y. Hostetler, 1991). There is evidence that the rate-limiting step in DPG biosynthesis is the conversion of PA into CDP-DG (G.M. Hatch, 1994). Consistent with this idea, the levels of CTP have been shown to regulate DPG biosynthesis in cardiac myoblasts (G.M. Hatch, 1996).

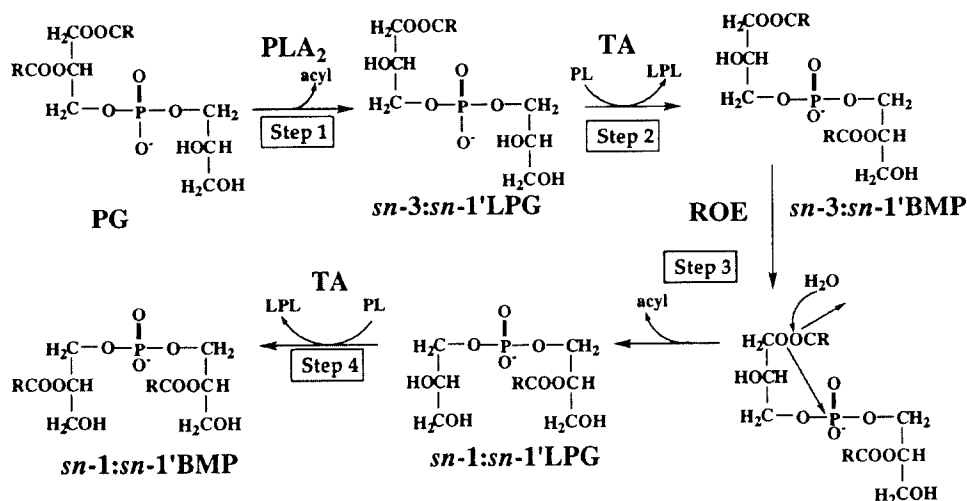


Fig. 10. Proposed pathway for the biosynthesis of bis(monoacylglycero)phosphate. Phospholipase A₂ (PLA₂) hydrolyzes PG to 1-acyl-lyso-PG (LPG). LPG is then acylated by a transacylase (TA), using a phospholipid (PL) as the acyl donor, to form bis(monoacylglycero)phosphate (BMP) that still retains the *sn*-3:*sn*-1' stereoconfiguration of the original PG and a lysophospholipid (LPL). The glycerol backbone of the *sn*-3:*sn*-1'-BMP is reoriented by an enzymatic activity (ROE) to yield *sn*-1:*sn*-1'-LPG (step 3). The final product, *sn*-1:*sn*-1'-BMP, is formed upon acylation of *sn*-1:*sn*-1'-LPG (step 4). The assignment of the acyl residues to the *sn*-2 positions of both glycerol moieties is based on their being primarily unsaturated and from degradation studies. It is believed that spontaneous rearrangement can occur so that the acyl residues end up on the *sn*-3 carbons as shown in Fig. 9. Figure from Amidon et al. [23] with permission.

Using techniques developed by Raetz and coworkers [25] M. Nishijima (1993) and coworkers isolated a temperature-sensitive mutant in PG-P synthase of CHO cells. The mutant had 1% of wild type CHO PG-P synthase activity at 40°C and a temperature-sensitive defect in PG and DPG biosynthesis. This mutant was used to show that DPG is required for the NADH-ubiquinone reductase (complex I) activity of the respiratory chain.

In yeast DPG synthesis has been genetically interrupted [26]. The yeast grows at temperatures between 16 and 30°C without DPG but fails to grow at 37°C on fermentable carbon sources such as glucose even though intact mitochondria are, therefore, not required for ATP synthesis. Thus, mitochondria must have some necessary function in yeast survival other than generating energy [26].

The fatty acyl content of phospholipids can also impact on mitochondrial function. Incubation of cardiomyocytes with palmitic acid increased the palmitic acid content of PA and PG and decreased DPG levels in mitochondria with a concomitant release of cytochrome *c* leading to apoptosis (W. Dowhan, 2001).

9. Remodeling of the acyl substituents of phospholipids

Phospholipids are made de novo with the fatty acid compositions present in the precursors DG and CDP-DG. Once the phospholipid is made, the fatty acid substituents

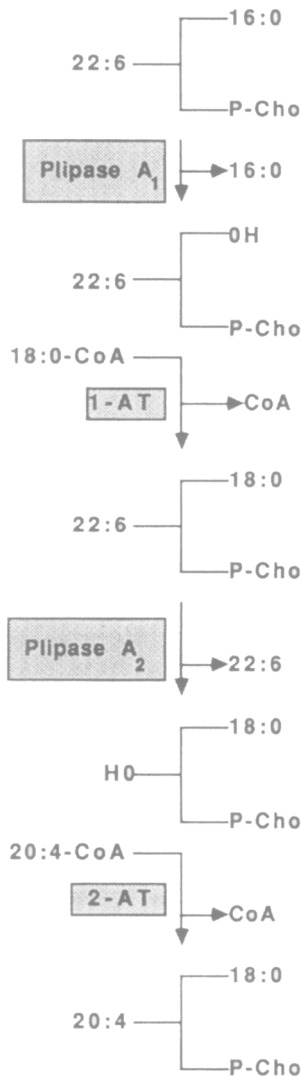


Fig. 11. Fatty acids at both the *sn*-1 and *sn*-2 positions of PC can be deacylated by phospholipases and reacylated by acyltransferases. Palmitic acid (16:0) can be removed from the *sn*-1 position and replaced with stearic acid (18:0). The fatty acid at the *sn*-2 position is depicted as docosahexaenoic acid (22:6) which can be replaced with 20:4 or 18:2. If the fatty acid at the *sn*-2 position were oleic acid, it could also be deacylated and reacylated. Alternatively, deacylation/reacylation could occur initially at the *sn*-2 position. Plipase, phospholipase; 1-AT, acyl-CoA : lyso-PC 1-acyltransferase; 2-AT, acyl-CoA : lyso-PC 2-acyltransferase; cho, choline.

can be remodeled via deacylation–reacylation reactions (Fig. 11). Remodeling can occur on either the *sn*-1 or *sn*-2 positions of the glycerolipid. For example, a major molecular species formed from the conversion of PE to PC is 16:0–22:6-PC (R.W. Samborski, 1990). This species of PC has a half-life of less than 6 h and appears not to be

significantly degraded but rather converted to other molecular species, particularly those with 18:0 on the *sn*-1 position and 20:4, 18:2 or 22:6 on the *sn*-2 position. Other studies have suggested that the main products of de novo PC and PE biosynthesis are 16:0–18:2, 16:0–18:1, 16:0–22:6 and 18:1–18:2. The major remodeled product is 18:0–20:4 for both PC and PE (H.H.O. Schmid, 1995). Why 18:0–20:4-PC and -PE are made by this circuitous route, rather than directly, is not known.

10. Regulation of gene expression in yeast

The pathways for the biosynthesis of phospholipids in yeast were largely elucidated by Lester and coworkers in the late 1960s (Fig. 12). These pathways are similar to those found in other eukaryotes except PS in yeast is made via a pathway similar to that found in *E. coli* where CDP-DG reacts with serine to yield PS and CMP.

Considerable interest in yeast as a model system has developed over the past two

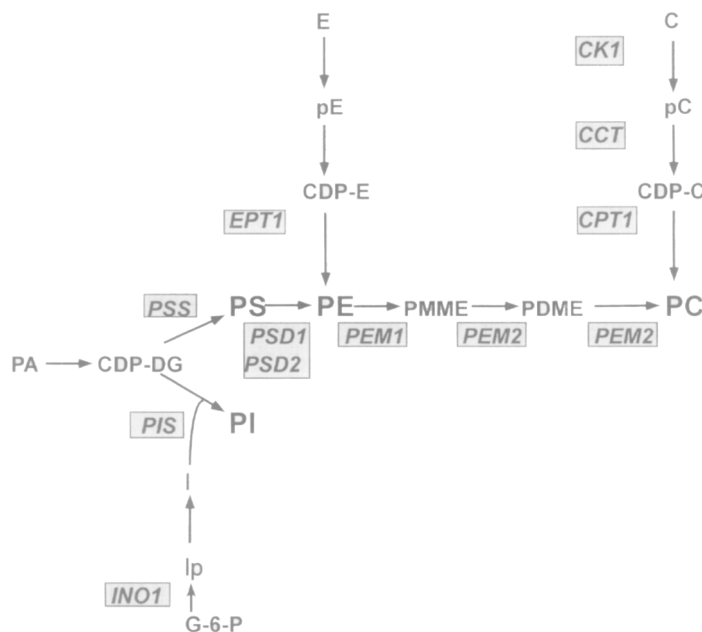


Fig. 12. The pathway for phospholipid biosynthesis in yeast and designation of the genes (italics in boxes) encoding the enzymes that catalyze the reactions. The abbreviations are: E, ethanolamine; pE, phosphoethanolamine; CDP-E, CDP-ethanolamine; C, choline; pC, phosphocholine; CDP-C, CDP-choline; PE, phosphatidylethanolamine; PMME, phosphatidylmonomethylethanolamine; PDME, phosphatidylmethylethanolamine; PC, phosphatidylcholine; PS, phosphatidylserine; PA, phosphatidic acid; CDP-DG, CDP-diacylglycerol; PI, phosphatidylinositol; I, inositol; Ip, inositol phosphate; G-6-P, glucose-6-phosphate. The genes encode the following enzymes: *INO1*, I-1-P synthase; *PIS*, PI synthase; *PSS* (also known as *CHO1*), PS synthase; *EPT1*, CDP-E:1,2-diacylglycerol ethanolaminephosphotransferase; *PEM1* (*CHO2*), PE methyltransferase; *PEM2* (*OPI3*), phospholipid methyltransferase; *CK1*, choline kinase; *CCT*, CTP:phosphocholine cytidyltransferase (abbreviated as CT elsewhere in this chapter); *CPT1*, CDP-C:1,2-diacylglycerol cholinephosphotransferase; *PSD1* and *PSD2*, PS decarboxylase.

decades. Reasons for choosing *Saccharomyces cerevisiae* include a large knowledge base in classical genetics, the ease of making mutant strains and the ability to grow large amounts of yeast. Whereas understanding the regulation of expression of phospholipid biosynthetic enzymes in animal cells is still in its infancy, considerable progress has been made in the yeast system [26–28]. When yeast cells are grown in the presence of choline and inositol, the expression of the enzymes involved in the conversion of PA and glucose-6-P to PI, PC and PE is depressed (Fig. 12).

Both positive and negative regulatory factors are involved in the regulation of expression of phospholipid biosynthetic enzymes in yeast. The *INO2* and *INO4* genes encode transcription factors that are required for the expression of inositol-1-P synthase (*INO1*). In vitro transcribed and translated proteins derived from *INO2* and *INO4* form a heterodimer that binds a specific DNA fragment of the *INO1* gene referred to as UAS_{INO} (S.A. Henry, 1994). Ino4p (the protein encoded by *INO4*) and Ino2p exhibit basic helix–loop–helix domains. The Ino2p–Ino4p heterodimer binds to UAS_{INO} of the *INO1* promoter that contains two copies of a binding site (CANNTG) for basic helix–loop–helix-containing proteins.

The *OPI1* gene encodes a protein that is a negative regulatory factor for phospholipid biosynthesis [27]. Opi1p contains a leucine zipper, a motif implicated in protein–DNA interactions and transcriptional control. *Opi1* mutants exhibit a two-fold increase in the constitutive expression of inositol-1-P synthase and other enzymes involved in PI, PC and PE biosynthesis. The mechanism by which Opi1p mediates its negative regulatory role is unknown. Opi1p does not interact directly with UAS_{INO} or with Ino2p or Ino4p. Phosphorylation of Opi1 by protein kinase C may be involved (G.M. Carman, 2001).

Recent experiments have identified other proteins that interact with Ino4p (J.M. Lopez, 2000) indicating that there is still much to learn about transcriptional regulation of phospholipid biosynthetic genes in yeast. How the regulatory genes (*INO2*, *INO4*, *OPI1*) are themselves regulated is just beginning to be studied (J.M. Lopes, 2001).

11. Future directions

Since the first edition of this book was published in 1985 there have been astonishing developments in phospholipid metabolism. Some of these advances have dictated that a separate chapter be devoted to the role of glycerophospholipids in signal transduction (Chapter 12). The purification of some enzymes involved and the use of genetic screens has allowed molecular biological techniques to be used to clone and express cDNAs and genes for eukaryotic phospholipid biosynthetic enzymes. In addition, genetically modified mouse models are being developed.

- (1) We can expect that crystal structures of some of the soluble proteins will be reported.
- (2) More genes that encode phospholipid biosynthetic enzymes will be cloned and characterized. Elements of the genes involved in regulation of transcription will be mapped and positive and negative transcription factors should be identified.
- (3) We can expect that more transgenic mice that over-express some of these enzymes,

as well as mice in which phospholipid biosynthetic genes have been disrupted, will be produced. Such studies should provide valuable insight into the role of these enzymes in whole animal physiology.

- (4) The yeast system will continue to be exploited for studies on gene function and expression as well as regulation of phospholipid biosynthesis.
- (5) There should be progress in understanding the regulation of PE, PI, PS and DPG biosynthesis.
- (6) In the process of testing hypotheses and asking fundamental questions about phospholipid biosynthesis, we can continue to expect the unexpected.

Abbreviations

CDP-DG	CDP-diacylglycerol
CHO	Chinese hamster ovary
CT	CTP : phosphocholine cytidylyltransferase
DPG	diphosphatidylglycerol (cardiolipin)
DG	diacylglycerol
ER	endoplasmic reticulum
MAM	mitochondria associated membrane
PA	phosphatidic acid
PC	phosphatidylcholine
PE	phosphatidylethanolamine
PEMT	phosphatidylethanolamine <i>N</i> -methyltransferase
PG	phosphatidylglycerol
PI	phosphatidylinositol
PS	phosphatidylserine
SREBP	sterol response element binding protein
TG	triacylglycerol

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Ether-linked lipids and their bioactive species

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1. Introduction

Naturally occurring ether lipids contain either *O*-alkyl or *O*-alk-1-enyl groupings. Those possessing the *O*-alk-1-enyl moiety with a *cis* double bond adjacent to the ether bridge are referred to as plasmalogens, as well as vinyl ethers. Both the *O*-alkyl and *O*-alk-1-enyl substituents are generally located at the *sn*-1 position of the glycerol moiety although di- and tetra-*O*-alkylglycerolipids have been described in some cells. Unlike the diverse types of acyl moieties present in glycerolipids, the predominant *O*-alkyl and *O*-alk-1-enyl ether-linked chains generally consist mainly of 16:0, 18:0, and 18:1 aliphatic groupings, but other types of chain lengths, degrees of unsaturation, and occasional branched-chains do exist as minor components. Except for intermediary metabolites and certain bioactive lipids, ether linkages in phospholipids of mammalian cells exist almost exclusively in the choline and ethanolamine glycerolipid classes. The majority of the *O*-alkyl moieties normally occur as plasmalyncholines¹, whereas the *O*-alk-1-enyl grouping is mainly associated with the plasmenylethanolamines with the exception of heart where plasmenylcholines predominate. Some neutral lipids such as alkyl-diacylglycerols (glyceryl ether diesters) and alkylacylglycerols, analogs of triacylglycerols and diacylglycerols, respectively, are also found in cells. Fig. 1 illustrates the chemical structures of the most common ether lipids found in mammals.

A number of books [1–6] and review articles [7–14] on ether lipids, some specifically emphasizing platelet-activating factor (PAF), are recommended as reading material. These sources provide a comprehensive listing of published papers.

2. Historical highlights

The early literature concerning ether-linked lipids has also been covered in detail [1,2,9,15]. Perhaps the first evidence, albeit circumstantial, to suggest the existence of *O*-alkyl lipids in nature was the isolation of an unsaponifiable fraction of lipids from starfish that was referred to as ‘astrol’, which was subsequently shown to have similar

¹ Plasmalyn designates the radical ‘1-alkyl-2-acyl-*sn*-glycero-3-phospho-’, whereas plasmenyl represents the radical ‘1-alk-1-enyl-2-acyl-*sn*-glycero-3-phospho-’; the prefix phosphatidyl is used only to denote the radical ‘1,2-diacyl-*sn*-glycero-3-phospho-’. ‘Radyl’ is used as a prefix in glycerolipid nomenclature when the aliphatic substituents are unknown at the *sn*-positions of the glycerol moiety or when either acyl, alkyl, or alk-1-enyl moieties would be of equal importance.

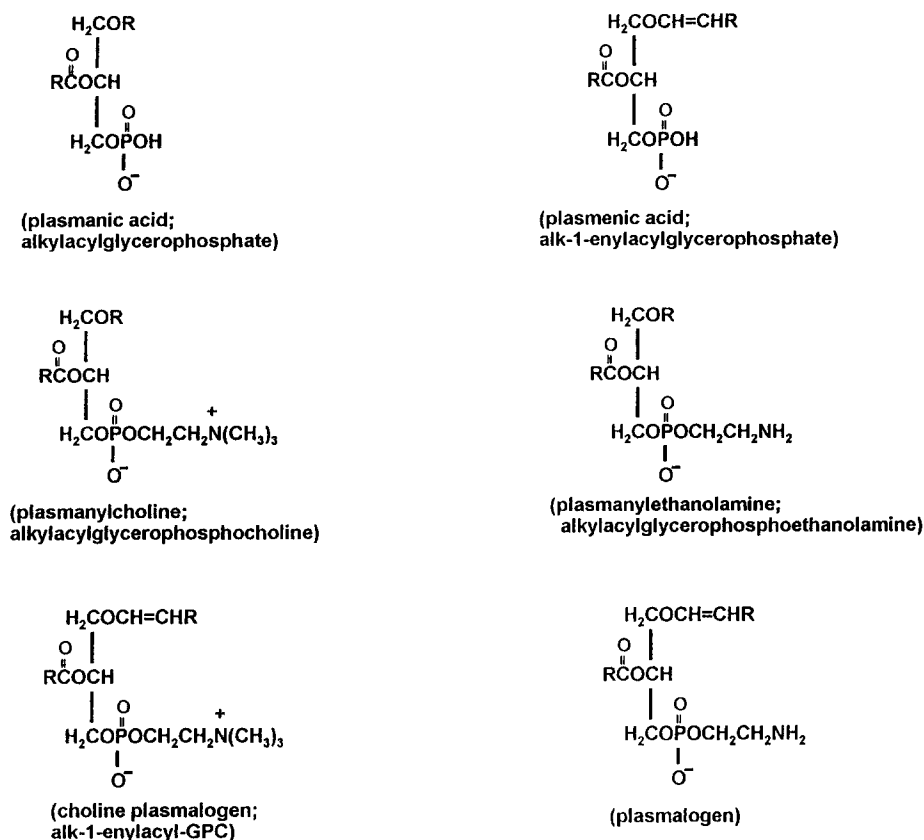


Fig. 1. Chemical structures of biologically significant types of ether-linked lipids found in mammalian cells.

properties to batyl alcohol, an alkylglycerol possessing an 18-carbon aliphatic chain at the *sn*-1 position of the glycerol moiety. During the same period the presence of alkyl ether lipids in liver oils of various saltwater fish was described by the Japanese scientists M. Tsujimoto and Y. Toyama (1922). The common names of the alkylglycerols, *chimyl* [16:0 alkyl], *batyl* [18:0 alkyl], and *selachyl* [18:1 alk-9-enyl] alcohols, are based on the fish species from which they were originally isolated. Complete proof of the precise chemical nature of the alkyl linkage at the *sn*-1 position in these glycerolipids was provided by W.H. Davies, I.M. Heilbron, and W.E. Jones (1933) from England.

The German scientists, R. Feulgen and K. Voit (1924) originally described plasmalogens in a variety of fresh tissue slices preserved in a HgCl_2 solution after being erroneously treated with a fuchsin-sulfurous acid reagent without the normal fixation and related histological processing with organic solvents. Only the cytoplasm of cells, but not the nuclei, was stained a red-violet color, which led to the conclusion that an aldehyde was present in the cell plasma. This substance was called 'plasmal'. If the histological preparations were treated with a lipid-extracting solvent before exposure to the dye, no colored stain appeared in the cytoplasm. This unknown precursor of

the cytosolic aldehyde that reacted with the dye was called plasmalogen, a name still retained as the generic term for all alk-1-enyl-containing glycerolipid classes.

It was not until the 1950s that the precise chemical structure of the alk-1-enyl linkage in ethanolamine plasmalogens was proven, primarily through the combined efforts of M.M. Rapport and G.V. Marinetti in the United States, G.M. Gray in England, E. Klenk and H. Debusch in Germany, and their various co-workers. The first cell-free systems to synthesize the alkyl ether bond were described independently in 1969 by F. Snyder, R.L. Wykle, and B. Malone and by A. Hajra. Shortly thereafter, studies by R.L. Wykle, M.L. Blank, B. Malone, and F. Snyder and by F. Paltauf and A. Holasek demonstrated that the *O*-alkyl moiety of an intact phospholipid could be enzymatically desaturated to the alk-1-enyl grouping (see Section 6.2.5). One of the most significant developments in the ether-lipid field occurred in 1979 when one of the most potent bioactive molecules known, an acetylated form of a choline-containing alkylglycerolipid called platelet-activating factor or PAF, was identified independently by three separate groups.

3. Natural occurrence

Chemical, chromatographic, and mass spectral methods for analyzing ether-linked glycerolipids have been reviewed [16,17]. Ether-linked phospholipids generally are isolated as a mixture with their ester-linked counterparts.

Ether-linked lipids occur throughout the animal kingdom and are even found as minor components in several higher plants. Some mammalian tissues, and avian, marine, molluscan, protozoan, and bacterial lipid extracts contain significant proportions of ether-linked lipids. Highest levels of ether lipids in mammals occur in nervous tissue, heart muscle, testes, kidney, preputial glands, tumor cells, erythrocytes, bone marrow, spleen, skeletal tissue, neutrophils, eosinophils, macrophages, platelets, and lipoproteins. The large quantities of ethanolamine plasmalogens associated with various lipoproteins from rat serum and human plasma (36% and 50%, respectively, of the total ethanolamine phosphatides) is of particular interest since the liver contains relatively low amounts of ether lipids and the plasmalogens are not acquired in the lipoproteins after their secretion. Although the dietary consumption of ether lipids by humans has largely been ignored by nutritionists, it is clear that certain meats and seafoods can contain relatively high amounts of these lipids.

Analogues of triacylglycerols have also been described. 1-Alkyl-2,3-diacyl-*sn*-glycerols are characteristically elevated in tumor lipids and 1-alk-1-enyl-2,3-diacyl-*sn*-glycerols (neutral plasmalogens) have also been detected in tumors and adipose tissue of mammals and in fish liver oil. In fact, even alkylacetylacylglycerols have been shown to be formed by human leukemic cells.

1-Alkyl-2-acyl-*sn*-glycero-3-phosphocholine (Fig. 1), a significant component of platelets, neutrophils, macrophages, eosinophils, basophils, monocytes, and endothelial, mast, and HL-60 cells (a human promyelocytic leukemic cell line), is a precursor of platelet-activating factor (PAF, 1-alkyl-2-acetyl-*sn*-glycero-3-phosphocholine; see Fig. 2). Thus, this precursor appears to be a constituent of all cells known to produce

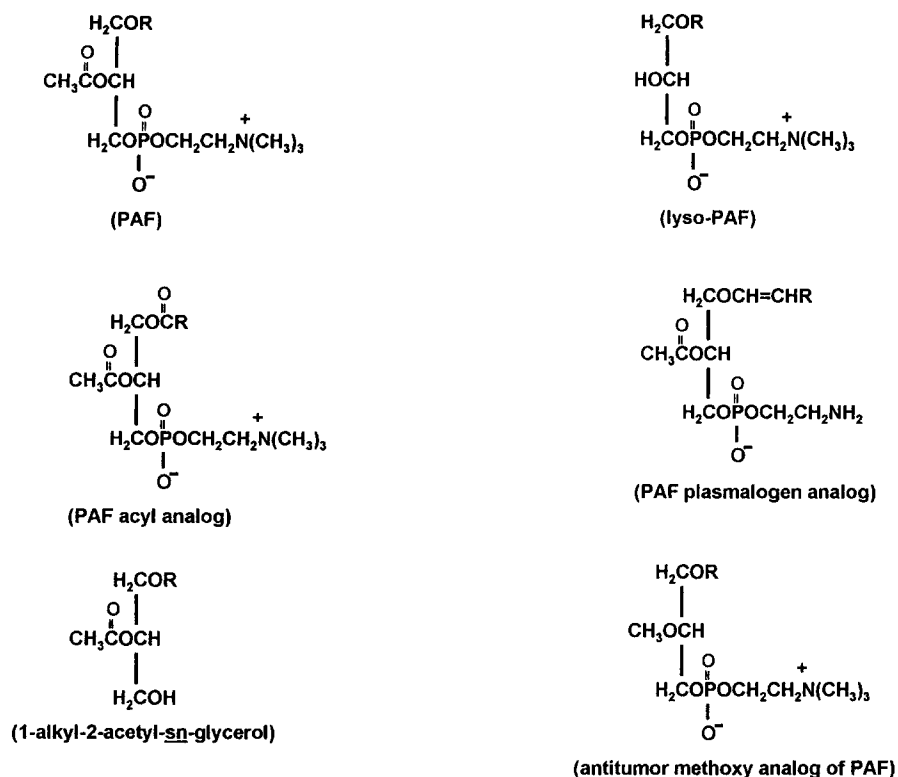


Fig. 2. Chemical structures of PAF and structurally related ether-linked glycerolipids possessing biological activities.

PAF by the remodeling pathway; in human neutrophils and eosinophils the alkyl subclass comprises 45 and 70 mol% of the choline-linked phosphoglycerides, respectively, while the ethanolamine-linked class contains 60–65 mol% plasmalogen. PAF is also found in saliva, urine, and amniotic fluid, which indicates that other cell types could be the source of PAF in these fluids.

Dialkylglycerophosphocholines have been reported as minor constituents of bovine heart and spermatozoa. Moreover, heart tissue is unique with respect to its plasmalogen content, since in some animal species, this is the only mammalian tissue known to contain significant amounts of choline plasmalogens instead of the usually encountered ethanolamine plasmalogens.

Halophilic bacteria contain an unusual dialkyl type of glycerolipid (a diphytanyl ether analog of phosphatidylglycerophosphate) that has an opposite stereochemical configuration from all other known ether-linked lipids, i.e., the ether linkages are located at the *sn*-2 and *sn*-3 positions. The biosynthetic pathway for the formation of the ether bond in halophiles is still unknown. Acidophilic thermophiles contain tetraalkyl glycerolipids with their two glycerol moieties linked across their membranes, which prevents them from being freeze-fractured.

Many anaerobic bacteria are highly enriched in plasmalogens. For example, *Clostridium butyricum* contains significant amounts of ethanolamine plasmalogens and *Megasphaera elsdenii* has been reported to contain very large quantities of plasmenyl ethanolamine and plasmenylserine. However, despite the large pool of plasmalogens in such anaerobes, no information has emerged about how they synthesize the alk-1-enyl ether bond.

4. Physical properties

Replacement of ester linkages in glycerolipids with ether bonds mainly affects hydrophobic–hydrophilic interactions. Nevertheless, the closer linear packing arrangement attainable with ether-linked moieties also is capable of influencing the polar head group region of phospholipids. The novel placement of the $\Delta 1$ double bond in plasmalogens can also exert effects on stereochemical relationships and therefore, the presence of an ether-linkage in phospholipids can modify both the configuration and functional properties of membranes.

In model membranes, ether-linked lipids have been shown to decrease ion permeability, surface potential, and lower the phase temperature of membrane bilayers when compared to their diacyl counterparts. *Clostridium butyricum* appears to be able to regulate the stability of the bilayer arrangement of membranes by altering the ratio of ether versus acyl type of ethanolamine phospholipids in response to changes in the degree of lipid unsaturation of the membranes. The experiments with bacteria indicate that the substitution of plasmenylethanolamine for phosphatidylethanolamine in biomembranes would have only small effects on lipid melting transitions, whereas the tendency to form non-lamellar lipid structures would be significantly increased.

5. Biologically active ether lipids

5.1. Platelet-activating factor

In 1979, the chemical structure of PAF was identified as 1-alkyl-2-acetyl-*sn*-glycero-3-phosphocholine (Fig. 2). The semisynthetic preparation tested in these initial experiments caused aggregation of platelets at concentrations as low as 10^{-11} M and induced an antihypertensive response when as little as 60 ng were administered intravenously to hypertensive or normotensive rats. It is now known that PAF, a phospholipid secreted by numerous cells, exerts many different types of biological responses (Table 1) and it has been implicated as a contributing factor in the pathogenesis of such diverse disease processes as asthma, hypertension, allergies, inflammation, and anaphylaxis, to name only a few.

PAF has been isolated and very well characterized from a number of cellular sources. Basophils, neutrophils, platelets, macrophages, monocytes and mast, endothelial, and HL-60 cells are among the highest producers of PAF when stimulated by agonists such as chemotactic peptides, zymosan, thrombin, calcium ionophores, antigens, bradykinin,

Table 1
Biological activities associated with platelet-activating factor

I. In vivo responses:	
1.	Bronchoconstriction ↑
2.	Systemic blood pressure ↓
3.	Pulmonary resistance ↑
4.	Dynamic lung compliance ↓
5.	Pulmonary hypertension and edema ↑
6.	Heart rate ↑
7.	Hypersensitivity responses ↑
8.	Vascular permeability ↑
II. Cellular responses:	
1.	Aggregation of neutrophils and platelets ↑
2.	Degranulation of platelets, neutrophils, and mast cells ↑
3.	Shape changes in platelets, neutrophils, and endothelial cells ↑
4.	Chemotaxis and chemokinesis in neutrophils ↑
III. Biochemical responses:	
1.	Ca ²⁺ uptake ↑
2.	Respiratory burst and superoxide production ↑
3.	Protein phosphorylation ↑
4.	Arachidonate turnover ↑
5.	Phosphoinositide turnover ↑
6.	Protein kinase ↑
	– protein kinase C
	– mitogen-activated protein kinase
	– G-protein receptor kinases
	– protein tyrosine kinase
7.	Glycogenolysis ↑
8.	Tumor necrosis factor production ↑
9.	Interleukin 2 production ↓
10.	Activation of immediate-early genes, e.g., <i>c-fos</i> and <i>c-jun</i> , <i>zif/268</i> ↑

ATP, C_{5a}, collagen, and disease states. The amount of PAF produced by various stimuli is dependent on the cell type and the specific agonist used. Most animal tissues also have the capacity to produce PAF by de novo synthesis (see Section 6.3.2).

Other acetylated glycerolipids that are structurally related to PAF include 1-alkyl-2-acetyl-*sn*-glycerols and the plasmalogen and acyl analogs of PAF that possess choline or ethanolamine moieties (Fig. 2). Both the alkylacetyl glycerols (perhaps via phosphorylation) and the choline plasmalogen analog of PAF can mimic the actions of PAF, perhaps through their interactions with the PAF receptor. Biological potencies of the PAF analogs range from 5- to 4000-fold weaker than PAF [18].

An unnatural chemically synthesized analog of PAF, 1-alkyl-2-methoxy-*sn*-glycero-3-phosphocholine (Fig. 2) and related derivatives, possesses unique highly selective antitumor activity [19]. Clinical studies in Europe have shown a promising therapeutic potential for the methoxy analog in treating certain types of human cancers. Although its mode of action has been difficult to ascertain, the primary site of action of these PAF analogs is the plasma membrane rather than the cell nucleus. The cytotoxic

activity of this antineoplastic phospholipid is apparently due to its ability to prevent the formation of membranes by blocking phosphatidylcholine synthesis (Chapter 8) via the inhibition of CTP:phosphocholine cytidyltransferase, the rate-limiting enzyme in phosphatidylcholine biosynthesis [20].

5.2. Other ether-linked mediators

In addition to PAF and eicosanoids, 1-alkyl-2-acyl-*sn*-glycero-3-phosphocholine yields 1-alkyl-2-acyl-*sn*-glycero-3-phosphate when acted upon by phospholipase D [21]. Both the alkylacyl- and diacylglycerols share in their ability to increase responses (priming) of neutrophils to other stimuli of arachidonic acid release and the oxidative burst. However, only the diacylglycerol primes for the formation of lipoxygenase products.

5.3. Oxidized phospholipids

Oxidation of the phospholipids of plasma lipoproteins generates bioactive phospholipid species with PAF-like activity [13]. A complex mixture of oxidation products is formed but the species that bind and act through the PAF receptor are alkyl ether-linked and contain short-chain oxidized residues in the *sn*-2 position derived from polyunsaturated acyl chains. Normally, PAF analogs containing *sn*-2 chains longer than four carbons have little activity but the introduction of an oxidized group at the end of the chain yields longer-chain active analogs. The PAF acetylhydrolase (Section 7.3.1) can remove the oxidized chains to inactivate these products. These oxidized species may play an important role in inflammation and development of atherosclerotic plaques and other cardiovascular disorders.

5.4. Receptors, overexpression, and knockout mice

The cDNA encoding the cell surface PAF receptor has been cloned and the primary structure sequenced from a number of cells/tissues including guinea pig lung, human neutrophils, HL-60 cells (granulocytic form), and human heart [13,14]. Human and guinea pig receptors consist of 342 amino acids with a C-terminal cytoplasmic tail possessing serine and threonine residues which could be potential sites for regulation via phosphorylation. The PAF receptor is typical of other members of the family of G-protein-coupled receptors, which span the membrane seven times (e.g., rhodopsin, β_1 and β_2 adrenergic, D_2 -dopamine, and M1–M5 muscarinic). Based on modeling and site-directed mutagenesis of the receptor it is proposed that the central portion of the receptor and the histidine residues 188, 248, and 249 form the PAF binding pocket. It is also deemed likely that there is a disulfide bond between the cysteine residues at positions 90 and 163. A mutation in the third transmembrane domain resulted in a constitutively active receptor. It has been shown in other studies that the third intracellular loop is necessary for initiating phosphatidylinositol turnover. The fate of the receptor-bound PAF is unknown.

The role of PAF *in vivo* has been examined by overexpressing the guinea pig receptor in mice [8,14]. These mice had an increased death rate in response to

endotoxin and surprisingly developed melanocytic tumors of the skin. Ishii and Shimizu [14] have also generated PAF receptor knockout mice. These mice had less severe anaphylactic responses including diminished cardiovascular instability, alveolar edema, and airway constriction than did wild-type mice. Unexpectedly, the receptor-deficient animals reproduced normally and remained susceptible to endotoxin. The existence of a second PAF receptor might explain these results. Further studies are required to resolve the somewhat contradictory observations. In humans lacking PAF acetylhydrolase, an enzyme that degrades PAF, an increased severity of asthma, coronary artery disease and stroke was observed. Several review articles [7,8,13,14] discuss the role of PAF receptors in signal transduction, cloning, sequencing, and related studies.

5.5. Receptor antagonists

A number of PAF antagonists are available that block binding of PAF to its receptor. Some of these antagonists are derived from plants, such as *Ginkgo biloba*, while others are structural analogs of PAF, and yet others are chemically synthesized compounds found through screening. Although some of the inhibitors effectively block PAF responses in certain systems, they have not proven highly effective as anti-inflammatory drugs. It is possible that the drugs do not gain access to all PAF receptors *in vivo*, or that the network of inflammatory mediators synergizes to overcome suppression of the PAF receptor.

6. Enzymes involved in ether lipid synthesis

In view of the vast literature about the enzymes involved in the metabolism of ether-linked lipids and PAF, the reader should consult the various reviews on this subject [5,10–13].

6.1. Ether lipid precursors

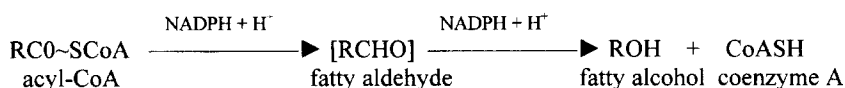
6.1.1. Acyl Co-A reductase

Fatty alcohol precursors of ether lipids are derived from acyl-CoAs via a fatty aldehyde intermediate in a reaction sequence catalyzed by a membrane-associated acyl-CoA reductase (Fig. 3A). A cytosolic form of the reductase from bovine heart has also been described.

Acyl-CoA reductases associated with membrane systems use acyl-CoA substrates, and in mammalian cells, they exhibit a specific requirement for NADPH. The fatty alcohols produced by the reductase can be oxidized back to the fatty acids by microsomes in the presence of NAD.

Although only traces of fatty aldehydes can normally be detected as an intermediate in these reactions, the use of trapping agents such as semicarbazide has documented that aldehydes are indeed formed as intermediates. Acyl-CoA reductase prefers saturated substrates over acyl-CoAs that are unsaturated; in fact, the enzyme in brain microsomes is not able to convert polyunsaturated moieties to fatty alcohols. Some evidence

A)



B)

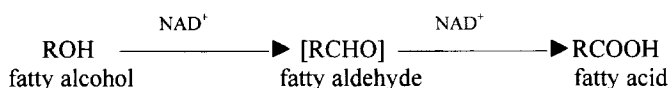


Fig. 3. Enzymatic synthesis (A) and oxidation (B) of long-chain fatty alcohols by (I) acyl-CoA reductase and (II) fatty alcohol oxidoreductase, respectively.

indicates that, at least in brain, acyl-CoA reductase is localized in microperoxisomes. Topographical studies of microperoxisomal particles have revealed the acyl-CoA reductase activity is located at the cytosolic surface of these membranes. In rabbit Harderian glands and *Euglena gracilis*, the NADPH-dependent reductase appears to be closely coupled with fatty acid synthase and it has been suggested that the fatty acid bound to acyl carrier protein, rather than acyl-CoA, is the substrate for this reductase.

6.1.2. Dihydroxyacetone-P acyltransferase

Presumably, dihydroxyacetone-P acyltransferase is present in all mammalian cells that synthesize alkylglycerolipids, since the acylation of dihydroxyacetone-P is an obligatory step in the biosynthesis of the ether bond in glycerolipids. On the other hand, the quantitative importance of the pathways utilizing dihydroxyacetone-P versus *sn*-glycerol-3-P in the biosynthesis of glycerolipid esters (Chapter 8) has never been firmly established in intact cells.

Current evidence indicates that dihydroxyacetone-P acyltransferase, as well as alkyl-dihydroxyacetone-P synthase, is localized in microperoxisomes. Nevertheless, many studies of these enzymes have been done with microsomal and/or mitochondrial preparations; however, it is well known that microperoxisomes sediment with microsomes and large peroxisomes sediment with mitochondria under the usual preparation conditions for these subcellular fractions. Investigations of the topographical orientation of dihydroxyacetone-P acyltransferase in membrane preparations from rabbit Harderian glands and rat brains indicate that unlike most other enzymes in glycerolipid metabolism, dihydroxyacetone-P acyltransferase appears to be located on the internal side of microsomal vesicles.

6.2. Ether lipids

6.2.1. O-Alkyl bond: mechanism of formation

Formation of the alkyl ether bond in glycerolipids is catalyzed by alkyl-dihydroxyacetone-P synthase (Fig. 4). This reaction, which forms alkyl-dihydroxyacetone-P as the first detectable intermediate in the biosynthetic pathway for ether-linked glycerolipids, is

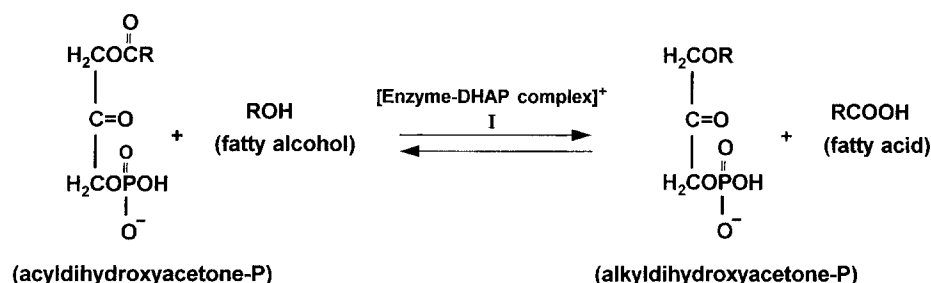


Fig. 4. The reaction that forms the *O*-alkyl bond is catalyzed by (I) alkyldihydroxy acetone-P synthase and is thought to proceed via a ping-pong mechanism. The abbreviation DHAP in this illustration designates dihydroxyacetone-P. Upon binding of acyl-DHAP to the enzyme, alkyl-DHAP synthase, the pro-*R* hydrogen at carbon atom 1 is exchanged by an enolization of the ketone, followed by release of the acyl moiety to form an activated enzyme–DHAP complex. The carbon atom at the 1 position of DHAP in the enzyme complex is thought to carry a positive charge that may be stabilized by an essential sulfhydryl group of the enzyme; thus, the incoming alkoxide ion reacts with the carbon 1 atom to form the ether bond of alkyl-DHAP. It has been proposed that a nucleophilic cofactor at the active site covalently binds the DHAP portion of the substrate.

unique in mammals since it is the only one known where a fatty alcohol can be directly substituted for a covalently linked acyl moiety. Alkyldihydroxyacetone-P synthase has been primarily investigated in microsomal preparations; however, as with dihydroxyacetone-P acyltransferase, there is general agreement that the synthase activity is peroxisomal. The alkyl synthase cDNAs from guinea pig and human liver reveal that both proteins contain a peroxisomal targeting signal 2 [22]. The importance of peroxisomes in ether lipid synthesis has been highlighted by the finding that patients with Zellweger syndrome (lacking peroxisomes) and related peroxisomal-deficient diseases have extremely low levels of plasmalogens and ether lipids.

Kinetic experiments with a partially purified enzyme from Ehrlich ascites cells have suggested the reaction catalyzed by alkyldihydroxyacetone-P synthase involves a ping-pong rather than sequential-type mechanism, with an activated enzyme–dihydroxyacetone-P intermediary complex playing a central role. The existence of this intermediate would explain the reversibility of the reaction, since the enzyme–dihydroxyacetone-P complex can react with either fatty alcohols (forward reaction) or fatty acids (back reaction). This unusual enzymatic mechanism is also consistent with other known properties of alkyldihydroxyacetone-P synthase. Acyldihydroxyacetone-P acylhydrolase does not appear to participate in this mechanism since its activity is not present in the purified synthase preparation.

A number of novel features characterize the reaction that forms alkyldihydroxyacetone-P. The pro-*R* hydrogen at C-1 of the dihydroxyacetone-P moiety of acyldihydroxyacetone-P exchanges with water, without any change in the configuration of the C-1 carbon. Cleavage of the acyl group of acyldihydroxyacetone-P occurs before the addition of the fatty alcohol, and either fatty acids or fatty alcohols can bind to the activated enzyme–dihydroxyacetone-P complex to produce acyldihydroxyacetone-P or alkyldihydroxyacetone-P, respectively. There is no evidence for a Schiff's base being formed. Nevertheless, a ketone function is an essential feature of the

substrate, acyldihydroxyacetone-P. In addition, mass spectrometric analyses have clearly shown that the oxygen of the ether bond is donated by the fatty alcohol and both oxygens of the acyl linkage of acyldihydroxyacetone-P are recovered in the fatty acid released.

The cDNAs for human and guinea pig alkyldihydroxyacetone-P synthase have been cloned and expressed. The apparent molecular mass of the enzyme from guinea pig is 65 kDa on polyacrylamide gel electrophoresis. The mature enzymes of both human and guinea pig have a predicted mass of 67 kDa. The enzyme is synthesized as a 658 amino acid precursor containing an N-terminal presequence of 58 amino acids encoding the peroxisomal targeting signal 2 motif, which is removed in the mature protein. In studies of the structure and mechanism of action of the enzyme, de Vet et al. [22] made the surprising finding that the enzyme contains a FAD binding domain. They demonstrated the presence of FAD in the enzyme and found that the FAD cofactor is required for activity of the enzyme. The FAD participates directly in catalysis and becomes reduced upon incubation with acyldihydroxyacetone-P. This finding suggests that the dihydroxyacetone-P moiety has been oxidized as the acyl chain is removed. Evidence indicated that the unidentified oxidized intermediate is not covalently linked to the enzyme but can be washed off the enzyme. Addition of fatty alcohol and synthesis of alkyldihydroxyacetone-P results in reoxidation of the FADH₂. Normally, acylhydrolase reactions proceed by acyl oxygen fission in which only one of the oxygens of the ester bond remains with the acyl chain; alkyl oxygen fission, where both oxygens of the ester bond remain with the released acyl chain, as catalyzed by the alkyl synthase, is very unusual. The proposed oxidized dihydroxyacetone-P intermediate is yet to be identified. These new findings and available systems may soon reveal the exact mechanism by which this exciting enzyme is able to synthesize an ether bond.

Alkyldihydroxyacetone-P synthase exhibits a very broad specificity for fatty alcohols of different carbon chain lengths. On the other hand, the specificity of the synthase for acyldihydroxyacetone-P possessing different acyl chains is less well understood, primarily because of their lack of availability.

6.2.2. *O*-Alkyl analog of phosphatidic acid and alkylacylglycerols

Once alkyldihydroxyacetone-P is synthesized, it can be readily converted to the *O*-alkyl analog of phosphatidic acid (Fig. 5) in a two-step reaction sequence. The NADPH-dependent oxidoreductase, located on the cytosolic side of peroxisomal membranes, is capable of reducing the ketone group of both the alkyl and acyl analogs of dihydroxyacetone-P. Dietary ether lipids can also enter this pathway, since alkylglycerols formed via the catabolism of dietary ether-linked lipids during absorption are known to be phosphorylated by an ATP:alkylglycerol phosphotransferase to form 1-alkyl-2-lyso-*sn*-glycerol-3-P (Fig. 5, Reaction IV), which can then be acylated by an acyl-CoA acyltransferase to produce plasmanic acid, the *O*-alkyl analog of phosphatidic acid. The latter can be dephosphorylated to alkylacylglycerols which occupy an important branch point in the ether lipid pathway in a manner analogous to the diacylglycerols. Reaction steps beginning with 1-alkyl-2-acyl-*sn*-glycerol in the routes leading to the more complex ether-linked neutral lipids and phospholipids (Fig. 5, Reactions V, VI, and VII) are thought to be catalyzed by the same enzymes as those involved in the

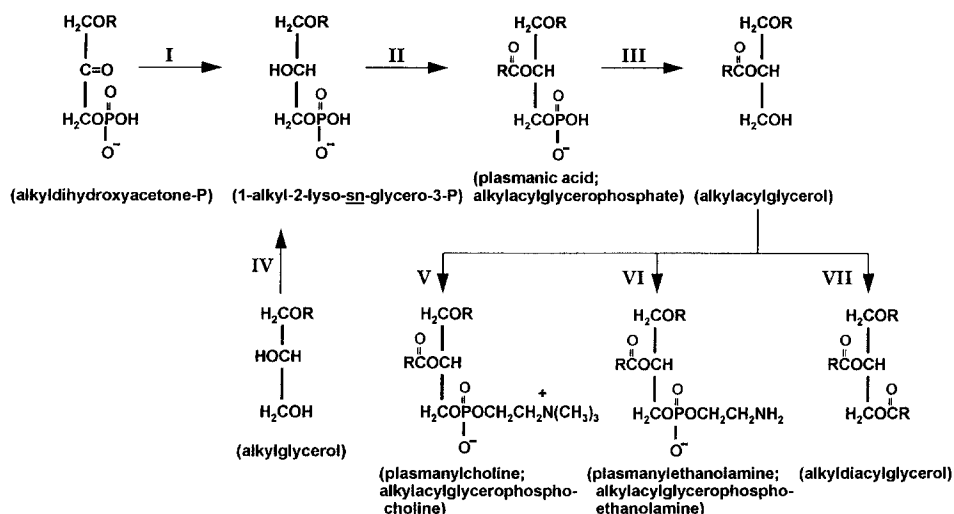


Fig. 5. Biosynthesis of membrane phospholipids from alkyl dihydroxyacetone-P, the first detectable intermediate formed in the biosynthetic pathway for ether-linked glycerolipids. Enzymes responsible for catalyzing the reactions shown are (I) NADPH:alkyl dihydroxyacetone-P oxidoreductase, (II) acyl-CoA:1-alkyl-2-lyso-*sn*-glycero-3-P acyltransferase, (III) 1-alkyl-2-acyl-*sn*-glycero-3-P phosphohydrolase, (IV) ATP:1-alkyl-*sn*-glycerol phosphotransferase, (V) CDP-choline:1-alkyl-2-acyl-*sn*-glycerol cholinephosphotransferase, (VI) CDP-ethanolamine:1-alkyl-2-acyl-*sn*-glycerol ethanolaminephosphotransferase, and (VII) acyl-CoA:1-alkyl-2-acyl-*sn*-glycerol acyltransferase.

pathways originally established by Kennedy and co-workers in the late 1950s for the diacylglycerolipids (Chapter 8).

6.2.3. Neutral ether-linked glycerolipid

Alkyl diacylglycerols, the *O*-alkyl analog of triacylglycerols, are produced by acylation of 1-alkyl-2-acyl-*sn*-glycerols in a reaction catalyzed by an acyl-CoA acyltransferase (Fig. 5, Reaction VII). The acyltransferase can also acylate 1-alk-1-enyl-2-acyl-*sn*-glycerols to form the 'neutral plasmalogen' analog of triacylglycerols. In addition, an acetylated *O*-alkyl analog of triacylglycerols has been shown to be synthesized from 1-alkyl-2-acetyl-*sn*-glycerols in HL-60 cells. The biological function of these ether-linked neutral lipids is unknown at the present time.

6.2.4. *O*-Alkyl choline- and ethanolamine-containing phospholipids

1-Alkyl-2-acyl-*sn*-glycerols, derived from the alkyl analog of phosphatidic acid by the action of a phosphohydrolase, also are utilized as substrates by cholinephosphotransferase (Fig. 5, Reaction V) and ethanolaminephosphotransferase (Fig. 5, Reaction VI) to form plasmalyncholines and plasmalyethanolamines, the alkyl analogs of phosphatidylcholine and phosphatidylethanolamine. Plasmalyncholine is the membrane source of lyso-PAF, the ether lipid precursor of the potent biologically active phospholipid, PAF, whereas plasmalyethanolamine is the direct precursor of ethanolamine plasmalogens.

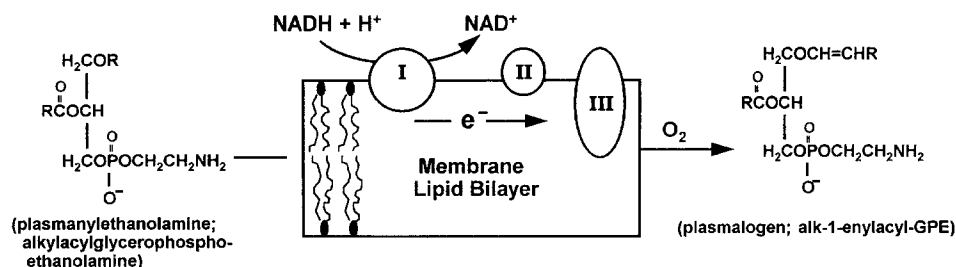


Fig. 6. Biosynthesis of ethanolamine plasmalogens by $\Delta 1$ -alkyl desaturase. Components of the enzyme complex responsible for this unusual type of desaturation between carbons 1 and 2 of the *O*-alkyl chain are (I) NADH cytochrome b_5 reductase, (II) cytochrome b_5 , and (III) $\Delta 1$ -alkyl desaturase. GPE, glycerophosphoethanolamine.

6.2.5. Ethanolamine plasmalogens

The $\Delta 1$ -alkyl desaturase, a microsomal mixed-function oxidase system, is responsible for the biosynthesis of ethanolamine plasmalogens from alkyl lipids (Fig. 6). The alkyl desaturase, which produces the alk-1-enyl grouping, is a unique enzyme, since it can specifically and stereospecifically abstract hydrogen atoms from C-1 and C-2 of the *O*-alkyl chain of an intact phospholipid molecule, 1-alkyl-2-acyl-*sn*-glycero-3-phosphoethanolamine, to form the *cis* double bond of the *O*-alk-1-enyl moiety. Only intact 1-alkyl-2-acyl-*sn*-glycero-3-phosphoethanolamine is known to serve as a substrate for the alkyl desaturase.

Like the acyl-CoA desaturases (Chapter 7), the $\Delta 1$ -alkyl desaturase exhibits the typical requirements of a microsomal mixed-function oxidase: molecular oxygen, a reduced pyridine nucleotide, cytochrome b_5 , cytochrome b_5 reductase, and a terminal desaturase protein that is sensitive to cyanide. The precise reaction mechanism responsible for the biosynthesis of the ethanolamine plasmalogens is unknown. However, it is clear from an investigation with a tritiated fatty alcohol, that only the 1*S* and 2*S* (*erythro*) labeled hydrogens are lost during the formation of the alk-1-enyl moiety of ethanolamine plasmalogens.

6.2.6. Choline plasmalogens

$\Delta 1$ -Alkyl desaturase does not utilize 1-alkyl-2-acyl-*sn*-glycero-3-phosphocholine as a substrate. In fact, biosynthesis of the significant quantities of choline plasmalogens that occurs in some heart tissues remains an enigma, although most available data strongly imply that they are derived from the ethanolamine plasmalogens. Considerable evidence has accumulated to indicate that a combination of phospholipase A_2 , lysophospholipase D, acyltransferase, phosphohydrolase, and cholinephosphotransferase activities participate in the conversion of plasmanylethanolamine to plasmenylcholine. Direct base exchange, coupled phospholipase C/cholinephosphotransferase reactions, and the methylation of the ethanolamine moiety could also contribute to the synthesis of plasmenylcholine [23,24]. Available evidence indicates that direct polar head group remodeling mechanisms (Fig. 7) or a combined enzymatic modification of the *sn*-2 and *sn*-3 positions of ethanolamine plasmalogens (Fig. 8) best explain how choline plasmalogens are formed.

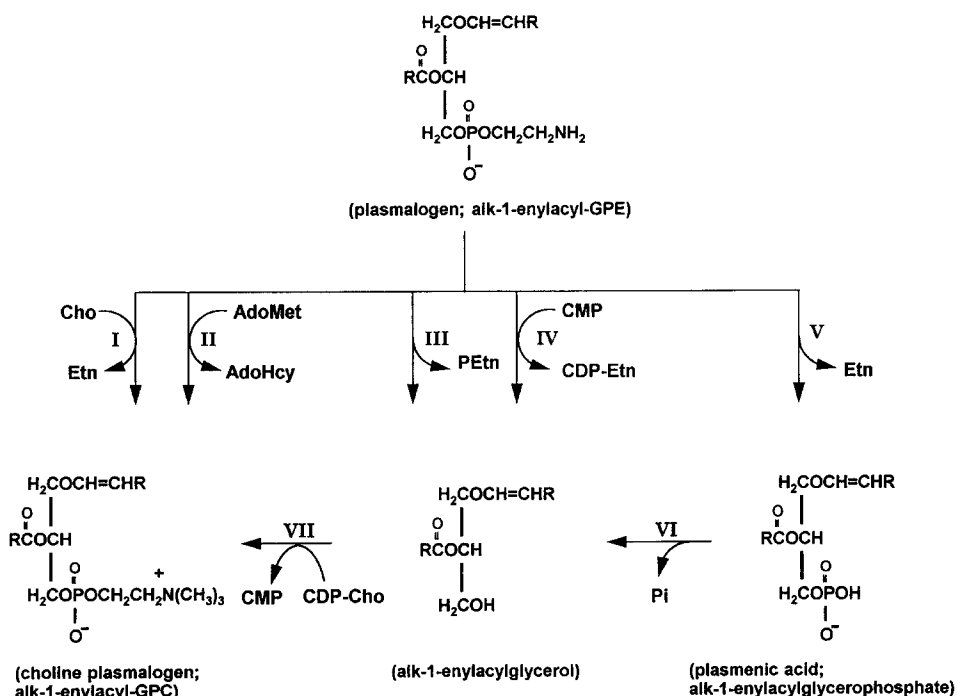


Fig. 7. Possible pathways for biosynthesis of choline plasmalogens via the modification of the *sn*-3 polar head group of ethanolamine plasmalogens are catalyzed directly by (I) a base exchange enzyme or (II) *N*-methyltransferase. A combination of other enzymatic reactions can also result in the replacement of the ethanolamine moiety of plasmenylethanolamine to produce plasmenylcholines; the enzymes responsible include (III) phospholipase C, (IV) the reverse reaction of ethanolamine phosphotransferase, (V) phospholipase D, (VI) a phosphohydrolase, and (VII) cholinephosphotransferase. Abbreviations: AdoMet, *S*-adenosyl-L-methionine; AdoHcy, *S*-adenosyl-L-homocysteine; Etn, ethanolamine; GPE, glycerophosphoethanolamine.

6.3. PAF and related bioactive species

6.3.1. Remodeling route

The remodeling pathway of PAF synthesis (Fig. 9) is thought to be the primary contributor to hypersensitivity reactions and for this reason this route has been implicated in most pathological responses involving PAF. Biosynthesis of PAF during inflammatory cellular responses or following various agonist stimulation occurs via the enzymatic remodeling of membrane-bound alkylacylglycerophosphocholines by replacing an acyl moiety with an acetate group. The enzymes responsible for catalyzing the hydrolytic deacylation step appear to be highly specific for the molecular species of alkylacylglycerophosphocholines possessing an arachidonoyl moiety at the *sn*-2 position. The initial reaction that produces lyso-PAF requires either the combined actions of a membrane-associated CoA-independent transacylase/phospholipase A₂ (Fig. 9, Reaction II; indirect route) or can be catalyzed in a single direct hydrolytic step by a phospholipase A₂ (Fig. 9, Reaction I). A CoA-dependent transacylase (reversal of an acyl-CoA acyltransferase

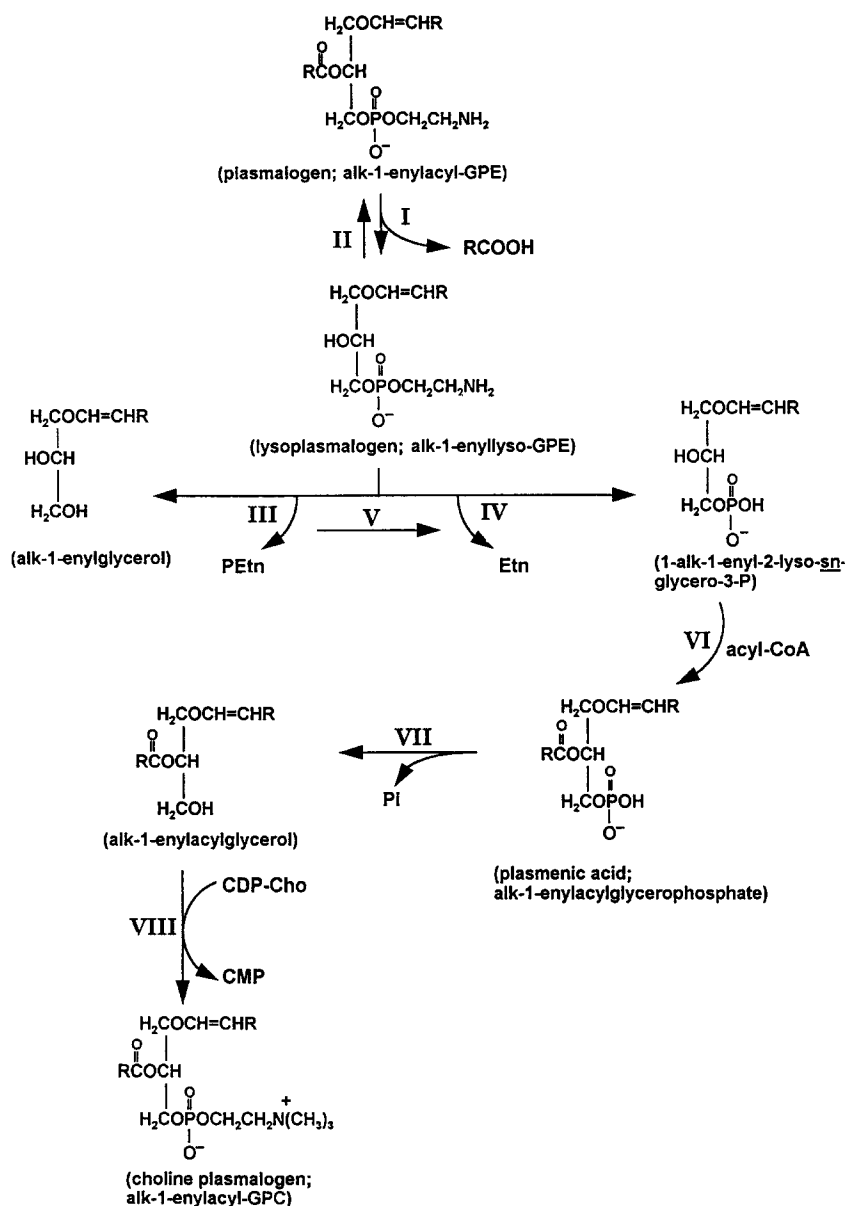


Fig. 8. Biosynthesis of plasmenylcholine via the modification of both the *sn*-2 acyl and *sn*-3 polar head group moieties of plasmenylethanolamine. (I) phospholipase A_2 , (II) CoA-independent transacylase, (III) lysophospholipase C, (IV) lysophospholipase D, (V) a phosphotransferase, (VI) acyl-CoA acyltransferase, (VII) phosphohydrolase, and/or (VIII) cholinephosphotransferase. Abbreviations: Etn, ethanolamine; Cho, choline; GPE, *sn*-glycero-3-phosphoethanolamine; GPC, *sn*-glycero-3-phosphocholine.

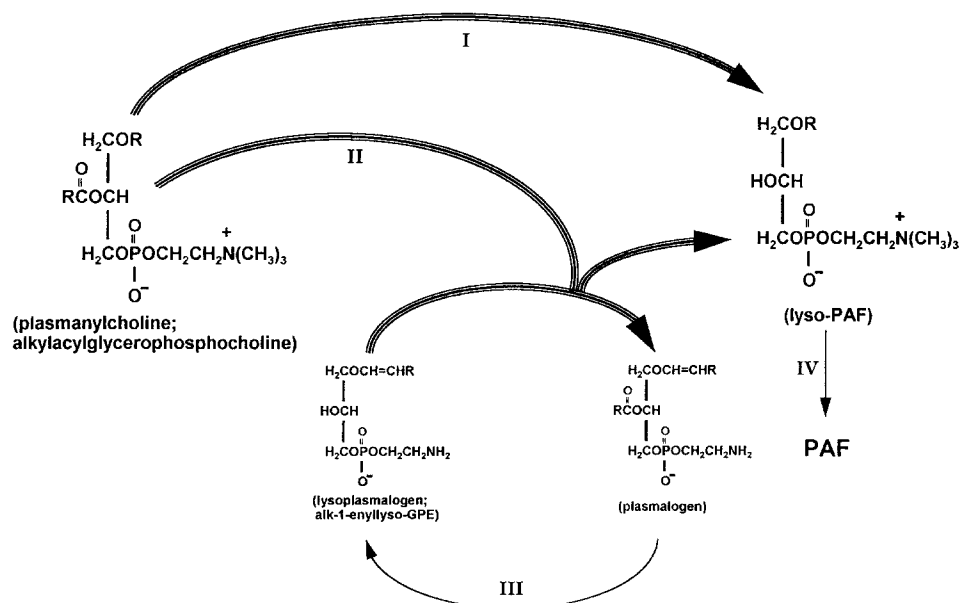


Fig. 9. Biosynthesis of PAF via the remodeling pathway. Lyso-PAF, the immediate precursor of PAF, can be formed from 1-alkyl-2-acyl-*sn*-3-glycerophosphocholine through the direct action of (I) a phospholipase A₂ or (II) a CoA-independent transacylase. The lysoplasmenylethanolamine (or other potential ethanolamine- or choline-containing lysoglycerophospholipids) is thought to be generated by (III) a phospholipase A₂ that exhibits a high degree of selectivity for substrates having an arachidonoyl moiety at the *sn*-2 position. The transacylase (II) appears to possess both acyl transfer and phospholipase A₂ hydrolytic activities, which could exist as a single protein or as a tightly associated complex of two distinctly different proteins. The lyso-PAF produced by either the transacylation (II) or direct phospholipase A₂ (I) reactions can then be acetylated to form PAF by (IV) an acetyl-CoA acetyltransferase.

reaction) is also capable of generating lyso-PAF (Fig. 10, Reaction I). Two reviews have focused on the different types of transacylase reactions involved in the remodeling of phospholipids [25,26].

A number of studies indicate that the 85 kDa cytosolic phospholipase A₂ is likely the phospholipase responsible for release of arachidonic acid and PAF synthesis in stimulated cells. It is highly selective for arachidonate as is the CoA-independent transacylase. One of the most convincing findings showing that this enzyme is responsible for initiating the remodeling pathway is that macrophages from cytosolic phospholipase A₂ knockout mice almost completely lose their ability to synthesize both PAF and eicosanoids [13,14]. Since the cytosolic phospholipase A₂ does not distinguish between the ester and ether linkage in the *sn*-1 position, acetylated products recovered from cells reflect the composition of the choline-containing phosphoglycerides. The activity is regulated by phosphorylation of the enzyme and by translocation from the cytosol to membranes, which requires concentrations of μM Ca²⁺; Ca²⁺ is not required for the hydrolytic mechanism of cytosolic phospholipase A₂.

The acetyltransferase responsible for the final step in the synthesis of PAF (Fig. 9,

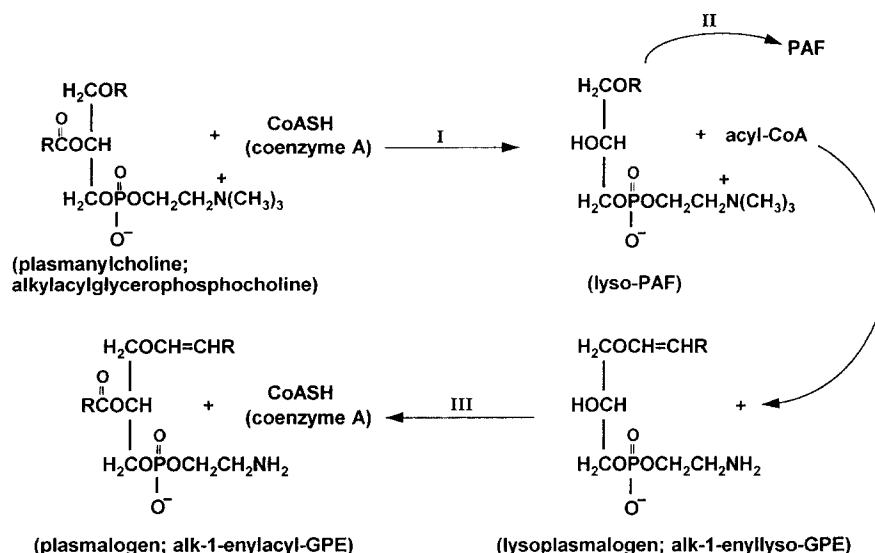


Fig. 10. Involvement of a CoA-dependent transacylase in the production of lyso-PAF for the synthesis of PAF and the remodeling of the *sn*-2 acyl group of membrane phospholipids. The enzymes responsible for catalyzing these reactions are (I) the CoA-dependent transacylase (with CoA as the acyl acceptor), (II) acetyl CoA:lyso-PAF acetyltransferase, and (III) an acyl-CoA:lysophospholipid acyltransferase. The reaction depicted for the CoA-dependent transacylase represents the reversal of the reaction catalyzed by acyl-CoA:lysophospholipid acyltransferase. GPE designates *sn*-3-glycerophosphoethanolamine.

Reaction IV) is membrane-bound and, like the CoA-independent transacylase, has neither been purified nor its cDNA cloned. This membrane-bound enzyme can also acetylate both the alk-1-enyl and acyl analogs of lyso-PAF and utilizes short-chain acyl-CoAs ranging from C_2 to C_6 as substrates. Several studies indicate that the enzyme is activated by phosphorylation even though the unphosphorylated enzyme appears to have a basal activity. In studies of human neutrophils, Nixon et al. [27] have concluded from studies with MAP kinase inhibitors and recombinant cytosolic phospholipase A_2 and MAP kinases that the acetyl-CoA:lyso-PAF acetyltransferase is specifically activated by the p38 stress-activated MAP kinase but not by p42 and p44 ERKs. In contrast, cytosolic phospholipase A_2 is activated in the cells by both the ERKs and p38 kinase. Related findings suggest that the production of phosphatidic acid by phospholipase D specifically activates the p38 kinase cascade but not the ERKs. Identification of the specific protein kinases responsible for the direct activation of cytosolic phospholipase A_2 and the acetyltransferase in intact cells is complicated by cross-talk between the kinases, including protein kinase C.

Regulation of the CoA-independent transacylase activity (Fig. 9, Reaction II) is poorly understood. It has been demonstrated that production of lyso-PAF via the transacylation step can occur in either a CoA-independent (Fig. 9) or CoA-dependent (Fig. 10) manner [20,21]. With the CoA-independent transacylase, ethanolamine lysoplasmalogens as well as other ethanolamine- or choline-containing lysoglycerophosphatides serve as the acyl acceptor for the selective transfer of an arachidonoyl moiety

from alkylacylglycerophosphocholine. CoA itself, instead of a lysophospholipid, is the acyl acceptor in the reaction catalyzed by the CoA-dependent transacylase (Fig. 10). This type of transacylation is thought to represent the reverse reaction of that catalyzed by an acyl-CoA:lyso-PAF acyltransferase but is not selective for arachidonate. In addition to participating in the formation of lyso-PAF, the transacylases also serve an important role in the remodeling of acyl moieties located at the *sn*-2 position of the choline- and ethanolamine-containing phospholipids.

The lysoplasmalogen or other lysophospholipid acceptors that are substrates for the transacylases appear to be formed by the direct action of a phospholipase A₂ on the appropriate membrane-associated phospholipid which simultaneously releases arachidonic acid for its subsequent metabolism to bioactive eicosanoid products (Chapter 13). In both the direct and indirect routes of lyso-PAF production, the action of a phospholipase A₂ is required; it is plausible that both routes participate in PAF synthesis to varying degrees depending on conditions. Since both eicosanoid and PAF mediators can be formed via the remodeling pathway and these mediators can act synergistically, the assessment of biological responses following cell activation can often be difficult to interpret.

6.3.2. *De novo* route

PAF biosynthesis via the *de novo* pathway [10,11] is thought to be the primary source of the physiological levels of PAF in cells and blood (Fig. 11). Both fatty acids and neurotransmitters can stimulate the *de novo* synthesis of PAF. The sequence of enzymatic reactions (Fig. 11) involved in the *de novo* route include (a) acetylation of 1-alkyl-2-lyso-*sn*-glycero-3-P by an acetyl-CoA-dependent acetyltransferase (Reaction I), (b) dephosphorylation of 1-alkyl-2-acetyl-*sn*-glycero-3-P (Reaction II), and (c) the transfer of phosphocholine from CDP-choline to 1-alkyl-2-acetyl-*sn*-glycerol by a dithiothreitol-insensitive cholinephosphotransferase (Reaction III) to form PAF. The acetyltransferases associated with the remodeling (Fig. 9) and *de novo* routes (Fig. 11) possess distinctly different properties and substrate specificities. Also, the dithiothreitol-insensitivity of this cholinephosphotransferase contrasts with the inhibitory effect of dithiothreitol on the cholinephosphotransferase that synthesizes phosphatidylcholine and plasmalogen from diacylglycerols and alkylacylglycerols, respectively. In addition, the two dissimilar cholinephosphotransferase activities that synthesize PAF and phosphatidylcholine exhibit different pH optima and respond differently to detergents, ethanol, temperature, and substrates. Although the enzymes in the *de novo* pathway exhibit a relatively high degree of substrate specificity, the *sn*-1 acyl analogs of the corresponding *O*-alkyl equivalents can also be utilized as substrates by the acetyltransferase, phosphohydrolase, and the dithiothreitol-insensitive cholinephosphotransferase.

6.3.3. PAF transacetylase

Two novel CoA-independent transacetylases that use PAF as the donor molecule (Fig. 12) are PAF:lysophospholipid transacetylase and PAF:sphingosine transacetylase [28]. Both transacetylases have no requirement for Ca²⁺, Mg²⁺, or CoA. The PAF:lysophospholipid transacetylase transfers the acetyl group from PAF to a variety

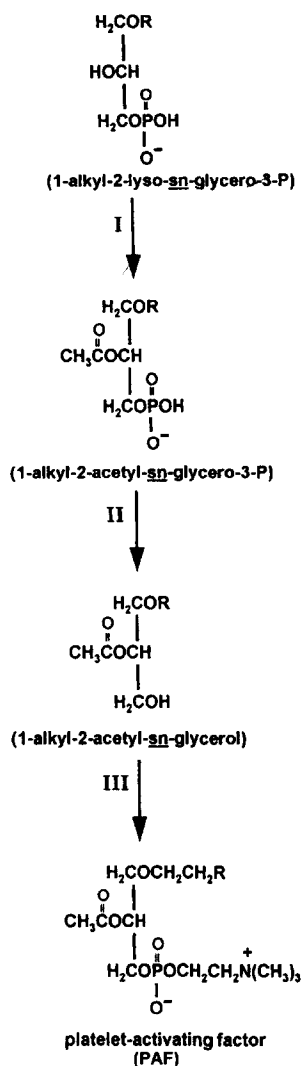


Fig. 11. Biosynthesis of PAF via the de novo pathway. The three-step reaction sequence in this route, beginning with 1-alkyl-2-lyso-*sn*-glycero-3-P as the precursor, is catalyzed by (I) acetyl-CoA:alkyllysoglycero-P acetyltransferase, (II) alkylacetylglycero-P phosphohydrolase, and (III) CDP-choline:alkylacetylglycerol cholinephosphotransferase.

of lysophospholipids to form a series of PAF analogs. Among all the lysophospholipids tested, acyllysoglycerophosphocholine is the most active acetyl group acceptor. In addition, *cis*-9-octadecen-1-ol can also serve as acetate acceptor, whereas alkylglycerol, acylglycerol, or cholesterol are inactive. Biochemical studies suggest that the CoA-independent transacetylase differs from the CoA-independent transacylase that transfers long-chain acyl moieties.

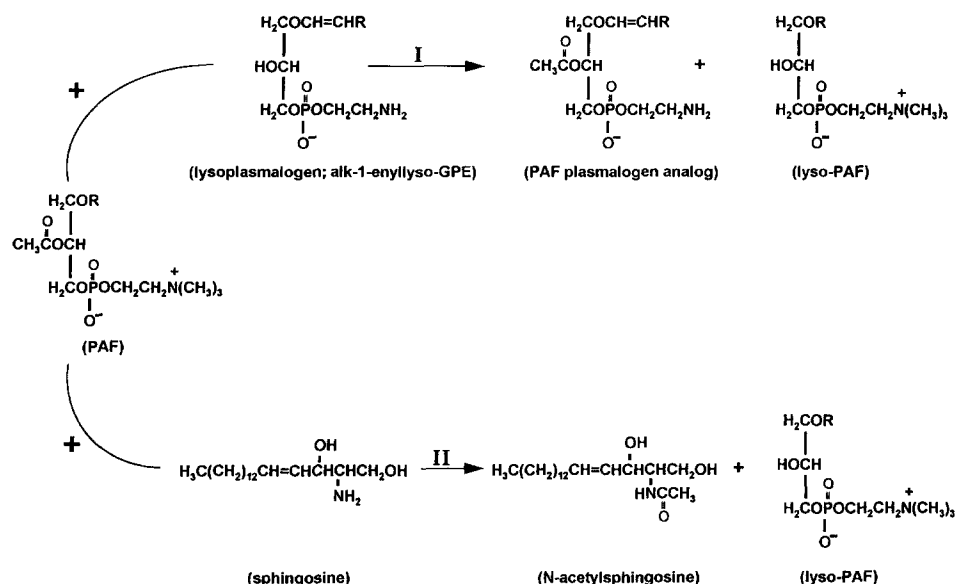


Fig. 12. PAF transacetylase transfers the acetate moiety of PAF to other selective substrates to produce a plasmalogen analog (I) and acetylsphingosine (II). GPE, glycerophosphoethanolamine.

The PAF: sphingosine transacetylase transfers the acetate group from PAF to sphingosine forming *N*-acetylsphingosine (C_2 -ceramide). The enzyme has a narrow substrate specificity and strict stereochemical configuration requirement. Ceramide, sphingosylphosphocholine, stearylamine, sphingosine-1-phosphate, or sphingomyelin are not substrates, whereas sphinganine has a limited capacity to accept the acetate from PAF. Only the naturally synthesized *D*-erythro isomer but not the synthetic *L*-erythro-, *D*-threo-, or *L*-threo-isomer of sphingosine can serve as a substrate. Both PAF: lysophospholipid transacetylase and PAF: sphingosine transacetylase have similar tissue distributions. The PAF: sphingosine transacetylase is located in mitochondria, microsomes, and cytosol with mitochondria having the highest specific activity. Physiological levels of C_2 -ceramide (in μM range) have been detected in both undifferentiated and differentiated HL-60 cells.

Rat kidney transacetylases from mitochondria/microsomes and cytosols have an apparent molecular mass of 40 kDa. Both purified enzymes from membranes and cytosols contain three catalytic activities; PAF: lysophospholipid transacetylase, PAF: sphingosine transacetylase, and PAF acetylhydrolase (PAF-AH). A search using a protein sequence data bank indicates that these sequences have homology with the sequences present in bovine PAF-AH II (Section 7.3.1).

The substrate specificity, kinetic parameters, and inhibitor effects suggest that the three individual catalytic activities of the transacetylase have different dependencies on the thiol-containing residue(s) of the enzyme, i.e., cysteine. Furthermore, the non-responsiveness of the purified cytosolic transacetylase to phosphatidylserine activation indicates that membrane and cytosolic transacetylase may be posttranslationally distinct.

Analysis of a series of site-directed mutant PAF-AH II proteins in CHO-K1 cells shows that lysophospholipid transacetylase is decreased, whereas PAF-AH activity is not affected in C120S and G2A mutants. Thus, Cys¹²⁰ and Gly² are implicated in the catalysis of the lysophospholipid transacetylase reaction in this enzyme. It appears that *N*-myristoylation is not required for PAF-AH activity.

Several lines of evidence indicate that transacetylase activity has a physiological role *in vivo*. With intact differentiated HL-60 cells, [³H]acetate from [³H]PAF can be incorporated into alk-1-enyl[³H]acetylgllycerophosphoethanolamine in the presence of ionophore A23187, but not in its absence. In endothelial cells stimulated by ATP, bradykinin, and ionophore A23187, acylacetylgllycerophosphocholine is the predominant product and the radiolabelled acetate group of PAF is incorporated into acylacetylgllycerophosphocholine in a time-dependent fashion.

In ATP-activated cells, PAF: acyllysoglycerophosphocholine transacetylase and formation of acylacetylgllycerophosphocholine are concurrently and transiently induced, while PAF: sphingosine transacetylase and PAF-AH activities remain unchanged. Evidence indicates that tyrosine protein kinase and protein kinase C are directly or indirectly involved in the activation of the transacetylase activity through protein phosphorylation. In addition, ATP induces the translocation of acyllysoglycerophosphocholine transacetylase from cytosol to membranes and also increases the specific enzyme activity on the membrane. Collectively, the three catalytic activities of the transacetylase are regulated in agonist-activated cells through posttranslational modifications (such as reversible phosphorylation/dephosphorylation, myristoylation, etc.) and translocation of the enzyme from cytosol to membranes.

7. Catabolic enzymes

7.1. Ether lipid precursors

7.1.1. Fatty alcohols

Fatty alcohols are oxidized to fatty acids via a fatty alcohol:NAD⁺ oxidoreductase, a microsomal enzyme found in most mammalian cells (Fig. 3B). The high activity of this enzyme probably accounts for the extremely low levels of unesterified fatty alcohols generally found in tissues or blood. Detection of fatty aldehydes, by trapping them as semicarbazide derivatives during oxidation of the alcohol, suggests that the fatty alcohol oxidoreductase catalyzes a two-step reaction that involves an aldehyde intermediate.

7.1.2. Dihydroxyacetone-P and acyldihydroxyacetone-P

Dihydroxyacetone-P can be diverted from its precursor role in ether lipid synthesis when it is converted to *sn*-glycerol-3-P by glycerol-3-P dehydrogenase. Another bypass that prevents the formation of alkyldihydroxyacetone-P occurs if the ketone function of acyldihydroxyacetone-P is first reduced by an NADPH-dependent oxidoreductase, since the product, 1-acyl-2-lyso-*sn*-glycerol-3-P, can then be converted to different diacyl types of glycerolipids. Obviously, the metabolic removal and/or formation of fatty

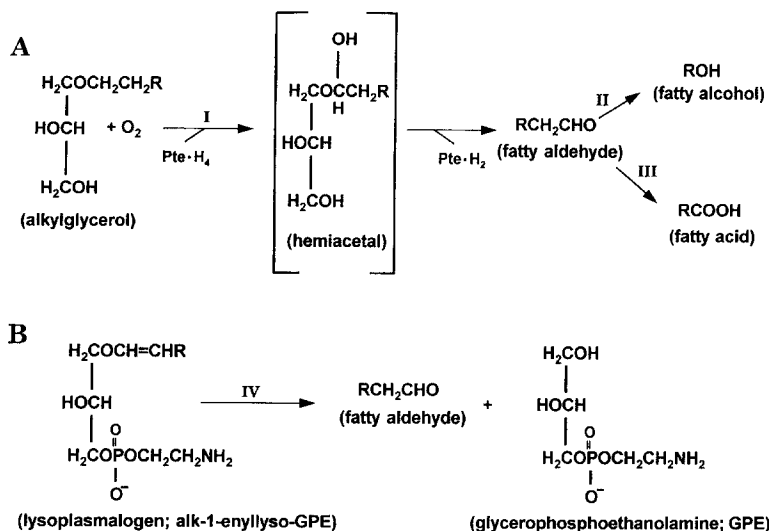


Fig. 13. Cleavage of the *O*-alkyl linkage in glycerolipids (A) is catalyzed by (I) tetrahydropteridine (Pte·H₄)-dependent alkyl monooxygenase. The fatty aldehyde product can be either reduced to a long-chain fatty alcohol by (II) a reductase or oxidized to a fatty acid by (III) an oxidoreductase. Removal of the *O*-alk-1-enyl moiety from plasmalogens (B) is catalyzed by a plasmalogenase. As with the *O*-alkyl monooxygenase, the fatty aldehyde can be converted either to the corresponding fatty alcohol or fatty acid.

alcohols, dihydroxyacetone-P, or acyldihydroxyacetone-P from the ether lipid precursor pool represent important control points for regulating the ether lipid pathway.

7.2. Ether-linked lipids

7.2.1. *O*-Alkyl cleavage enzyme

Oxidative cleavage of the *O*-alkyl linkage in glycerolipids is catalyzed by a microsomal tetrahydropteridine (Pte·H₄)-dependent alkyl monooxygenase (Fig. 13A). The required cofactor, Pte·H₄, is regenerated from the Pte·H₂ by an NADPH-linked pteridine reductase, a cytosolic enzyme. Oxidative attack on the ether-linked grouping in lipids is similar to the enzymatic mechanism described for the hydroxylation of phenylalanine. Fatty aldehydes produced via the cleavage reaction can be either oxidized to the corresponding acid or reduced to the alcohol by appropriate enzymes.

Alkyl cleavage enzyme activities are highest in liver and intestinal tissue, whereas most other cells/tissues possess very low activities. Tumors and other tissues that contain significant quantities of alkyl lipids generally have very low alkyl cleavage enzyme activities, which is consistent with the overall premise that the level of ether-linked glycerolipids is inversely proportional to the activity of the alkyl cleavage enzyme.

Structural features of glycerolipid substrates utilized by the alkyl cleavage enzyme are (a) an *O*-alkyl moiety at the *sn*-1 position, (b) a free hydroxyl group at the *sn*-2 position, and (c) a free hydroxyl or phosphobase group at the *sn*-3 position. If the

hydroxyl group at the *sn*-2 position is replaced by a ketone or acyl grouping, or when a free phosphate is at the *sn*-3 position, the *O*-alkyl moiety at the *sn*-1 position is not cleaved by the Pte-H₄-dependent monooxygenase. Thus, 1-alkyl-2-lysophospholipids (e.g., lyso-PAF) are substrates for the cleavage enzyme, but they are attacked at much slower rates than are alkylglycerols.

7.2.2. Plasmalogenases

Plasmalogenases (Fig. 13B) are capable of hydrolyzing the *O*-alk-1-enyl grouping of plasmalogens or lysoplasmalogens. The products of this reaction are a fatty aldehyde and either 1-lyso-2-acyl-*sn*-glycero-3-phosphoethanolamine (or choline) or *sn*-glycero-3-phosphoethanolamine (or choline), depending on the chemical structure of the parent substrate. Plasmalogenase activities have been described in microsomal preparations from liver and brain of rats, cattle, and dogs, but their biological significance is poorly understood.

7.2.3. Phospholipases and lipases

In general, the *sn*-2 and *sn*-3 ester groupings associated with either the alkyl or alk-1-enyl glycerolipids are hydrolyzed by lipolytic enzymes with the same degree of substrate specificity as their acyl counterparts (Fig. 14). However, the ether linkages themselves are not hydrolyzed by lipases or phospholipases and the presence of an ether linkage at the *sn*-1 position of the glycerol moiety generally reduces the overall reaction rate to the extent that certain lipases have been successfully used to remove diacyl contaminants in the purification of some ether-linked phospholipids. The only lipolytic enzyme (other than those that cleave the ether linkages) known to exhibit an absolute specificity for ether-linked lipids is lysophospholipase D. The uniqueness of lysophospholipase D is that it exclusively recognizes only 1-alkyl-2-lyso-*sn*-glycero-3-phosphobases or 1-alk-1-enyl-2-lyso-*sn*-glycero-3-phosphobases as substrates; thus, lyso-PAF is a substrate for this novel enzyme (Fig. 14, Reaction II).

7.3. PAF and related bioactive species

7.3.1. Acetylhydrolase

PAF acetylhydrolase (AH) enzymes (Fig. 14, Reaction I) are a specific group of Ca²⁺-independent phospholipases A₂ that remove the acetyl moiety at the *sn*-2 position of PAF [29–31]. Mammalian PAF-AHs can be classified into intracellular and extracellular types. Intracellular PAF-AHs consist of at least three groups of enzymes, namely, PAF-AH I, PAF-AH II, and erythrocyte-type PAF-AH. Extracellular PAF-AH occurs as plasma AH.

The erythrocyte enzyme appears to be a homodimer comprised of the 25-kDa polypeptide and is different from PAF-AH II. PAF-AH is a serine esterase and requires reducing agents for maximal activity. The most likely role of the erythrocyte PAF-AH *in vivo* is to hydrolyze the products of oxidative fragmentation of membrane phospholipids.

PAF-AH I, rich in brain and exclusively located in cytosols, is an unusual G-protein-like (α_1/α_2) β heterotrimer complex composed of 45(β)-, 30(α_2)-, and 29(α_1)-kDa

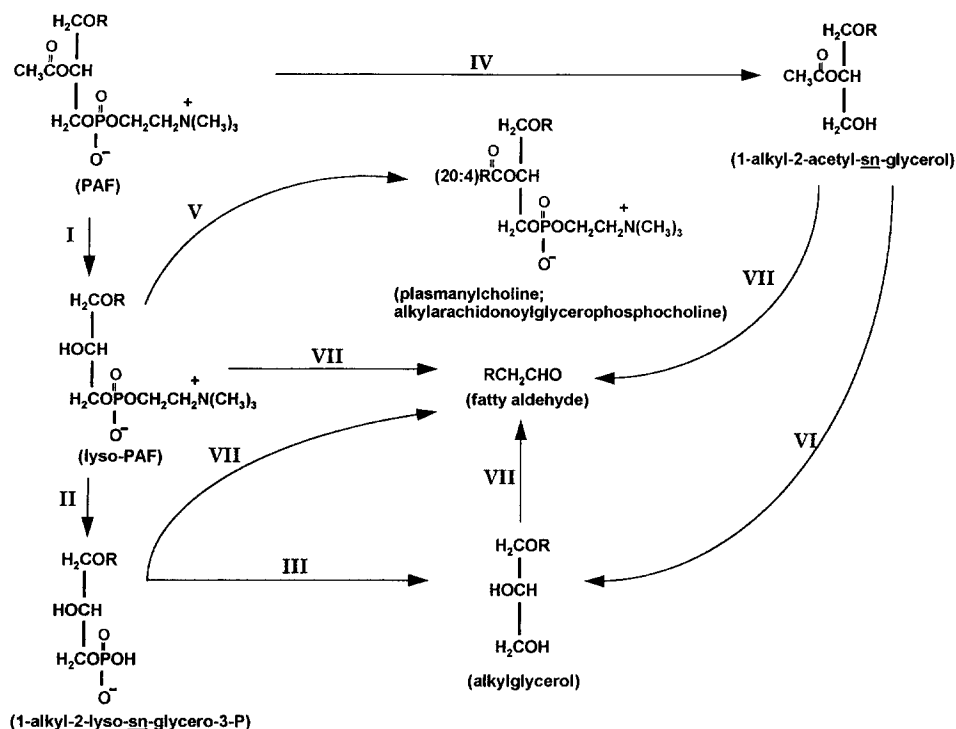


Fig. 14. Catabolism of PAF and its metabolites can be catalyzed by the following enzymes: (I) PAF acetylhydrolase, (II) lysophospholipase D, (III) phosphohydrolase, (IV) phospholipase C, (V) CoA-independent or CoA-dependent transacylase, and/or (VI) alkylacetylgllycerol acetylhydrolase. The *O*-alkyl linkage in those products that contain free hydroxyl groups can be cleaved by (VII) the *O*-alkyl Pte·H₄-dependent monooxygenase.

subunits. The tertiary fold of α_1 subunit is similar to that found in p21^{ras} and other GTPases. The active site is made up of a trypsin-like triad of Ser 47, His 195, and Asp 193. A sequence of ~30 amino acids adjacent to the active serine residue exhibits significant similarity to the first transmembrane region of the PAF receptor. The catalytic 30(α_2)-kDa subunit is highly homologous (63.2% identity) to that of the 29(α_1)-kDa subunit, especially (86%) in the catalytic and PAF receptor homologous domains.

The 45(β) kDa subunit, which is not essential for catalytic activity, exhibits striking homology (99%) with a protein encoded by the causal gene (*LIS-1*) for the onset of Miller–Dieker lissencephaly, a human brain malformation manifested by a smooth cerebral surface and impaired neural migration. In addition, the 45-kDa subunit contains a 7-tandem WD-40 repeat in its primary structure. This repeat is thought to be important for interactions with other protein components, especially with pleckstrin-homology (PH) domains. Therefore, the hydrolysis of PAF may induce conformational changes in the heterotrimeric PAF-AH I complex that affect the ability of the 45-kDa subunit to interact with cytoskeletal proteins.

Furthermore, the α_1 subunit appears to be expressed specifically in neurons of fetal and neonatal brain of rats. Significant levels of the α_1 subunit are not expressed in any adult rat tissues. In contrast, α_2 and β transcripts and proteins are almost constantly expressed from fetal stages through adulthood. The catalytic subunits switch from the α_1/α_2 heterodimer to the α_2/α_2 homodimer and along with the above-mentioned data suggest that PAF-AH I in brain is involved in brain development through regulation of neuronal migration.

PAF-AH type II (PAF-AH II), expressed most abundantly in liver and kidney, is a monomeric 40-kDa protein and a member of the serine esterase family. This enzyme exhibits broader substrate specificity than PAF-AH I. PAF-AH II hydrolyzes oxidized phospholipids as effectively as PAF, whereas PAF-AH I is more specific for PAF. Furthermore, unlike PAF-AH I, PAF-AH II is distributed in both the membrane and soluble fractions.

PAF-AH II is a *N*-myristoylated enzyme, the first reported among lipases and phospholipases. It translocates between cytosol and membranes in response to the redox state. When overexpressed in cells, it protects against oxidative stress-induced cell death. PAF-AH II may function as an antioxidant phospholipase and promote the hydrolysis of oxidized phospholipids produced during reactive oxygen species-induced apoptosis.

Plasma PAF-AH is unique since it is mainly associated with the high density and low density lipoproteins. Plasma PAF-AH also hydrolyzes PAF and structurally related oxidized phospholipids with up to 9 carbon *sn*-2 acyl chains. The cDNA for this enzyme encodes a 44-kDa secretory protein that contains a typical signal sequence and a serine esterase/neutral lipase consensus motif GX SXG. Serine 273 (of the GX SXG motif), Asp 296, and His 351 are required for catalysis. Plasma PAF-AH displays ~40% homology with intracellular PAF-AH II, but little homology/similarity with PAF-AH I over the whole amino acid sequence.

Pretreatment of animals with recombinant plasma PAF-AH blocks PAF-induced inflammation. Furthermore, deficiency of plasma PAF-AH is an autosomal recessive syndrome that is associated with severe asthma in Japanese children. A point mutation of exon 9 of the plasma PAF-AH gene results in the production of an inactive protein. In addition, an increase in enzyme activity has been reported in humans or rats with hypertension, and in the plasma of patients with atherosclerosis. The level of plasma PAF-AH decreased markedly near the end of pregnancy in rabbits. It has been proposed that this is a component of the mechanism for initiating the onset of labor. Since PAF stimulates the contraction of uterine muscle, a decrease in PAF-AH activity will allow the accumulation of PAF.

8. Metabolic regulation

Regulatory mechanisms that control the metabolism of ether-linked lipids are still poorly understood. In fact, most progress in this area has concerned PAF metabolism, primarily because of the high degree of interest in this potent mediator. Nevertheless, a variety of factors are known to influence the overall rates of ether lipid metabolism, but such studies have mainly been of the descriptive type and none have addressed the molecular

enzymatic mechanisms involved. Regulatory controls that must be considered in the metabolism of ether-linked lipids are those that influence (a) the enzymes responsible for catalyzing the biosynthesis and catabolism of the ether lipid precursors (fatty alcohols and dihydroxyacetone-P), (b) alkyl-dihydroxyacetone-P synthase which is responsible for the synthesis of alkyl-dihydroxyacetone-P, and (c) branch point enzymes, e.g., those steps that utilize diradylglycerols.

Glycolysis plays an important role in controlling the levels of ether lipids at the precursor level. For example, the high glycolytic rate of tumors generates significant quantities of dihydroxyacetone-P, which could explain the relatively high levels of ether lipids found in cancerous cells.

Factors responsible for the regulation of biosynthetic and catabolic enzyme activities that catalyze specific reaction steps in the metabolic pathways for ether-linked lipids appear to be very complex. Although the rate-limiting steps are poorly understood, two important intermediary branch points in the biosynthesis of ether-linked lipids involve 1-alkyl-2-lyso-*sn*-glycero-3-P and 1-alkyl-2-acyl-*sn*-glycerols. The 1-alkyl-2-lyso-*sn*-glycero-3-P can be ultimately converted to either PAF via de novo route or to 1-alkyl-2-acyl-*sn*-glycerols following an acylation and dephosphorylation step. The alkylacylglycerols represent a branch point since they are the direct precursors of plasmalyncholines, plasmalylethanolamines, and alkyl-diacylglycerols. Conditions that influence either branch point would have profound effects on the proportion of the different types of ether-linked lipids formed. Since the choline- and ethanolamine-phosphotransferases appear to be able to utilize both diacyl- and alkylacyl-glycerols, it is apparent that the availability of specific diradylglycerols is crucial in controlling the diacyl and alkylacyl species composition of membranes. Most of the catabolic enzymes of ether lipid metabolism have received far less attention than those associated with the biosynthetic pathways.

Studies of the regulation of PAF metabolism are still in the early stages of development. Rate-limiting steps in the de novo pathway of PAF biosynthesis are the acetyl-CoA:1-alkyl-2-lyso-*sn*-glycero-3-P acetyltransferase and the cytidylyltransferase that forms CDP-choline for the cholinephosphotransferase catalyzed step (fig. 10, Chapter 8). Any factor that stimulates these rate-limiting reactions (e.g., activation of cytidylyltransferase by fatty acids) is also known to enhance the de novo synthesis of PAF.

In the remodeling pathway, it is clear that arachidonic acid can influence the formation of PAF at the substrate level since cells depleted of alkylarachidonoylglycerophosphocholines lose their ability to synthesize PAF. Therefore, the transacylase/phospholipase A₂ step (Fig. 9, Reaction II) as well as a specific phospholipase A₂ (Fig. 9, Reactions I or III) can be rate-limiting. Regulation of the acetyl-CoA:lyso-PAF acetyltransferase in the remodeling pathway (Fig. 9, Reaction IV) appears to be controlled by a phosphorylation/phosphohydrolase system.

Acetylhydrolase and other catabolic enzymes in PAF metabolism also have an important regulatory role in controlling PAF levels since it is known that the activity of acetylhydrolase can drastically change during various diseases, pregnancy, and macrophage differentiation.

9. Functions

9.1. Membrane components

Cellular functions of ether-linked glycerolipids are especially poorly understood, but their ability to serve as both membrane components and as cellular mediators is now well established. Both the alkyl and alk-1-enyl phospholipids that contain long-chain acyl groups at the *sn*-2 position appear to be essential structural components of many membrane systems. Some species of the ether lipids associated with membranes act as storage reservoirs for polyunsaturated fatty acids. The apparent protective nature of ether-linked groups against lipolytic actions is reflected by their ability to slow the rate of hydrolysis of acyl moieties at the *sn*-2 position by phospholipase A₂. The preferential sequestering of polyunsaturated fatty acids in ether-linked phospholipids has been observed even in essential fatty acid deficiency. The plasmalogens of inflammatory cells are highly enriched in arachidonate; in human neutrophils 80% of the cellular arachidonate is found in the plasmalogen ethanolamine fraction. There is one report with photosensitized cells exposed to long wavelength ultraviolet radiation that has suggested that plasmalogens might have a role in protecting membranes against certain forms of oxidative stress. However, the true function(s) of plasmalogens remains a puzzle.

9.2. Cell mediators

The multifaceted responses generated by PAF *in vivo* and in target cells and the ubiquitous distribution of PAF-related enzymes in mammalian cells has emphasized the important role of bioactive ether-linked lipids as diverse regulators of metabolic and cellular processes. Also, identification of the PAF receptor as a member of the general family of G-protein-coupled receptors further strengthens the potential importance of PAF as a mediator in cell signalling pathways. Activation of phosphatidylinositol-specific phospholipase C by the binding of PAF to its receptor elevates the intracellular levels of Ca²⁺ and diacylglycerols, events that activate protein kinase C (Chapter 12). The latter catalyzes the phosphorylation of specific proteins and it is clearly documented that 20- and 40-kDa proteins are phosphorylated in PAF-treated rabbit platelets. However, an array of kinase cascades and cellular alterations need to be considered in any proposed biochemical mechanism for explaining the diverse actions of PAF.

10. Future directions

There are many unanswered questions about the dual role that ether lipids serve as membrane components and as cellular signaling molecules. Although it is clear that arachidonic acid is closely associated and tenaciously retained by ether lipids in membranes, even in essential fatty acid deficiency, much remains to be elucidated about the enzymatic systems and regulatory controls that affect the release of this sequestered pool of arachidonic acid for its subsequent conversion to bioactive eicosanoid metabolites.

The mechanisms that account for the synergistic actions of PAF and eicosanoids are still poorly understood. In addition, the significance of ether lipids as a dietary nutrient has received little attention even though they occur in a variety of foods and it is known that ether lipid supplements are readily incorporated into cellular lipids.

It is noteworthy that organisms living in harsh environments of high temperatures or high salt and low pH contain only ether lipids suggesting that they serve as the Teflon of lipids. Invasive cells such as neutrophils and eosinophils also have high levels of ether lipids. Based on these observations it is tempting to speculate that the high levels of ether lipids found in most tumor cells may contribute to their invasiveness.

Now that the alkyl synthase has been cloned and sequenced and shown to contain a flavin, further elucidation of the mechanism of ether bond synthesis is anticipated. Despite the advances made in cloning and sequencing of the PAF receptor, the exact mechanisms for explaining how PAF participates in signal transduction, the generation of second messengers, and gene expression remain poorly understood. Moreover, even the binding site of PAF to its receptor has not yet been rigorously identified and the presence of intracellular PAF binding sites still are unresolved issues. Certainly, the significance of PAF in physiological and disease processes needs to be more firmly established.

A major enigma is the function of plasmalogens. Despite the large quantities of ethanolamine plasmalogens found in nervous tissue and other cells, their cellular role or the molecular mechanism and regulatory controls for the alkyl desaturase responsible for their formation are still unknown. The alkyl desaturase has yet to be purified, cloned and carefully compared to fatty acid desaturases. Likewise, the biosynthesis of choline plasmalogens is still not fully understood, although compelling evidence exists to indicate that they originate from ethanolamine plasmalogens via remodeling mechanisms.

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Adipose tissue and lipid metabolism

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1. Introduction

The development of adipose tissue and the biochemistry of the adipocyte are research areas that have intrigued investigators for decades. Originally considered as simply a storage organ for triacylglycerol, interest in the biology of adipose tissue has increased substantially within the last decade, coming to the forefront in areas such as molecular genetics, endocrinology and neurobiology. Recent advances have demonstrated that the adipocyte is not a passive lipid storage depot but a dynamic cell that plays a fundamental role in energy balance and overall body homeostasis. Moreover, the fat cell functions as a sensor of lipid levels, transmitting information to a neural circuit affecting hunger, satiety and sleep.

This chapter will focus on the biochemistry of the adipocyte. Adipocytes make up approximately one-half of the cells in adipose tissue, the remainder being blood and endothelial cells, adipose precursor cells of varying degrees of differentiation, and fibroblasts. The reader is referred to excellent reviews by G. Ailhaud (1992) and P. Cornelius (1994) that focus exclusively on the differentiation process. While touching on adipose cell biology, this chapter will focus on the biochemistry of triacylglycerol metabolism.

2. Adipose development

2.1. Development of white and brown adipose tissue *in vivo*

The study of white adipose tissue (WAT) development in mammals has been facilitated by the use of experimental animal models. Rodents, guinea pigs, rabbits, pigs, as well as humans, have all been evaluated for the development of white adipose tissue. In general, WAT is not detected at all in mice or rats during embryogenesis, but in pigs and humans it is evident during the last third of gestation. In humans, small clusters of adipocytes are present that increase in size during gestation. Larger clusters of fat cells are associated with tissue vascularization and a general increase in cluster size is positively correlated with larger blood vessels. Paracrine/autocrine factors play a significant role in both capillary growth and adipose conversion.

After birth, sex- and site-dependent differences in fat deposits are well known in humans and several animal species. The diet plays a critical role in the degree of lipid

filling within an adipocyte. However, controversy surrounds the question of new fat cell development following a long-term fast. In general, starvation conditions lead to a loss of adiposity and some apparent diminution in the number of fat cells. Refeeding restores lipid levels and the apparent number of adipocytes. Consequently, fasting/refeeding typically has little effect on the number of adipocytes in the body. It is generally accepted that adipose precursor cells are present throughout life and that removal of adipocytes, either by diet or surgical methods, will ultimately result in a restoration of adipose levels.

In contrast to white adipose tissue, brown adipose tissue (BAT) develops during fetal life and is morphologically and biochemically identifiable at birth. Using the uncoupling protein 1 as a BAT marker (specific for brown adipose tissue mitochondria, see Section 3.6), brown fat development has been shown to occur maximally during the last third of gestation. Two conditions have been shown to enhance the development of brown fat hyperplasia in rodents: cold-acclimation and hyperphagia. Both conditions result in a metabolic demand for high-energy expenditure, either in the form of increased heat need or increased metabolism. Brown adipose tissue is common in rodents, camels and hibernating animals such as bears and marmots. The oxidation of triacylglycerol stores in brown adipose tissue depots during hibernation or fasting provides certain animals with a source of water and energy during nutrient deprivation. In humans, although still somewhat controversial, it is generally accepted that brown fat is not present to any significant extent and that white adipose tissue carries out the body's energy storage functions.

In general it is assumed that WAT and BAT develop from different immediate precursor cells. However, a common precursor for both cannot be ruled out and the possible transformation of BAT into WAT has been considered. The conversion from BAT to WAT would be correlated with a decrease in BAT-specific gene products such as the uncoupling protein. However, the reverse does not appear to take place. That is, cold-adapted rodents do not lose WAT and redevelop BAT in response to a low-temperature challenge.

2.2. In situ models of adipose conversion

Ailhaud and colleagues have described the adipoblast to adipocyte conversion as a multistep process initiating with the determination of pluripotent proliferative cells to the adipocyte pathway (Table 1). To better characterize the differentiation process and examine the molecular basis of adipose development, a number of murine, hamster, and rat model cell lines (3T3-L1, 3T3-F442A, 10T $\frac{1}{2}$, Ob1771) have been established. In general, committed preadipocytes express few markers associated with mature fat cells and are still capable of DNA replication and cell division. Hormonal stimulation by IGF-1 (or high concentrations of insulin), the addition glucocorticoids (e.g., dexamethasone), as well as a phosphodiesterase inhibitor (isobutylmethylxanthine), are frequently used to induce terminal differentiation in culture [1]. Following differentiation the mature adipocytes exhibit a massive triacylglycerol accumulation and are responsive to hormonal stimulation.

For decades, brown fat metabolism has been studied with tissue explants. While

Table 1
Stages of adipose conversion

Stages	Cell type	Characteristics
Stage 1	Mesenchymal/pluripotent	Multipotential — ability to differentiate into muscle, cartilage or fat
	<i>Determination</i>	
Stage 2	Adipoblasts	Unipotential — can only differentiate into adipocytes
	<i>Commitment</i>	
Stage 3	Preadipocytes	No lipid accumulation, early transcription factors, e.g. C/EBP β , and early markers of differentiation expressed, e.g. lipoprotein lipase
	<i>Terminal differentiation</i>	
Stage 4	Adipocytes	Lipid accumulation and expression of late transcription factors, e.g. PPAR γ and C/EBP α , and late markers of differentiation, e.g. PEPCK, aP2, FATP

white adipose tissue is important for the storage of energy in the form of triacylglycerol, brown fat functions to dissipate energy in the form of heat through the action of a specific mitochondrial proton transporter, the uncoupling protein 1. While the 3T3-L1 or 3T3-F442A cell lines provided a convenient method to study white adipose metabolism, similar brown fat models were, until recently, lacking. However, by expressing the SV40 early genes under control of the strong fat-cell specific adipocyte fatty acid-binding protein (aP2) promoter in transgenic animals, brown fat tumors developed due to t-antigen-induced oncogenesis [2]. Such tumors were used to derive hibernoma cell lines (rapidly growing brown fat cells) exhibiting the properties of brown fat. The brown fat hibernomas express the mRNA for the uncoupling protein upon stimulation with cAMP, cAMP analogs, or a variety of β_2 and β_3 -receptor agonists. Such cell lines have been extremely useful for the study of brown fat gene expression and metabolism.

2.3. Transcriptional control during development

To characterize the molecular basis for differential gene expression, a number of laboratories have identified transcription factors regulating genes expressed in adipocytes. Of those genes most actively studied, the adipocyte fatty acid-binding protein (aP2) gene and the insulin-stimulatable glucose transporter gene (GLUT4) have proven particularly useful. The aP2 gene is expressed in an adipose-specific manner and is up-regulated at least 50-fold as a consequence of adipose conversion [3]. The aP2 gene is regulated by glucocorticoids, insulin, and polyunsaturated fatty acids while the insulin-stimulatable glucose transporter is regulated primarily by insulin, cAMP and fatty acids [4,5]. Using these and other adipose genes as templates, three different transcription factor families have been identified as critical components of the adipocyte differentiation program.

2.3.1. C/EBP family of transcription factors

The CCAAT/enhancer-binding proteins (C/EBP) are a family of transcription factors strongly implicated in the control of genes involved in intermediary metabolism. Ori-

nally cloned by McKnight and colleagues, the C/EBPs are leucine-zipper transcription factors, a family of proteins whose sequences are characterized by the presence of a basic region followed by a leucine-rich motif. Leucine-zipper proteins are capable of forming coil-coil interactions with other similar types of factors. As such, the C/EBPs form homo- and heterodimers with other family members, thereby allowing for their binding to *cis*-regulatory elements within the promoter/enhancers of genes regulated by C/EBPs.

A number of genes involved in adipose lipid metabolism are regulated by the C/EBP family of transcription factors. Binding sites for the C/EBP proteins reside within the promoters of the aP2, stearoyl-CoA desaturase and insulin-stimulatable glucose transporter genes [4]. Transient transfection studies have revealed that the C/EBP sites within the promoter of the glucose transporter gene are functional and that these transcription factors play the central role in regulating the expression of the gene in differentiated adipocytes. Expression of antisense C/EBP α RNA in 3T3-L1 preadipocytes blocked the expression of C/EBP α and concomitant expression of several adipocyte genes including the insulin-stimulatable glucose transporter and adipocyte fatty acid-binding protein. Moreover, in such antisense C/EBP α -expressing cells, the accumulation of cytoplasmic triacylglycerol was blocked, suggesting that a global inhibition of genes expressing proteins of adipose lipid metabolism was occurring. Consistent with a central role for C/EBP α in lipid metabolism, mice bearing a targeted disruption in the C/EBP α allele fail to accumulate triacylglycerol in both adipose and liver.

Three members of the C/EBP family of transcription factors are expressed in adipocytes: α , β and δ . The temporal expression of the three isoforms during 3T3-L1 differentiation suggests that the C/EBP genes are subject to exquisite regulatory controls. For example, within the C/EBP α promoter resides a C/EBP binding site that suggests that C/EBP β and/or C/EBP δ may be responsible for the activation of expression of the C/EBP α gene. In addition, insulin regulates the transcription of the C/EBP α , β , and δ genes in fully differentiated 3T3-L1 adipocytes. Insulin addition to 3T3-L1 adipocytes represses the expression of C/EBP α while inducing the expression of C/EBP β and C/EBP δ . Furthermore, glucocorticoids reciprocally regulate expression of the C/EBP α and δ genes in 3T3-L1 adipocytes and white adipose tissue [6]. This observation may provide a mechanistic connection between the accumulation of central adipose tissue and hypercortisolemia associated with Cushing's syndrome.

2.3.2. PPAR/RXR family of transcription factors

While the C/EBP family of transcription factors has been implicated as central to the control of gene expression in the differentiated adipocyte, a different family of DNA-binding proteins is apparently instrumental in regulating the differentiation of preadipocytes into mature fat cells. With the adipocyte fatty acid-binding protein gene as a template, Spiegelman and colleagues employed transgenic animal technology to map the region of the adipocyte fatty acid-binding protein gene necessary and sufficient to direct the expression of a chloramphenicol acetyl transferase transgene in a fat-cell-specific manner. Surprisingly, they found that while the region of DNA necessary for C/EBP action was essential for regulation in the mature adipocyte, a distinct 518

bp enhancer region some 5.4 kb upstream of the start of transcription was required for fat-cell-specific expression. By using DNA gel mobility shift analysis, an adipose-specific factor (ARF6) was identified which bound to a DNA element (ARE6) within the upstream enhancer. Importantly, the ARE6 element exhibited sequence similarity to the consensus nuclear hormone response elements. The ARE6 DNA element was similar to that which bound a heterodimer between the nuclear retinoid X receptor (specific for 9-*cis* retinoic acid ligands) and the peroxisome proliferator activated receptor (PPAR).

PPARs belong to a family of nuclear transcription factors that function in a ligand-dependent manner. Once bound by a ligand, the receptors heterodimerize with retinoid X receptors (RXR). They can activate transcription in target genes by recognizing and binding to specific DNA recognition elements termed peroxisome proliferator activated receptor response elements, PPRE (direct repeat of AGGTCA spaced by one or two nucleotides). This PPRE element is identical to the ARE6 DNA element originally identified by Spiegelman and his colleagues. To date, three different PPAR isoforms α , δ/β and γ , and splice variants have been identified that are encoded by separate genes [7]. The tissue-specific expression pattern of these transcription factors is indicative of their function in those tissues [8].

PPAR α target genes involved in fatty acid catabolism (β and ω oxidation pathways) and is most abundant in liver although also found in kidney, heart and brown adipose tissue [8]. The δ isoform is most widely distributed, and is found in a variety of tissues including heart, kidney, brain, intestine, muscle, spleen, lung and adrenal. The γ isoform is the most highly restricted in its expression pattern with primary sources being adipose, macrophage and mammary tissue. Alternate promoter usage coupled with differential mRNA splicing result in two closely related PPAR γ isoforms that differ by only 30 amino terminal amino acid residues [9]. The $\gamma 1$ isoform is found in adipose tissue and to a lesser extent in liver, kidney and heart. The $\gamma 2$ isoform is found almost exclusively in white adipose tissue. During the adipocyte differentiation program, the level of expression of the three different PPAR isoforms is temporally regulated. Low expression levels of PPAR α in white adipose tissue as compared to brown adipose tissue suggests that its role in differentiation is a minor one.

In cultured cell lines, PPAR γ , which is expressed during the late stages of adipocyte differentiation, has been shown to be the most adipogenic of the three isoforms. A balance between the expression of PPAR γ and retinoic acid receptor forms (RAR and RXR) controls heterodimer formation during adipogenesis [7]. Some of the genes regulated by PPAR γ in adipose tissue include adipocyte fatty acid-binding protein (aP2), lipoprotein lipase, fatty acid transport protein (FATP1), acyl-CoA synthetase, stearyl-CoA synthetase, and phospho-enol pyruvate carboxykinase [10].

A variety of hydrophobic ligands including polyunsaturated and oxidized fatty acids have been suggested as natural ligands for PPARs. Moreover, the antidiabetic glitazone drugs are targeted towards PPARs [11]. Some eicosanoids, in particular, 15-deoxy $\Delta^{12,14}$ -prostaglandin J2, 15 HETE and 9- and 13-hydroxyoctadecadienoic acids derived from linoleic acid (Chapter 13), have been identified as potent natural ligands for PPAR γ [2]. Other arachidonic acid metabolites derived from the lipoxygenase pathway, namely 8-S-hydroxyeicosatetraenoic acid (8SHETE) and leukotriene B4 have also been suggested as ligands for PPAR α [8]. The expression of PPAR γ in adipose tissue

is upregulated by hormones like insulin and glucocorticoids while cytokines, tumor necrosis factor α in particular, have been shown to decrease the expression of PPAR γ and C/EBP α [8].

The PPAR γ activity is regulated in concert with a nuclear coactivator termed PGC-1. While PGC-1 was originally believed to function only in white and brown adipocytes, PGC-1 is also expressed in hepatocytes where it plays a primary role in controlling the expression of genes involved in gluconeogenesis.

2.3.3. *SREBP family of factors*

Sterol regulatory element-binding proteins (SREBPs) belong to a family of transcription factors that regulate genes involved in cholesterol and fatty acid metabolism [10]. SREBPs consist of three major isoforms: SREBP1a, 1c (ADD1) and 2. The SREBP1 gene has two alternate promoters which result in the synthesis of two proteins (SREBP1a and 1c) of different lengths. SREBP1a is a more potent activator of transcription because of its longer acidic amino terminal transactivation domain and its capacity to induce a wider range of target genes. SREBP2 is encoded by a separate gene.

SREBPs are basic helix–loop–helix leucine zipper (bHLH-LZ) DNA-binding proteins. These proteins have dual DNA specificity. They can recognize and bind to inverted repeat sequences known as E box motif (5'-CANNTG-3') as well as direct repeat sterol regulatory elements. SREBPs contain a tyrosine residue within their basic domain which confers a conformation change within the protein allowing for the recognition of sterol regulatory elements or related sites [12].

SREBPs are unique transcription factors because they are initially synthesized as precursor membrane-bound proteins present in the endoplasmic reticulum and the nuclear envelope. Both their amino and carboxyl terminals are cytosolic. Cleavage at two specific sites within the amino terminal by proteases releases the transcription activation and DNA-binding domain which represents the mature SREBP [12]. The cleavage process is regulated by the cellular concentration of sterol which directly regulates the activity of the proteases and by a chaperone protein, SREBP-cleavage activating protein (SCAP), which is involved in trafficking the precursor SREBPs to the Golgi network for cleavage [13,14] (see Chapter 15).

All cultured cell lines express SREBP1a and 2 [12]. Liver and adipose tissue express predominantly SREBP 2 and 1c. Overexpression of SREBP1a in cultured cells and animal livers results in an increase in the expression of cholesterol and fatty acid metabolism genes. An increase in the levels of cholesterol and triacylglycerols was observed in the liver; however, there was no change in the serum levels of these metabolites. A decrease in the mass of triacylglycerols present in white adipose tissue was also noted [14].

Overexpression of SREBP1c in adipose cells led to the development of hyperglycemia, fatty liver and an increase in the levels of serum triacylglycerols while overexpression of SREBP2 led to a significant increase in the accumulation of primarily cholesterol. In mature adipocytes, insulin has been shown to upregulate the expression of SREBP1c which can then induce the expression of genes involved in lipogenesis, e.g. lipoprotein lipase and fatty acid synthetase [15] and possibly enzymes necessary for synthesis of an endogenous PPAR γ ligand.

Targeted disruption of the SREBP1 gene (lack of SREBP1a and c) has resulted in negligible effects on white adipose tissue mass and function possibly due to the upregulation of SREBP2. No difference in the expression of genes involved in fatty acid catabolism or triacylglycerol synthesis was observed [10]. SREBP2 knockout mice are embryonic lethal.

3. Biochemical aspects of lipid metabolism

3.1. Lipid delivery to adipose tissue

The primary function of adipose tissue is to serve as a storage site for the excess energy derived from food consumption. This energy can then be utilized by the organism to fulfill subsequent metabolic requirements during times of little or no consumption. In the case of white adipose tissue, these requirements entail efficient storage of large amounts of energy in a form that can be mobilized readily to supply the needs of organs and tissues elsewhere in the body. Lipids, particularly fatty acids, are an exceptionally efficient fuel storage species. The highly reduced hydrocarbon tail can be readily oxidized to produce large quantities of reduced coenzymes and subsequently ATP. At the same time, the very hydrophobic nature of the fatty tail precludes concomitant storage of excess water that would increase the mass and spatial requirements of the organism considerably. Also, the relatively straight, chain-like structure of the fatty acid permits dense packing of many molecules into each cell, maximizing the use of storage space available. Brown adipose also stores energy in lipid form, but more frequently produces heat by oxidizing fatty acids within the adipocyte, rather than supplying free fatty acids for use by other cell types.

3.2. Fatty acid uptake and trafficking

At the adipose tissue beds, fatty acids are liberated from triacylglycerol-rich lipoproteins through the action of lipoprotein lipase (Chapter 20). Released fatty acids are bound by albumin and are the donors of lipid for fatty acid uptake. Two schools of thought have dominated hypotheses dealing with fatty acid uptake. First, local protonation of fatty acids due to the relative acidity at the plasma membrane, coupled with the low aqueous solubility of fatty acids at neutral pH and high permeability of fatty acids into the hydrophobic environment of the plasma membrane, creates a sufficient driving force for diffusion across the outer and inner leaflets of the membrane. A second, and drastically different viewpoint, is that there are protein cofactors called lipid transporters that facilitate the transfer process.

Several studies by Hamilton and others have demonstrated that fatty acids can diffuse through membranes at rates, and with properties, consistent with biological transport. Adsorption, flip-flop, and desorption of fatty acids at the plasma membrane functions to transfer protons across the membrane. Acidification of micelles, blood neutrophils, and hepatocytes accompanies fatty acid uptake, consistent with a diffusional mechanism and indicates that protein-mediated transfer is not necessary.

In contrast, a number of plasma membrane fatty acid-binding proteins have been identified and argued to facilitate transbilayer flux. Abumrad and colleagues have identified a murine homologue of human cell surface antigen CD36 as a lipid-binding protein involved in lipid transport and termed the protein fatty acid translocase. Fatty acid translocase is also found in muscle and cardiac myocytes and has been genetically linked to hyperlipidemia and hypertension. Lodish and co-workers have identified the fatty acid transporter family of plasma membrane proteins and used expression cloning to demonstrate fatty acid transport activity. FATP1 and FATP4, exhibit very long-chain acyl-CoA synthetase activity suggesting that esterification of very long-chain fatty acids at the plasma membrane may be coupled to uptake.

Once inside the cell, free fatty acids are minimally soluble in the aqueous cytoplasm. The charged carboxylate group provides enough electrostatic hindrance to prevent association with the neutral triacylglycerols, whereas the hydrocarbon tail reduces solubility in water. At high enough concentrations fatty acids exert a detergent-like effect that would disrupt membranes and/or they could cluster together in micelles in the crowded cytoplasm. To alleviate this problem, the adipocyte and other lipid-metabolizing cell types have evolved intracellular fatty acid-binding proteins, a family of small, soluble, highly abundant proteins that bind and sequester free fatty acids [16]. Adipocytes express two fatty acid-binding proteins, the products of the FABP4 and FABP5 genes.

The adipocyte fatty acid-binding protein (aP2) has become a paradigm for in vitro studies of protein-lipid interactions. Ligands for this protein are long-chain (>14 carbon) fatty acids and/or retinoic acid. Its small size (≈ 15 kDa), high solubility and stability have facilitated purification and characterization of many features of the protein. Crystal structures of the adipocyte fatty acid-binding protein have been solved at high resolution for wild type and site-directed mutant forms, both in the absence and presence of bound ligands. Despite a widely varying degree of primary sequence similarity, the intracellular lipid-binding proteins as a family share a virtually superimposable tertiary structure consisting of ten antiparallel β -strands arranged in a flattened barrel [16] (Fig. 1). A single lipid ligand is bound inside the barrel within a large interior water-filled cavity, and held in place by the concerted effect of general surface contacts and specific electrostatic interactions between highly conserved cavity residues and the ligand's polar head group.

3.3. Glucose transport and the generation of the triacylglycerol backbone

The immediate backbone precursor for acylglycerol formation is primarily glycerol 3-phosphate, derived from glycolysis or glycerolgenesis within adipocytes. Fat cells express specific glucose transporters on the plasma membrane to ensure a ready supply of glycolytic intermediates for triacylglycerol synthesis. There are two types of glucose transport proteins in adipose: GLUT1 and GLUT4 [17]. Both are structurally similar with 12 membrane-spanning α -helices with intracellular amino and carboxyl termini. Both proteins are expected to have a large, hydrophilic intracellular loop separating transmembrane domains six and seven, as well as an extracellular loop containing *N*-glycosylation site(s) demarcated by transmembrane domains one and two. The

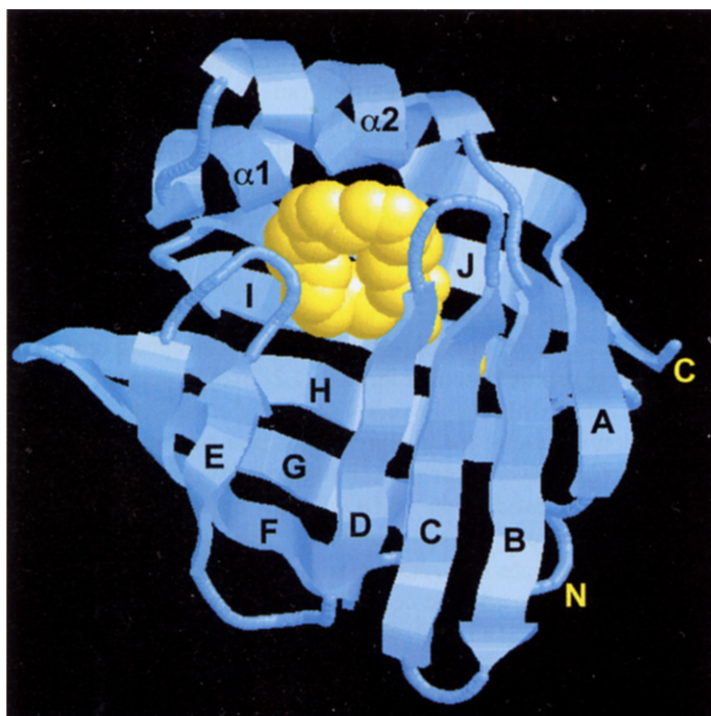


Fig. 1. Diagram of arachidonate buried within the cavity of crystalline adipocyte lipid-binding protein. The protein is depicted as a ribbon drawing with the 10 β -strands labeled A through J and the two α -helices indicated by $\alpha 1$ and $\alpha 2$. The bound arachidonate is illustrated by the yellow space-filling model. N- and C-termini are labelled as N and C, respectively. Note that the carboxyl function of arachidonate is found buried within the cavity, oriented away from the surface of the protein. The entire contact surface of the fatty acid is found within the binding cavity, sequestered from the surrounding milieu.

majority of GLUT1 has been shown by Cushman and co-workers to be present in the plasma membrane of cells unstimulated by insulin, constitutively facilitating transport of glucose down a concentration gradient. However, the bulk of insulin-stimulated glucose transport results from the activity of GLUT4. In the basal state, GLUT4 is largely found in small, intracellular vesicles, but rapidly translocates to the plasma membrane following insulin stimulation. In addition, insulin promotes a change in the rate of intracellular GLUT4 recycling and trafficking which results in a net 10- to 15-fold stimulation of hexose transport in response to insulin [11]. Once inside the cell, facilitative transport of glucose by GLUT1 and GLUT4 is rendered unidirectional by the action of hexokinase. Glucose 6-phosphate can only proceed to the glycolytic pathway because adipocytes do not express significant levels of glucose 6-phosphatase.

3.4. Fatty acid and triacylglycerol biosynthesis

Adipocytes readily convert the products of glycolysis into fatty acids via the de novo biosynthetic pathway (Chapter 6). Briefly, surplus citrate is transported from the

mitochondrion and cleaved to produce cytoplasmic acetyl-CoA. Cytoplasmic acetyl-CoA is acted upon by acetyl-CoA carboxylase that produces malonyl-CoA. The next steps of the fatty acid biosynthetic pathway are carried out by the multifunctional fatty acid synthase enzyme that utilizes NADPH to catalyze multiple condensations of malonyl-CoA with acetyl-CoA or the elongating lipid, eventually generating palmitate.

In adipocytes, two pathways exist for the production of phosphatidic acid. In one, glycerolphosphate is sequentially esterified with two acyl-CoAs to produce 1-acylglycerolphosphate and 1,2-diacylglycerolphosphate. In the second pathway, dihydroxyacetone phosphate is esterified with acyl-CoA to produce acyl dihydroxyacetone phosphate. An acyl dihydroxyacetone phosphate reductase subsequently produces 1-acylglycerolphosphate that leads to the production of 1,2-diacylglycerolphosphate. The resultant phosphatidic acid is dephosphorylated generating 1,2-diacylglycerol and triacylglycerol formed through the activity of diacylglycerol acyltransferase (DGAT).

The DGAT catalyzed reaction is crucial for it represents a branch point for hydrocarbon flow towards either the triacylglycerol or phospholipid pathways. However, this view has been modified through the production of DGAT null mice. DGAT-deficient mice are viable, synthesize triglycerols normally and are resistant to diet-induced obesity. This implies that either alternate biosynthetic pathways are being utilized or additional DGAT isoforms are present in adipose cells. To address this, Farese and colleagues have identified a second DGAT (DGAT2) with kinetic properties distinct from the original DGAT (now termed DGAT1). Moreover, additional DGAT-like sequences are present in the murine and human genome suggesting an unappreciated complexity in diacylglycerol metabolism.

3.5. Triacylglycerol mobilization

Lipolysis refers to the process by which triacylglycerol molecules are hydrolyzed to free fatty acids and glycerol. During times of metabolic stress (i.e. during fasting or prolonged strenuous exercise when the body's energy needs exceed the circulating nutrient levels), the adipocyte's triacylglycerol droplet is degraded to provide free fatty acids to be used as an energy source by other tissues. Numerous stimuli are capable of eliciting the lipolytic response in adipocytes. However, ultimately the same pair of enzymes, hormone-sensitive lipase and monoacylglycerol lipase, is responsible for catalyzing the hydrolysis of the triacylglycerol ester bonds.

Complete hydrolysis of triacylglycerol involves the breakage of three ester bonds to liberate three fatty acids and a glycerol moiety (Fig. 2). The same enzyme, hormone-sensitive lipase, is responsible for facilitating hydrolysis of the esters at positions 1 and 3 of the triacylglycerol. A second enzyme, 2-monoacylglycerol lipase, catalyzes hydrolysis of the remaining ester to yield a third free fatty acid and glycerol. Glycerol must be shuttled back to the liver for use in oxidation or gluconeogenesis. Glycerol has no alternative fate in the adipocyte; adipocytes do not express a glycerol kinase and so are unable to reuse glycerol. Glycerol is effluxed out of adipocytes via an aquaporin-type of transport molecule. Mono- and diacylglycerols can be re-esterified by the endoplasmic reticulum acyltransferases. During a lipolytic stimulus, re-esterification is thought to be minimized so that the net direction of these reactions is toward lipolysis.

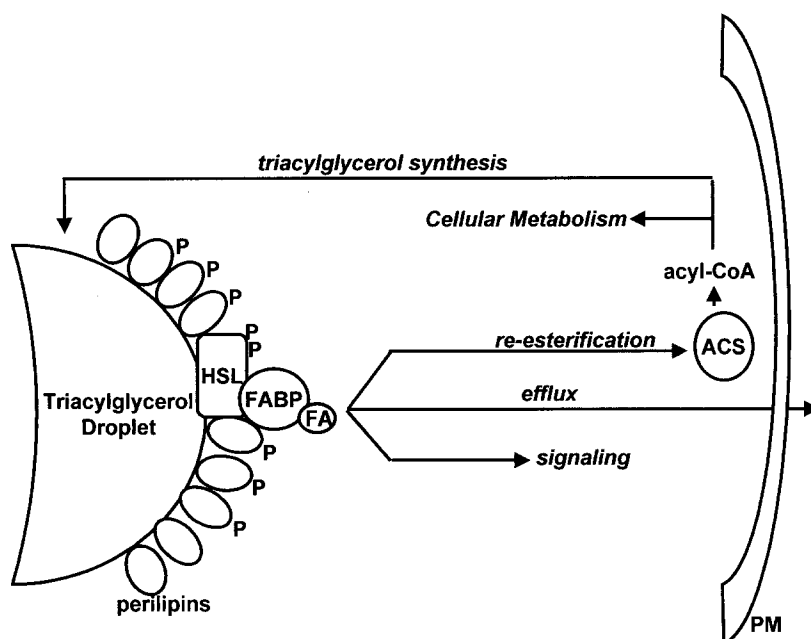


Fig. 2. Schematic representation of the key steps in lipolysis. Triacylglycerol and diacylglycerol depots, surrounded by lipid-associated proteins such as perilipin, are hydrolyzed by hormone-sensitive lipase (HSL) releasing fatty acids (FA) and generating monoacylglycerol. Monoacylglycerol lipase catalyzes the last step in lipolysis formation of fatty acid and glycerol. Glycerol is released from the adipocyte while fatty acids can be bound by intracellular lipid-binding proteins (FABP) and shuttled out of the cell, to signaling depots, or to sites of re-esterification by acyl-CoA synthetase (ACS) and subsequent metabolism or regeneration of triacylglycerol. PM, plasma membrane; P, phosphorylation.

However, under maximal lipolytic conditions, substantial recycling of fatty acids occurs such that on average about two fatty acid molecules are released per glycerol molecule. Outside the adipocyte, fatty acids are immediately bound to serum albumin and carried in the bloodstream to the liver, muscle and other tissues for oxidation.

To avoid futile cycling of fatty acids (and concomitant loss of large amounts of energy), and to maintain proper energy balance between storage and expenditure, triacylglycerol synthesis and hydrolysis are carefully regulated. This regulation is present on several levels, including hormonal secretions from the endocrine system, neurotransmitter secretions from the sympathetic nervous system, intracellular G-protein-mediated signal cascades, gene expression, post-translational modification and product inhibition (Fig. 3). The proximal target of regulatory action is the enzyme responsible for initiating fatty acid mobilization: hormone-sensitive lipase (HSL).

HSL catalyzes the rate-limiting step in lipolysis and is regulated through the stimulation of β -adrenergic receptors by catecholamines or, more recently discovered, by natriuretic peptides [18] and the activation of cAMP-dependent protein kinase. HSL is an 84 kDa protein modeled to be organized into N-terminal (1–300) and C-terminal (300–767) domains. The regulatory and catalytic activity of the enzyme lies within

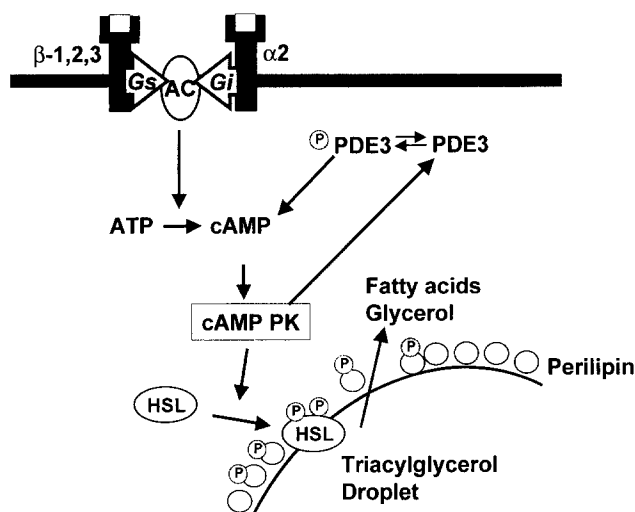


Fig. 3. Activation of lipolysis via adrenoreceptor-coupled systems. Binding of lipolytic agonists to β -adrenoreceptors (β_1 , β_2 and β_3) couples to the G-protein which in turn activates adenylyl cyclase (AC) thereby producing cAMP. cAMP activation of protein kinase A (cAMP PK) results in phosphorylation and activation of hormone-sensitive lipase (HSL) and perilipin and subsequent translocation of HSL to and movement of perilipin around the lipid droplet. cAMP PK also phosphorylates and activates the cGMP-inhibited cAMP phosphodiesterase (PDE), providing a feedback system to lower intracellular cAMP. α_2 -adrenoreceptor activation results in coupling with G_i and a decrease in adenylyl cyclase activity. Dynamic interplay between β and α_2 adrenoreceptors regulates the activity of adenylyl cyclase and, therefore, cAMP PK.

the predicted α/β hydrolase fold of the C-terminal domain. Three residues (Asp703, His733 and Ser423) form the catalytic triad. Phosphorylation of the C-terminal domain results in a modest increase in specific activity of the enzyme and is correlated with a translocation of the protein from the cytoplasm to the surface of the lipid droplet.

Multiple phosphorylation sites on HSL have been identified, and its regulation by phosphorylation/dephosphorylation makes it unique among lipases. Two phosphorylation sites originally thought to be basal and regulatory sites [19], have been extensively examined. Ser565 is a basal phosphorylation site, while Ser563, once thought to be the site of activation, is now of unknown function. More recently two additional serine phosphorylation sites have been mapped and linked to translocation. Ser659 and Ser660 have been determined to be regulatory sites essential for the activation of HSL and mutation of these sites inhibits the translocation of HSL to the lipid droplet [20].

Phosphorylation and translocation of HSL to the droplet surface is thought to be functionally linked to association with one or more droplet-associated proteins. Present at the lipid droplet are a number of proteins, including perilipins, ADRP, lipotransin, and adipophilin. These proteins are thought to form a functional complex surrounding the lipid droplet, both protecting it from uncontrolled hydrolysis and assisting in amassing lipid. Phosphorylation of perilipin is correlated with hormone-stimulated lipolysis suggesting that phosphorylated perilipin has reduced affinity for droplets and dissociates from the surface concomitant with HSL association. Consistent with this

model, perilipin null mice are found to have constitutively active HSL, are hyperphagic, but maintain normal body weights, suggesting that perilipin acts as a barrier to HSL at the droplet surface. Perilipin null mice also reverse the obese phenotype when bred into db/db mice.

The N-terminal domain of HSL has recently been shown to be a docking domain for interaction with the adipocyte fatty acid-binding protein (aP2). Using a combination of yeast two hybrid analysis co-immunoprecipitation, and deletion analysis, it has been shown that the adipocyte fatty acid-binding protein forms a complex with the N-terminal domain. Moreover, the FABP stimulates the activity of HSL and relieves product inhibition by fatty acids. In experiments using aP2 null mice, lipolysis is markedly reduced while fatty acids accumulate intracellularly suggesting that the fatty acid-binding protein docks onto HSL, binds a product fatty acid and facilitates the intracellular trafficking of lipolyzed fatty acids in an efflux shuttle.

Multiple lines of HSL null mice have been developed that exhibit somewhat different characteristics. However, there are some generalizations that can be made. HSL-deficient mice exhibit reduced, but not eliminated hormone-stimulated lipolysis. This suggests that there may be additional lipases linked to hormone stimulation although compensatory mechanisms not normally associated with adipose tissue *in vivo* have not been ruled out. The HSL null mice are sterile due to the lack of neutral cholesterol ester hydrolase activity inherent to HSL and essential for spermatozoa production.

3.5.1. Catecholamines and adrenoreceptors in adipocytes

The catecholamines epinephrine and norepinephrine (adrenaline and noradrenaline) originate in the inner medullar region of the adrenal glands. Stimulation of the adrenal by the sympathetic nervous system leads to secretion of catecholamines into the bloodstream. In addition, adipose tissue is itself directly innervated by the sympathetic nervous system. Various types of metabolic stress trigger the sympathetic nervous system to release its neurotransmitter, norepinephrine, directly into adipose, where its effects on the adipocyte are mediated by specific plasma membrane adrenoreceptors. Rapid reflex responses are primarily stimulated by the sympathetic nervous system, whereas more long-term (i.e. on the scale of hours, days, and weeks) and/or basal effects are subject to regulation by catecholamine secretion.

The effects of catecholamines and the mechanisms that mediate them have been extensively studied in adipocytes. Adipocytes express a combination of five different adrenoreceptor isoforms: α_1 , α_2 , β_1 , β_2 , and β_3 [21]. Lipolysis is signaled by β -adrenergics. An anti-lipolytic signal is transduced by the α_2 -adrenergics, and the α_1 -adrenergics are involved in a separate pathway. In short, although lipolysis is the observed outcome of catecholamine stimulation, it is the steady state result of competition between two opposing pathways triggered by the same signal.

The mechanisms of signal transduction are reasonably well known. Binding of catecholamines to the β -adrenoreceptors activates adenylyl cyclase via a stimulatory G-protein (G_s) (Fig. 3). Adenylyl cyclase catalyzes the conversion of ATP to cAMP. cAMP binds the regulatory subunit of protein kinase A, releasing the active catalytic subunit. Active protein kinase A in turn phosphorylates the HSL and perilipins which increase translocation to and from the triacylglycerol droplet, respectively. The same signal

bound to the α_2 -adrenoreceptor affects an inhibitory G-protein (G_i), which inhibits the activity of adenylyl cyclase. Disappearance of cAMP eventually causes cAMP to dissociate from the regulatory subunit of protein kinase A, which then inactivates the catalytic subunit by reassociation. In the absence of continued phosphorylation, dephosphorylation mediates reverse translocation of HSL and perilipins to the basal state.

With simultaneous activation of opposing pathways, the relative contribution of each receptor type becomes very important. Small mammals, such as rats and hamsters, express mainly β_1 and β_3 while rats express very little of the α_2 isotype. Large mammals (e.g. humans and monkeys) express almost exclusively β_1 and β_2 and a significant amount of α_2 receptor. It appears that the β_3 receptor is expressed to a greater extent in brown adipocytes than in white adipocytes.

A second observed pattern of receptor regulation was demonstrated by the use of agonists and antagonists for each receptor isotype. At very low agonist concentrations, only α_2 -receptor activity is observed (i.e. anti-lipolysis). As the agonist concentration is increased, β_1 becomes active and initiates lipolysis. Only under much more stimulatory agonist conditions do β_3 receptors become active. β_2 , in animals that express it, seems to be active under conditions more similar to β_1 . Affinity for ligands and level of expression of receptors are two methods utilized by adipocytes to regulate catecholamine effects. The interplay between the various isotypes is responsible for the adrenergic balance of lipolysis and anti-lipolysis. In general, α_2 -mediated anti-lipolysis modulates resting adipocyte activity, whereas during stress-induced norepinephrine release, increased binding to the β -adrenergics overcomes the α_2 inhibitory effect and β -mediated lipolysis prevails.

3.5.2. Glucagon

Although catecholamines are perhaps the strongest physiological lipolytic stimulus, other hormones also play an important role in mediating energy balance. One such hormone is glucagon that is one of three polypeptide hormones secreted by endocrine tissues located within the pancreas. Glucagon is secreted into the circulation in response to low blood glucose levels and the result of its action is mobilization of stored energy.

Stimulation by glucagon takes place by a virtually identical pathway to stimulation by catecholamines. Glucagon binds extracellularly to a specific seven-transmembrane-domain receptor, activating adenylyl cyclase via a stimulatory G-protein. Protein kinase A is subsequently activated and phosphorylates HSL, which begins to hydrolyze triacylglycerol stores. Protein kinase A also phosphorylates (and activates) enzymes in the glycogen degradation pathway, and inhibits *de novo* fatty acid synthesis by phosphorylation of acetyl-CoA carboxylase in concert with AMP-activated protein kinase (Chapter 6).

Because the same regulatory pathway is activated, the same feedback mechanisms used to modulate chronic catecholamine effects are equally significant for prolonged glucagon stimulation. At some level of cAMP production, the cAMP response element-binding protein transcription factors become phosphorylated by protein kinase A and upregulate cAMP-responsive gene expression leading to increased receptor expression.

However, protein kinase A phosphorylation of the cell surface receptors leads to uncoupled G-protein activity and heterologous desensitization to both the glucagon and catecholamine signals. It is also relevant that protein kinase A can phosphorylate and thereby activate cGMP-inhibited phosphodiesterase [22], which cleaves cAMP and probably helps modulate its effects to minimize desensitization.

3.5.3. *Steroid and thyroid hormone*

In addition to the major metabolic regulators in adipocytes (catecholamines, glucagon, and insulin), many diverse types of hormones have effects on adipocyte metabolism. The most notable results are effected by glucocorticoids, sex steroids, and thyroid hormones.

Glucocorticoids are steroid hormones secreted by the adrenal cortex in response to stress or starvation. Glucocorticoids display a permissive effect on lipolysis stimulated by catecholamines. Glucocorticoid response elements have been observed in the upstream regions of β_1 and β_2 adrenoreceptor genes and, in fact, an increase in numbers of expressed β -receptors in response to glucocorticoids has been reported. Glucocorticoid response elements have also been identified in the upstream regions of the C/EBP family of transcription factors. Additionally, the activity of the stimulatory G-protein is enhanced by glucocorticoids. These effects are consistent with the finding that adrenalectomy reduces G_s protein and mRNA levels, and that subsequent administration of a glucocorticoid such as dexamethasone can restore those levels. Glucocorticoids, therefore, probably ensure maintenance of catecholamine-induced lipolysis by enhancing transcription of the genes involved in that signal cascade.

Sex steroids (primarily estrogen in females, which is synthesized by the ovaries, and testosterone in males, synthesized by the testes) like glucocorticoids, also affect gene transcription by binding to nuclear Zn-finger transcription factors that recognize steroid response elements. In female rats, ovariectomy was shown to diminish lipolysis by decreasing the effectiveness of the adenylyl cyclase catalytic activity. Lipolysis was restored to normal levels in these animals by administration of estrogen but not progesterone. Castrated male rats exhibited decreased lipolysis that appeared to be caused both by defective adenylyl cyclase catalysis and a decreased number of β -adrenergic receptors, again implying desensitization to catecholamines. Normal lipolytic levels could be restored by administration of testosterone.

The circulating thyroid hormones thyroxine and its more potent derivative triiodothyronine are secreted from the thyroid gland in response to hypothalamus/pituitary stimuli. The effect of elevated thyroid hormone is increased lipolysis, which appears to be mediated by an increase in β_1/β_2 -adrenergic receptor expression and a decrease in inhibitory G-protein expression. These alterations effectively sensitize the adipocyte to catecholamine stimulation.

3.5.4. *Insulin and anti-lipolysis*

Pancreatic β cells secrete the polypeptide hormone insulin in response to elevated blood glucose levels (hyperglycemia). Insulin is the most important physiological stimulus for energy storage. Its effect directly counteracts the effects of glucagon and the catecholamines. Insulin receptors are found in many diverse cell and tissue types, not the least significant of which is adipose.

The insulin receptor is an integral membrane protein that functions as a tetramer composed of two α and two β subunits. The β subunits each span the plasma membrane once, and the α subunits are covalently attached to the β subunit extracellularly by disulfide bonds. The insulin binding site is external. The intracellular domains contain many tyrosine phosphorylation sites and the receptor is itself a tyrosine kinase. Ligand-binding induces autophosphorylation of several intracellular domains, activating the kinase activity of each β subunit. A complex series of interactions follows in which the insulin receptor phosphorylates some of its substrates directly (insulin receptor substrate-1, IRS-2) or recruits various adaptor proteins such as Shc and Grb2 that transmit the insulin signal [23].

Insulin-binding to the adipocyte insulin receptor simultaneously stimulates lipogenesis and inhibits lipolysis. Insulin action effectively clears fatty acids and glucose from the blood both by increasing uptake and storage, and by decreasing mobilization of stored energy. The mechanisms by which these effects are accomplished are highly complex and have not been entirely elucidated although some aspects of the process are clear. The insulin receptor tyrosine kinase is capable of inducing phosphorylation and activation of the cGMP-inhibited phosphodiesterase and several protein serine-phosphatases (most likely protein phosphatases 1, 2A and 2C). Phosphatidyl inositol 3-kinase has been demonstrated as an essential component in the insulin-stimulated activation of the cGMP-inhibited phosphodiesterase but, as previously mentioned, protein kinase A can also fulfill this role in the absence of insulin. Thus, insulin inhibits the cAMP cascade (including activation of hormone-sensitive lipase) through cleavage of cAMP and direct dephosphorylation of protein kinase A-activated substrates. Dephosphorylation also activates acetyl-CoA carboxylase, the enzyme that catalyzes the first committed step in *de novo* fatty acid synthesis, and fatty acyl-CoA synthetase, the first enzyme in the triacylglycerol synthetic pathway. Glucose transport is stimulated via GLUT4 translocation to the plasma membrane, and lipoprotein lipase secretion (and, therefore, fatty acid uptake) is enhanced. In addition, insulin reduces dramatically the number of cell surface β -adrenergic receptors, which further desensitizes the adipocyte to lipolytic stimuli.

The concerted insulin-induced actions of fatty acid/glucose uptake and triacylglycerol synthesis reduce blood glucose. Eventually the diminished glucose levels signal the pancreas to stop secreting insulin and initiate secretion of glucagon. Intermediate stress such as fright or strenuous exercise is capable of stimulating lipolysis via sympathetic nervous system secretion of norepinephrine. In well fed, resting adipocytes, insulin effects are supported and strengthened by the anti-lipolytic action of α_2 -adrenergics. Additional anti-lipolytic influence is exerted by adenosine and certain prostaglandins (notably prostaglandin E_2 in mature adipocytes and prostacyclin in preadipocytes). Adenosine effects are modulated through the A_1 adenosine receptor, a member of the family of purinergic receptors identified in various tissues. Prostaglandins also bind specific cell surface receptors (Chapter 13). Both types of receptors are known to act through an inhibitory G-protein, and so transduce a signal similar to insulin. Thus, the delicate balance between energy storage and mobilization is ensured by complex (and differential) interplay of many regulatory systems and factors.

Table 2
Comparison of major features of white and brown adipose tissue

Major feature	White adipose	Brown adipose
Vascularization	Some, limited	Extensive
Distribution	Extensive, many sites	Restricted
Sympathetic innervation	Some, limited	Extensive
Fatty acid role(s)	Synthesis, storage, signaling	Storage, oxidation, signaling
Uncoupling protein	Present	Highly expressed
Thermogenesis	Present	Highly developed
Insulin effects	Extensive	Extensive
Adrenoreceptors	Primarily α_2 , β_1 , β_2 , β_3	Primarily α_1 , β_1 , β_3
Droplet size	Large, single	Small, multiple
Mitochondria	Few	Many, densely packed

3.6. Brown fat lipid metabolism

White and brown adipocytes are obviously labeled as a result of their difference in color. The variation in appearance between these tissues is a direct reflection of the very different role each performs in the organism and results from specific morphological differences on a cellular level. Whereas the purpose of white fat is to store and release energy in the form of free fatty acids, the essential function executed by brown fat is the expenditure of fatty acid-derived energy for maintenance of the organism's thermal stability (Table 2).

Brown fat derives its color from extensive vascularization and the presence of many densely packed mitochondria (due to the heme cofactors in the mitochondrial enzyme cytochrome oxidase). Brown fat is traversed by many more blood vessels than is white fat. These blood vessels assist in delivering fuel for storage and oxidation, and in dispersing heat generated by the numerous mitochondria to other parts of the body. Brown adipocytes differ in appearance from white adipocytes by the presence of many small triacylglycerol droplets, as opposed to a single large droplet (i.e. multilocular, rather than unilocular). Regulation of brown fat activity is accomplished primarily through the action of the sympathetic nervous system. The blood vessels and each individual brown adipocyte are directly innervated by sympathetic nervous system nerve endings that exert control by release of norepinephrine. Stimulation by the sympathetic nervous system in response to external temperature decrease is essential to the maintenance of brown fat function, and atrophy occurs when regular sympathetic nervous system activity declines [24].

Brown adipocytes also differ from white adipocytes at the molecular level. The major adrenergic receptor subtype expressed by brown adipocytes is the β_3 , but α_1 and β_1 are also found. The most notable difference between brown and white adipocytes is the production of uncoupling protein by the former and its relative absence in the latter [21,24]. Brown adipocytes also express a type II 5'-deiodinase enzyme, which converts the thyroid hormone thyroxine to its more potent form, triiodothyronine, and are capable of secreting triiodothyronine into circulation.

Uncoupling proteins confer to the brown adipocyte the ability to catabolize fatty acids

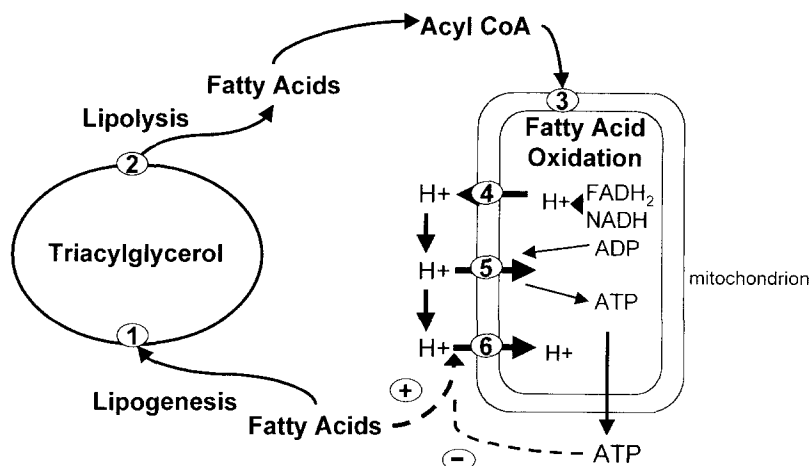


Fig. 4. Thermogenesis in the brown fat mitochondrion. The major fuel pathways of brown fat are represented. Triacylglycerol levels are balanced by the processes of lipogenesis (1) and lipolysis (2). When excess fuel is present, or when heat is needed, fatty acids produced by lipolysis are activated with CoA and transferred to the mitochondrion. Via β -oxidation (3), the long-chain fatty acids are degraded to acetyl-CoA and reduced coenzymes (NADH and FADH₂). The coenzymes transfer their reducing equivalents across the mitochondrial inner membrane (4) against the concentration gradient. Typically the proton gradient is dissipated by the action of the proton-ATPase (5) which uses the energy to drive ATP synthesis. However, brown fat mitochondria possess the uncoupling protein (6), which allows for proton transport across the membrane, down the concentration gradient, with the change in free energy lost as heat. The uncoupling protein is positively regulated by fatty acids and inhibited by purines.

inefficiently (that is, without the usual concomitant ATP production) and dissipate the heat generated by this excessive catabolic activity to other tissues via the bloodstream (Fig. 4). This process is known as thermogenesis and is characterized on two levels. Obligatory thermogenesis occurs in all cell types as the result of ubiquitous nominal inefficiencies in metabolism. Facultative thermogenesis occurs specifically in response to stimuli such as cold (non-shivering thermogenesis in adipose, shivering thermogenesis in muscle) or overfeeding (diet-induced thermogenesis). Facultative thermogenesis, particularly non-shivering thermogenesis, is the specific role of brown adipose. It should be noted that uncoupling protein isoforms are expressed ubiquitously throughout the body suggesting that thermogenesis and/or proton dissipation are common regulatory themes.

3.6.1. Triacylglycerol synthesis and storage

Brown adipose tissue, in its cold/epinephrine-activated state (as opposed to an atrophied or quiescent state), demonstrates increases in blood flow, lipoprotein lipase activity, triacylglycerol synthesis, 5'-deiodinase activity, and triiodothyronine-enhanced uncoupling protein gene expression. The processes of fatty acid uptake and triacylglycerol synthesis are essentially the same in both brown and white fat. However, norepinephrine release by the sympathetic nervous system in acute cold exposure stimulates brown adipose tissue to enhance expression and secretion of lipoprotein lipase to its sites in the vascular

epithelium. Lipoprotein lipase releases fatty acids from passing chylomicrons and very low-density lipoprotein (Chapter 20), causing an influx of fatty acids into the brown adipocytes.

Free fatty acids and norepinephrine inhibit acetyl-CoA carboxylase. Increased fatty acid uptake, lipolysis and esterification occur simultaneously in brown adipose tissue. BAT fatty acyl-CoA synthetase and acyltransferases, associated primarily with the endoplasmic reticulum, catalyze triacylglycerol formation as in white adipose. Triacylglycerol synthesis is decreased during fasting, and increases sharply in an insulin- and norepinephrine-dependent fashion, upon refeeding. Probably the increased triacylglycerol synthesis is required by active brown adipocytes to accommodate the enhanced fuel influx, which is in turn required for thermogenesis. Concomitant synthesis of triacylglycerol and degradation of fatty acids probably constitutes a futile process that is itself thermogenic.

3.6.2. *Fatty acid oxidation, bioenergetics and thermogenesis*

Fatty acids utilized by BAT for thermogenesis are derived from several sources including dietary triacylglycerol (via chylomicrons), very low-density lipoprotein triacylglycerol from the liver, free fatty acids from white adipose bound to circulating albumin, hydrolysis of internal acyl-CoA molecules, and hydrolysis of internal triacylglycerol stores by HSL. In fact, the capacity of BAT for lipolysis actually exceeds its capacity for thermogenesis, such that it becomes an exporter of fatty acids at very high norepinephrine concentrations. Norepinephrine stimulates HSL via β_1 and β_3 adrenoreceptors as described for white adipose (Section 3.5.1). Increased synthesis of thyroxine 5'-deiodinase, responsible for increased levels of triiodothyronine and uncoupling protein, is mediated by α_1 adrenoreceptors [21,24].

Non-shivering thermogenesis is induced by heat loss when the temperature of the environment is significantly below the temperature of the organism. Thermogenesis can be suppressed by fever, exercise, and environmental temperatures similar to body temperature. Heat generation occurs through the dissipative relaxation of the proton gradient independent of ATP production (Fig. 4). The uncoupling protein facilitates proton movement down a concentration gradient across the mitochondrial membrane without simultaneous production of ATP by the proton-dependent ATP synthetase. Since the rate of ATP synthesis is usually the limiting factor of respiration and is dependent on utilization of energy from proton movement along the gradient, dissipation of the gradient by uncoupling protein uncouples oxidation from its rate limitations. Unlimited oxidation produces the large amounts of heat that are distributed by brown adipose during thermal distress.

The original uncoupling protein (termed UCP1) was discovered nearly 25 years ago, and is now recognized to be one member of a burgeoning UCP multigene family. The proteins are found in several tissues and across species. UCP1 is expressed predominantly in brown fat and is a ~306 amino acid (~33 kDa) protein predicted to span the inner mitochondrial membrane several times, projecting its C-terminus into the intermembrane space [25]. Its activity is regulated by free fatty acids, which interact with the protein in the membrane and probably lower the membrane potential for proton translocation, facilitating gradient dissipation. Uncoupling protein also has

a highly pH-dependent C-terminal purine nucleotide binding site that may serve as a regulator of the protein's activity as well. Small changes in pH drastically affect ADP- and ATP-binding to this site, and ADP/ATP-binding has been shown to inhibit proton translocation in reconstituted phospholipid vesicles.

The oxidative fuel for thermogenesis is exclusively fatty acids even if glucose is available. This is interesting because insulin facilitates uptake of large amounts of glucose during thermogenesis — much more than the cell requires for synthesis of glycerol backbones. During thermogenesis, norepinephrine activates key regulated glycolytic enzymes such as phosphofructokinase and pyruvate dehydrogenase, thus upregulating glycolysis as well as fatty acid oxidation. It has been postulated that upregulation of glycolysis may be essential for ATP production by substrate-level phosphorylation. Since ATP synthesis is uncoupled from oxidation, the cell's ATP requirements must be met another way. In addition, the cell continues to utilize large amounts of reduced cofactors to produce heat. These too can be replenished by an elevated glycolytic rate.

The transcription of uncoupling protein mRNA is upregulated by norepinephrine activation of adrenoreceptors and increases in cAMP. This upregulation can be enhanced by the presence of triiodothyronine and abolished if the deiodinase activity of the cell is inhibited. BAT has a nuclear receptor for triiodothyronine that functions as a transcription factor, and probably binds upstream of the uncoupling protein gene to activate transcription. The presence of both thyroid response elements and cAMP response elements is likely to be required.

4. Molecular cell biology of adipose tissue

4.1. Energy balance and basal metabolic rate

The balance of food intake and energy expenditure is critical for survival. Each organism represents a unique energy equation based upon its feeding habits, exercise patterns, body composition, and environmental conditions. The net result of the organism's solution to this equation determines its basal metabolic rate (BMR), which is defined in the laboratory setting as the output of a metabolite per unit time, measured at rest after an overnight fast.

Intake and storage of fatty acids must be counterbalanced by an equivalent expenditure of stored energy to maintain constant body mass. The capacity of an organism for expenditure is indexed by its BMR. So, what factors determine individual BMR? Prolonged exposure to harsh environmental conditions, such as extreme cold, leads to an elevated BMR in rodents via thermogenesis and dissipation of heat by brown adipose. Starvation or semi-starvation lowers BMR while regular strenuous and/or prolonged exercise enhances BMR. Resting muscle metabolizes primarily fatty acids, so lean body mass enhances BMR in the fed state. Obesity also elevates BMR due to the grossly increased lipolytic rates observed for an enlarged, excessively proliferated adipose tissue. However, the increased lipolysis is nonetheless overbalanced by consumption in this syndrome. Also, adipocytes of obese individuals proliferate more readily in culture than

adipocytes derived from lean individuals, suggesting fat tissue may form more easily in obesity. Increased body mass in the absence of a concomitant increase in fat-free (i.e. metabolically active) mass does not enhance BMR, since futile cycling of fatty acids between the triacylglycerol-esterified and non-esterified states is energetically not very costly. The distribution of fat, however, is a relevant factor for BMR. Sex hormones, for example, tend to direct proliferation of abdominal (visceral) adipocytes in males, but preferentially direct deposition of adipose to the lower body (gluteal–femoral region) of females. Visceral adipose displays inherently reduced sensitivity to the anti-lipolytic effects of insulin and, therefore, elevated lipolysis that contributes to higher BMR, whereas lower body adipose tends to the opposite. In addition, female adipocytes produce and secrete estrogen, which stimulates further production of preadipocytes, amplifying the estrogen effect on adipose deposition. The net effect, as in obesity, is an increased body cell mass independent of fat-free mass and a decreased BMR in females relative to males of comparable mass.

4.2. The hypothalamus-adipocyte circuit and the *ob* gene

In lean individuals, energy balance is maintained by equilibrium between consumption and expenditure. The central control for this complex mechanism is localized in the brain, and specifically to the hypothalamus. The idea that fat cell-derived secreted proteins were responsible for regulating energy intake was first proposed in the 1950s. Studies of mouse models for obesity led to the hypothesis that the adipocyte was capable of sensing when its stores were replete and transmitting a signal to that effect. The theory was expanded as several secreted products have been identified that link the biology of the fat cell to energy homeostasis (Fig. 5). The most notable of these products is leptin, the product of the *ob* gene.

There are five mouse models of obesity identified as resulting from single gene mutations (Table 3). Of these five, *ob*, and its protein product leptin, have received the most attention. Administration of leptin to obese *ob/ob* mice halts hyperphagia and reduces adiposity without dramatically adverse effects on overall body homeostasis [27]. This adipocyte-derived hormone circulates at levels that are proportional to the body fat content and has been extensively investigated for its role in regulation of body fat stores, food intake, satiety and energy expenditure. As part of its mechanism of action, leptin traverses the blood–brain barrier into cerebrospinal fluid and binds to

Table 3
Genetic models for obesity

Gene name	Allele	Protein	Physiological function
Lethal yellow	Ay	Agouti	Antagonist of melanocortin receptors
Obese	ob	Leptin	Hormone — regulates food intake
Diabetes	db	Leptin receptor	Expressed in brain and peripheral tissues — binds leptin
Fat	fat	Carboxypeptidase E	Processing and sorting neuropeptides and prohormones
Tubby	tub	Tubby protein	Possible transcription factor in hypothalamus

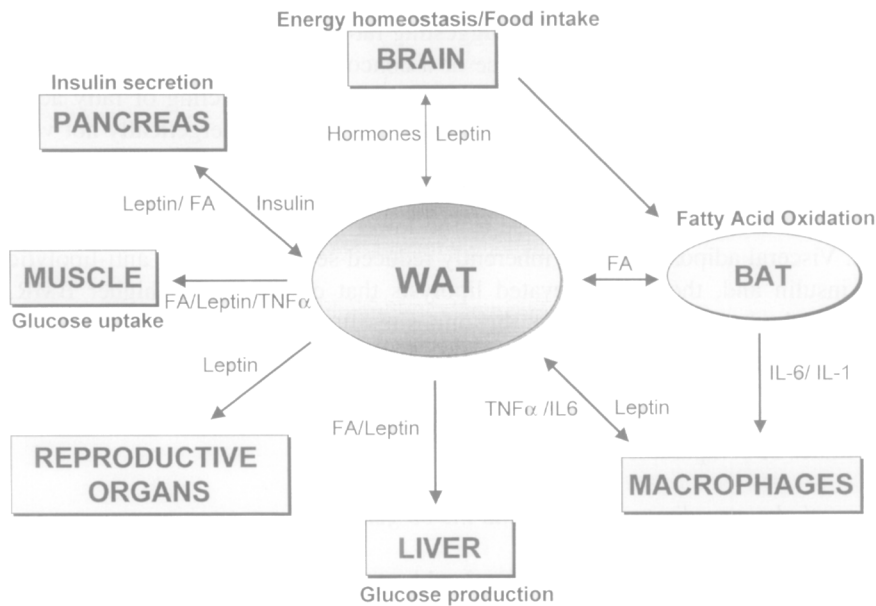


Fig. 5. The adipocyte circuit. Depicted is the network connecting the adipocyte to different target tissues by the factors it secretes. The expression and secretion of leptin by the adipocyte in response to energy balance and food intake directly affect hypothalamic function, liver (glucose production), muscle (glucose uptake) and pancreas (insulin secretion) function, macrophages and reproductive organs. Numerous endocrine factors (catecholamines, glucagon, insulin, and cytokines) secreted by these tissues also affect brown and white fat metabolism. FA, fatty acids; TNF α , tumor necrosis factor; IL-6, interleukin-6; IL-1, interleukin-1.

receptors (products of the *db* gene) on the hypothalamus, thereby controlling feeding behavior. Although the brain expresses the highest number of leptin receptors there are also leptin receptors ubiquitously expressed in peripheral tissues like adipose tissue [28] mammary gland, ovarian follicles, placenta and bone [29]. Therefore, fluctuations in leptin levels not only affect metabolism as a whole but also have a wider impact on reproduction, bone formation and the immune system. Leptin receptors are members of the class 1 cytokine receptor family and are linked to JAK-STAT signaling systems [28].

The short-term regulation of leptin expression is influenced by a number of factors that regulate C/EBP α and PPAR γ activity on the leptin promoter. Consumption of a high fat diet, insulin, glucose and glucocorticoids induce *ob* expression [29], while fasting, cold exposure, β_3 agonists, catecholamine stimulation and thiazolidinediones all decrease leptin expression [28,29]. Leptin works to suppress appetite by affecting the synthesis and/or action of a complex system of neuropeptides including neuropeptide Y, MCH, POMC, alpha-MSH, all of which directly or indirectly influence food intake.

Insulin is an additional factor that has been implicated in energy homeostasis. Insulin, although an anabolic hormone when acting directly upon liver, muscle, and adipose, mediates a completely opposite effect via the brain. Ventromedial hypothalamic lesions in rodents are associated with obesity and hyperphagia [26]. Administration of insulin directly to the ventromedial hypothalamus of rats has been shown to cease feeding and

initiate weight loss, effects that disappeared when insulin was removed. In addition, feeding was stimulated by direct administration of anti-insulin antibodies to this region. Moreover, the presence of insulin receptors in the hypothalamus and correlation of plasma insulin levels to body mass provide additional support for the putative involvement of insulin in regulation of food intake and body fat stores. One possible vehicle for this proposed insulin-signaling mechanism is the hypothalamic neurotransmitter neuropeptide Y. Administration of neuropeptide Y directly to the hypothalamus strongly stimulates feeding, an effect opposite from that of insulin. Synthesis and release of neuropeptide Y are known to be stimulated by caloric deprivation and there is evidence that insulin may reduce neuropeptide Y mRNA levels. However, it is evident that insulin, whose effects are widespread in various tissues, is not specific enough to solely exert appetite control on the time scale in which satiety and cessation of feeding occur.

4.3. Cytokine control of adipose lipid metabolism

Adipocytes have profound sensitivity to tumor necrosis factor- α (TNF- α), interferons α , β and γ , and interleukins 1, 6 and 11 [30] (Fig. 5). In general, such cytokines inhibit lipogenesis and triacylglycerol storage by adipocytes, activate lipolysis and antagonize insulin. Additionally, many cytokines interfere with proliferation of preadipocytes in 3T3-L1 or 3T3-F442A cell lines and/or diminish adipogenesis in vivo.

TNF- α is expressed at high levels in adipose tissue derived from several genetically defined rodent models of obesity and insulin resistance. In *db/db* obese mice, TNF- α expression is elevated but not the expression of other cytokines such as interleukin-1 or interleukin-6. Measurements of insulin receptor tyrosine kinase activity in obese *fa/fa* rats shows a reduced function, which could be restored by administering soluble TNF- α receptor to sequester the secreted TNF- α protein. TNF- α interferes with insulin signaling by affecting the phosphorylation of the insulin receptor, thus contributing to the development of insulin resistance in these animals. In cultured 3T3L1 cells, TNF- α inhibits the expression of C/EBP α and PPAR γ , thus leading to reduced expression of genes that are involved in triacylglycerol accumulation and metabolism. Interleukin-6 is secreted by white adipose tissue under basal conditions and its expression is increased by TNF- α and by other conditions linked to wasting disorders (cancer, cachexia, HIV).

TNF- α and the interferons decrease lipogenesis, down-regulating the mRNA levels of the key lipogenic enzymes acetyl-CoA carboxylase and fatty acid synthase; the interferons diminish mRNA levels of fatty acid synthase but not of acetyl-CoA carboxylase. TNF- α , interleukin-1, and some interferons also increase lipolysis, although the mechanism is probably post-transcriptional, since Northern blot analysis shows that TNF- α and the interferons decrease the level of HSL mRNA [30].

The adipocyte is involved in expressing and secreting a diverse range of factors (Table 4) which have a direct impact on adipocyte metabolism. These biomolecules have varied physiological functions in other target tissues thus placing the adipocyte as a central player in regulating energy homeostasis overall.

Table 4
Summary of several adipocyte secreted factors

Adipose secreted factor	Expression in the obese state	Function in vivo
Acylation stimulating protein (ASP)	Decreased	Stimulates triacylglycerol synthesis
Adiponectin/AdipoQ/Acrp 30	Decreased	Associated with insulin resistance
Adipsin	Decreased	Activation of alternative complement pathway
Angiotensinogen	Increased	Regulates blood pressure
Insulin-like growth factor 1	Increased	Proliferation/mediates effects of growth hormone
Interleukin-6	Increased	Immune response/glucose and lipid metabolism
Leptin	Increased	Energy expenditure, reproduction satiety factor
Plasminogen activator inhibitor	Increased	Cardiovascular function/wound healing
Prostaglandin E ₂	Increased	Antilipolytic, suppresses cAMP production
Prostaglandin F ₂	Increased	Inhibits adipogenesis
Prostaglandin I	Increased	Adipogenic in preadipocytes
Transforming growth factor β	Increased	Involved in proliferation, differentiation and apoptosis
Tumor necrosis factor	Increased	Contributes to insulin resistance and type 2 diabetes

5. Future directions

Obesity is an increasing worldwide public health concern. In the United States, six out of ten adults are overweight according to a recent National Health and Nutrition survey and approximately, 300,000 deaths annually are a direct result of obesity or complications that result from the disease. While obesity may seem on the surface to simply be the excess consumption of food compared to energy expenditure, it is a complex metabolic disorder centered on adipose lipid metabolism and cellular signaling systems linked to it. Adipocytes are now generally accepted to be a complex cell type involved in generating a number of signals which include cytokines, hormones and growth factors that not only affect itself and the neighboring cells but also impact target tissues involved in energy metabolism. When Albert Lehninger authored the second edition of his famous text *Biochemistry* in 1975, the topic of adipose tissue was given one and a half thin pages while discussions of liver metabolism occupied four such pages. That was not too bad considering that discussions of the metabolic functions of skeletal muscle or brain were given less than one page. At the time, Lehninger's text was the most common entry point for students interested in the topic of adipose metabolism. In the last two decades our appreciation and awareness of adipocytes as a dynamic cell type with connections to both the endocrine and nervous systems has increased dramatically. As described within, adipocytes play the central role in maintenance of the energy balance. As such, pathophysiologic conditions such as obesity and non-insulin-dependent diabetes mellitus, two of the most common disease states of the Western Hemisphere, bring adipose tissue to center stage.

Over the next decade, more transgenic and knockout mouse models will be developed to test hypotheses concerning regulation of adipose triacylglycerol metabolism (L. Chan, 2000; C. Londos, 2001) and its relationship to the overall energy balance. In addition, known genetic alleles linked to control of lipid synthesis will be explored in detail

providing additional insights into control points affecting lipid storage and oxidation (K. Reue, 1999). The combination of natural variants and laboratory model systems provides a wealth of experimental systems designed to explore lipid metabolism. Adipocytes will continue to be a favorite experimental system for the analysis of hormone action. The major effects of insulin on fat-cell metabolism have made adipocytes the system of choice for probing the mechanistic basis of insulin-stimulated glucose transport, adrenergic receptor activation of lipolysis and the connections of cytokine biology to wasting diseases. Genomic and proteomic technology platforms now support the analysis of fat cells under a variety of hormonal, metabolic and genetic backgrounds. The ability to integrate complex gene expression patterns with protein analysis into a comprehensive view of lipid metabolism and its control is within reach. As such, it remains to be seen in 20 to 25 years how many pages fat cell metabolism will warrant in the modern textbooks.

Abbreviations

ACS	acyl-CoA synthetase
ADRP	adipose differentiation-related protein
alpha-MSH	alpha-melanocyte-stimulating hormone
aP2	adipocyte fatty acid-binding protein
ARE	adipocyte response element
ARF	adipocyte regulatory factor
BAT	brown adipose tissue
bHLH-LZ	basic helix-loop-helix leucine zipper
BMR	basal metabolic rate
C/EBP	CCAAT/enhancer-binding protein
DGAT	diacylglycerol acyltransferase
FABP	fatty acid-binding protein
FATP	fatty acid transport protein
GLUT	glucose transporter
HETE	hydroxyeicosatetraenoic acid
HSL	hormone-sensitive lipase
IGF-1	insulin-like growth factor 1
IRS-1	insulin receptor substrate-1
MCH	melanin-concentrating hormone
NUC1	nuclear transcription factor 1
PDE	phosphodiesterase
PGC-1	nuclear receptor coactivator 1
PKA	cyclic AMP-dependent protein kinase A
POMC	pro-opiomelanocortin
PPAR	peroxisome proliferator-activated receptor
PPRE	peroxisome proliferator-activated receptor response elements
RXR	retinoid X receptor
SCAP	SREBP-cleavage activating protein

SRE	sterol regulatory elements
SREBP	sterol regulatory element-binding proteins
SV40	simian virus 40
TNF α	tumor necrosis factor
UCP	uncoupling protein
WAT	white adipose tissue

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Phospholipases

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1. Overview

1.1. Definition of phospholipases

Phospholipases (PLs) are a ubiquitous group of enzymes that share the property of hydrolyzing a common substrate, phospholipid. Nearly all share another property; they are more active on aggregated substrate above the phospholipid's critical micellar concentration (cmc). As shown in Fig. 1, phospholipases have low activity on monomeric substrate but become activated when the substrate concentration exceeds the cmc. The properties of phospholipids that define the aggregation state (micelle, bilayer vesicle, hexagonal array, etc.) are described in Chapter 1.

The phospholipases are diverse in the site of action on the phospholipid molecule, their function and mode of action, and their regulation. The diversity of function suggests that phospholipases are critical to life since the continual remodeling of

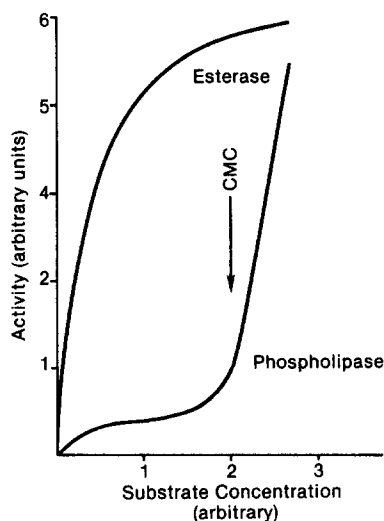


Fig. 1. Dependence of phospholipase and nonspecific esterase activity on substrate concentration. Esterase exhibits Michaelis–Menten kinetics on soluble substrates, whereas a phospholipase becomes fully active above the cmc of the substrate.